



AIRTS-FORECAST – ANALYSIS OF THE CORRELATION BETWEEN ENVIRONMENTAL CONDITIONS AND AIR POLLUTION IN TOULOUSE USING STATISTICAL AND MACHINE LEARNING METHODS

A report for the 8th-semester supervised project from École Nationale d'Ingénieurs de
Tarbes (ENIT)

CONDUCTED BY

TOLOCZKO ROSS Tiago

REINOSO URABAYEN Lucas

UNDER THE SUPERVISION OF

Prof. PhD. HOUÉ-NGOUNA Raymond (HDR)



ACKNOWLEDGEMENTS

This project was a challenging educational endeavor for both of us, and every person involved was key to the success achieved during this study.

First, we would like to express our deepest gratitude to Dr. Raymond Houé Ngouna (HDR), who guided us through this journey of discovering time-series analysis and the application of machine learning to real-world problems. His invaluable feedback has not only improved this project but also shaped our personal academic trajectories as we look toward our upcoming End-of-Studies Projects (PFE).

Special thanks to Prof. Carmen Martin, our semester supervisor, whose unwavering support for us as students, even amidst great personal difficulties, was truly appreciated.

We would also like to thank the Human Sciences (SHS) and English professors for equipping us with the skills necessary to effectively present our work. Even the best technical project is meaningless without a proper presentation, and for their guidance in this area, we are deeply grateful.

A special thanks goes to our families, who supported us from a distance during our studies abroad. Balancing rigorous academic work with personal well-being is only possible with their continued assistance.

Thank you to all our international friends, especially from European Project Semester and SPIR, who shared the enjoyable moments outside of the university this semester, helping us maintain a sense of balance and humanity amidst all the machine learning studies.

We also extend our special gratitude to all the professors who have guided us through our courses this academic year, notably those explicitly mentioned in this report, whose teachings we will carry forward into our engineering practice.

Finally, to every student, professor, staff member, and anyone else who supported us, even if only through good thoughts or prayers, we offer our deepest thanks. The smallest good deed shall not be forgotten.



SUMMARY

LIST OF PICTURES	6
ABBREVIATIONS.....	9
INTRODUCTION	11
1. CONTEXT AND CHARTER.....	12
1.1 BACKGROUND	12
1.2. OBJECTIVES OF THE PROJECT	13
1.2.1 Context [WHY].....	14
1.2.2 Content [WHAT FOR].....	14
1.2.3 Key Performance Indicators [WHAT]	15
1.2.4 Constraints: Budget [HOW MUCH] and deadline [WHEN]	15
1.2.5 Means [HOW]	16
1.3. PROJECT MANAGEMENT	17
Agile Methodology	17
1.4. LEGAL AND ETHICAL CONSIDERATIONS	18
1.4.1 Stakeholders	18
1.4.2 Environmental and Social Responsibility	19
1.4.3 Data protection in the context of France's RGPD	19
2. THEORETICAL BACKGROUND.....	21
2.1 TIMESERIES ANALYSIS – THEORY AND REMARKS.....	21
2.1.1 Stationarity.....	21
2.1.2 Seasonal-Trend Decomposition	23
2.1.3 Autocorrelation Function.....	24
2.2 CLASSICAL STATISTICAL MODELS	25
2.2.1 Autoregressive Integrated Moving Average (ARIMA).....	25
2.2.2 Seasonal Autoregressive Integrated Moving Average (SARIMA)	27
2.2.3 Holt-Winters Exponential Smoothing	27
2.3 DEEP LEARNING MODELS.....	28
2.3.1 Deployment of ANNs for study.....	30



2.3.2	Recurrent Neural Networks	32
2.3.3	LSM and its variants	34
2.3.4	Optimization	36
2.4	POLLUTANTS THEORETICAL CYCLES AND SOURCES.....	37
2.1.1	Nitrogen dioxide (NO ₂) and NO _x	37
2.4.1	Ozone	38
2.4.2	Fine and ultrafine particulate matter	38
3	DATA EXPLORATION AND BASIC STATISTICS	40
3.1	Pollution data description	40
3.1.1	Data gathering methods	40
3.1.2	Time series plots	41
3.1.3	Power Spectral Density analysis.....	42
3.1.4	Moving average, trend and stationarity.....	43
3.1.5	STL Decomposition	46
3.1.6	Memory analysis.....	48
3.1	ENVIRONMENTAL DATA DESCRIPTION	48
3.2.1	Data Gathering Methods	48
3.1.2	Data exploration methods	50
4.	TRADITIONAL STATISTICAL METHODS	52
4.1	ARIMA METHOD.....	53
4.2	SARIMA METHOD.....	53
4.3	HOLD WINTER'S METHOD.....	54
4.3	RESULTS COMPARISON	55
5.	DEEP LEARNING METHODS.....	57
5.1	HYPERPARAMETERS BAYESIAN OPTIMISATION	57
5.2	UNIVARIATE PREDICTION OF POLLUTANT TREND	59
5.2	EFFECT OF ADDED ENVIRONMENTAL FEATURES	60
5.3	RESULTS COMPARISON	63
	CONCLUSION.....	66



REFERENCES.....	69
APPENDIX – OTHER PLOTS.....	72
1. OTHER POLLUTION EDA PLOTS	72
Hourly trend	72
Hourly trend – Differentiation orders	73
Trend and stationarity comparison	73
STL Decomposition	75
ACF and PACF plots	77
2. OTHER ENVIRONMENTAL EDA PLOTS.....	78
Exogenous variables time series plots	78
Spatial average plots	80
Yearly average plots.....	82
3. OTHER DEEP LEARNING PLOTS	84
3.1 Univariate Models.....	84
3.2 Multivariate Models.....	88

LIST OF PICTURES

Figure 1.1: Global geographical distribution of studies employing Deep Learning techniques for air quality prediction. Source: LAWRENCE; BHATMANABHAN, 2026	13
Figure 1.2: a) Agile method used in this project and (b) the workflow of model architectures deployment.	18
Figure 2.1: Decomposition Plot of Daily Carbon Dioxide concentration data in ppm. Source: CLEVELAND; et al., 1990.	23
Figure 2.2: a) Logical activation perceptron and (b) sigmoid activation artificial neuron. Source: STAUNDEMEYER; MORRIS, 2019.	29
Figure 2.3: Techniques used in this project, based on the definition sequence by Le Cun, 2019.	31
Figure 2.4: A multilayer feed-forward neural network with one input layer, two hidden layers, and an output layer. Using neurons with sigmoid threshold functions, these neural networks are able to express non-linear decision surfaces. Source: STAUNDEMEYER; MORRIS,	32
Figure 2.5: An Elman neural network, an example of connected RNN with context cells. Source: STAUNDEMEYER; MORRIS, 2019.	32
Figure 2.6: a) A simple fully recurrent neural network with a two-neuron layer. The same network unfolded over time with a separate layer for each time step is shown in (b). The latter representation is a feed-forward neural network. Source: STAUNDEMEYER; MORRIS,	33
Figure 2.7: a) A standard LSTM memory block. The block contains (at least) one cell with a recurrent self-connection (CEC) and weight of '1'. The state of the cell is denoted as s_c . Read and write access is regulated by the input gate, y_{in} , and the output gate, y_o . Source: STAUNDEMEYER; MORRIS, 2019.	34
Figure 2.8: Example of a full net with 8 input units, 2 memory cell blocks of size 2 and four output units. Source: HOCHREITER; SCHMIDHUBER.	35
Figure 2.9: An illustration of the hidden activation function proposed by Cho et al., 2014. The update gate z selects whether the hidden state is to be updated with a new hidden state $\tilde{h}$. The reset gate r decides whether the previous hidden state is ignored (CHO; 36	
Figure 3.1: Hourly trend plot of PM2.5.	41
Figure 3.2: Differentiated plots of PM2.5 timeseries.	42
Figure 3.3: (a) Representative periodograms for three selected pollutants, highlighting their dominant spectral peaks. The dashed lines represent 8h, 12h, 24h, half-a-year and a year periods. (b, c) Periodograms of the first- and second-order differenced time series	43

Figure 3.4: Evaluation of underlying trends and local stationarity in the PM2.5 time series utilizing rolling mean analysis.	44
Figure 3.5: Evaluation of underlying trends and local stationarity utilizing rolling mean analysis for a representative pollutant from each chemical class, comparing 24-hour and 4,380-hour (semi-annual) temporal windows.	45
Figure 3.6: Comparison between transformed data for 24 hour- and 4380 hour-rolling average window.	46
Figure 3.7: STL Decomposition for PM2.5.	47
Figure 3.8: STL Decomposition Comparison for first- and second-order differentiation for PM2.5.	47
Figure 3.9: ACF and PACF for raw PM2.5 timeseries and its derivative.	48
Figure 3.10: Toulouse territory comprised between the bounds of the collected data, shown in Google Earth.	49
Figure 3.11: a) Timeseries evolution of dewpoint over two meters in Toulouse from 2021 to 2026. b) average spatial distribution of the temperature over the concerned period; c) anual mean.	51
Figure 4.1: Best ARIMA model (in orange) applied to forecasting the pollutant concentration series (in black)	53
Figure 4.2: Best SARIMA model (in orange) applied to forecasting the pollutant concentration series (in black)	54
Figure 4.3: Best Holt-Winters model (in orange) applied to forecasting the pollutant concentration series (in black)	55
Figure 4.4: Comparison between classical models	56
Figure 5.1: PM2.5 concentration forecasts across all evaluated models. The vertical blue dashed line delineates the start of the test dataset on March 21, 2025. The solid black line represents the observed ground truth data collected through March 21, 2026, while the dashed red line denotes the models' predictions. Optimal model performance is indicated by a high degree of overlap between the predicted and observed values. The discrepancy between the two curves, representing the residual prediction error, is highlighted by the light red shaded area.	60
Figure 5.2: Pearson correlation matrix between pollutants and environmental features.	61
Figure 5.3: PM2.5 concentration forecasts across all multivariate evaluated models. The vertical blue dashed line delineates the start of the test dataset on March 21, 2025. The solid black line represents the observed ground truth data collected through March 21, 2026, while the dashed red line denotes the models' predictions. Optimal model performance is indicated by a high degree of overlap between the predicted and	

observed values. The discrepancy between the two curves, representing the residual prediction error, is highlighted by the light red shaded area.	62
Figure 5.4: Comparative performance of the deep learning architectures, grouped by target pollutants. Solid bars denote univariate configurations, while hatched bars represent the multivariate models.	63
Figure 5.5: Comparative evaluation of univariate versus multivariate forecasting configurations, grouped by deep learning architecture. Performance enhancements are quantified by the relative reduction in forecasting error and the absolute gain in the Coefficient of Determination (R^2), where greater positive values indicate superior predictive accuracy.	64
Figure 5.6: Heatmap matrix illustrating the comparative forecasting performance between the univariate deep learning architectures and the classical statistical models' performances across all evaluated pollutants. The color gradient quantifies the relative performance, red being the worse ones and green being the better ones.	65
Figure 0.1: Foreseen stakeholders' chart for the future realization of this project as presented by the students, implicating various sectors of civil society.	68



ABBREVIATIONS

ACF – Autocorrelation Function

ADAM – Adaptive Moment Estimation

ADF – Augmented Dickey-Fuller

AI – Artificial Intelligence

ANN – Artificial Neural Networks

ARIMA – Autoregressive Integrated Moving Average

Bi-LSTM – Bidirectional Long Short-Term Memory [Recurrent Neural Network]

BP – Backpropagation

BPTT – Backpropagation through time

CEC – Constant Error Carousel

CNN – Convolutional Neural Network

CO₂ – Carbon Dioxide

COPD – Chronic Obstructive Pulmonary Disease

D2M – 2 meters dewpoint temperature

DL – Deep Learning

DTF – Discrete Fourier Transform

EDA – Exploratory Data Analysis

ENIT – École Nationale des Ingénieurs de Tarbes

EU – European Union

FAL – Forecast Albedo

FFNN – Feedforward neural networks

GDPR – General Data Protection Regulation

GRU – Gated Recurrent Unit

LSTM – Long Short-Term Memory [Recurrent Neural Network]

MAE – Mean Absolute Error

MAPE – Mean Absolute Percentage Error

ML – Machine Learning

MSE – Mean Squared Error

NH₃ – Ammonia

NO₂ – Nitrogen dioxide



NO_x – Nitrogen oxides

O₃ - Ozone

PACF – Partial Autocorrelation Function

PAH – Polycyclic aromatic Hydrocarbons

PM – Particulate Matter

PPA – Plan de Protection de l'Atmosphère [Atmospherical Protection Plan]

PSD – Power Spectral Analysis

R² – Correlation Coefficient

RGPD - Règlement Général sur la Protection des Données

RMSE – Root Mean Square Error

RNN – Recurrent Neural Network

RSE - Responsabilité Sociale de l'Entreprise

SARIMA – Seasonal Autoregressive Integrated Moving Average

SGD – Stochastic Gradient Descent

SKT – Skin Temperature

SO₂ – Sulfur Dioxide

SP – Surface Pressure

STL – Seasonal-Trend decomposition using Loess [smoothing]

STL1 – Soil temperature at level 1

STL2 – Soil temperature at level 2

STL3 – Soil temperature at level 3

STL4 – Soil temperature at level 4

T2M – temperature at 2 meters above the soil

U10 – u component of wind at 10 meters from soil

UTTOP - Université de Technologie Tarbes-Occitanie-Pyrénées

V10 – v component of wind at 10 meters from soil

VOC – Volatile Organic Compounds

ZAS – Zone Administrative de Surveillance

INTRODUCTION

As an integral component of the engineering curriculum at the École Nationale d'Ingénieurs de Tarbes (ENIT), eighth-semester students undertake a comprehensive technical project designed to synthesize their academic training with applied industry challenges. Under the supervision of Dr. Raymond Houé Ngouna, and motivated by the rapid convergence of Artificial Intelligence (AI) and critical environmental risk management, this project investigates the predictive modeling of atmospheric pollution. Specifically, we analyze the spatiotemporal dynamics of key pollutants and their correlation with exogenous meteorological variables (e.g., wind velocity, atmospheric pressure, and temperature) utilizing advanced machine learning methodologies.

The primary objective of this research is to develop robust forecasting models capable of accurately predicting extreme pollution events to support proactive environmental decision-making. To achieve this, we deployed a dual-tiered methodological approach. We initially established classical statistical baselines, utilizing classical models informed by Autocorrelation Function (ACF) analysis. Subsequently, we transitioned to a sophisticated deep learning framework, evaluating a suite of Recurrent Neural Network (RNN) architectures, including GRU, LSTM, Bi-LSTM, and hybrid configurations. These models were tested under both univariate and multivariate conditions, with hyperparameter optimization rigorously managed via the Optuna framework (AKIBA; *et al.*, 2019).

This report is structured to systematically detail this analytical progression:

- **Section 1** outlines the project context, fundamental objectives, and the established stakeholder charter.
- **Section 2** establishes the theoretical framework, encompassing atmospheric pollution dynamics, time-series data processing, and the mathematical principles governing both the classical and deep learning algorithms utilized in this study.
- **Section 3** details the exploratory data analysis (EDA) of the pollutant time series, highlighting the critical identification of seasonal trends and local stationarity.
- **Sections 4 and 5** present the implementation, optimization, and comparative results of the classical and deep learning models. These sections empirically demonstrate the superior capacity of neural networks to capture complex, high-frequency emission events compared to traditional baselines.
- **Section 6** synthesizes these findings, contextualizing the models' performance within the broader scope of governmental decision-support systems and proposing strategic avenues for future product development and research.

1. CONTEXT AND CHARTER

1.1 BACKGROUND

In recent years, air quality has become a major healthcare and societal concern, as it is estimated that 6.7 million deaths arise each year due to outdoor pollution, with two billion households exposed (WHO, 2022). For instance, fine and ultrafine particulate matter (PM) is particularly of concern, since it penetrates the respiratory system, causing permanent damage (SCHRAUFNAGEL, *et al.*, 2019). Beyond carbon dioxide (CO₂), gaseous pollutants such as nitrogen and sulfur oxides (NO_x and SO₂), primarily generated from engine combustion, are also linked to acidic rain and lung diseases (GÓMEZ-GARCIA; PITCHON; KIENNEMANN, 2005). Even ground-level ozone (O₃) is found to be a hazard, contributing to photochemical smog (WHO, 2022).

These concerns have been considered by several public policies, regarding industry, energy, transportation, urban planning, among other sectors. As an example, in the French Region of Occitanie, we find the Atmospheric Protection Plan (Plan de protection de l'atmosphère, PPA), in its capital, Toulouse. These measures have been held since 1996 and have attained success in recent decades.

The escalating impacts of global pollution migrated also to the engineering domain. Several engineering and applied sciences studies have been developed to answer global pollution climates, for example, the adoption of catalytic filters for gasoline engines (ROOT, *et al.*, 2019) and the ongoing air-filtering systems research (TRINH, *et al.*, 2025).

Considering all these areas, given the importance of data science for the engineering studies and career, and the current trends in deep learning and environmental concerns, we have decided, with Dr. Raymond HOUÉ NGOUNA, to conduct the project which is detailed along this report. The problem is posed as follows: how can modern data science techniques, classical prediction models and deep learning alike, be applied to data pollution and help as an aid to decision making?

This project carries also an innovative tone since the application of deep learning techniques on pollution concentration forecast is currently being covered in many countries, the most studies being found in China, India, South Korea, Turkey, and Taiwan (LAWRENCE; BHATMANABHAN, 2026). France sits well behind, in 15th place (Figure 1), and quick research in the academic AI tool Consensus results in no publications in this country related to the usage of recurrent neural networks in pollution data analysis.

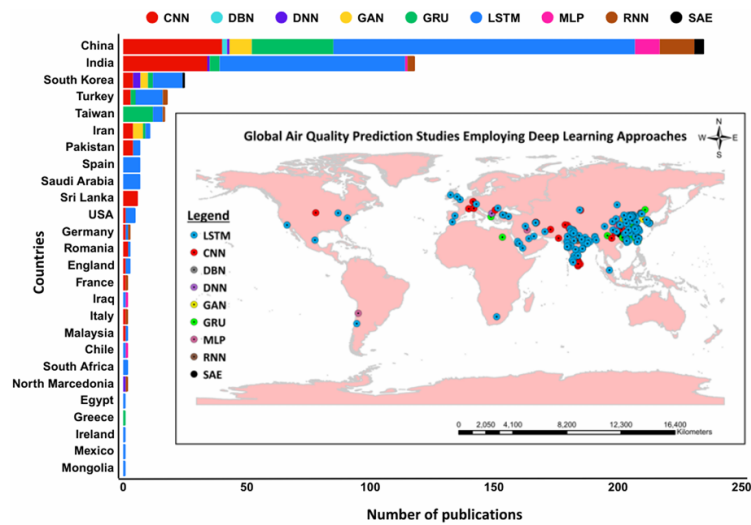


Figure 1.1: Global geographical distribution of studies employing Deep Learning techniques for air quality prediction. Source: LAWRENCE; BHATMANABHAN, 2026

Following the project delimitations, as well as considerations regarding environmental and legal concerns, we shall pose the theoretical foundations to solve the problem: the place of data science in the modern engineering praxis, how to get data, the foundation of classical statistics and deep learning (DL) prediction, the dangers of air pollution, and benchmark methods in decision-making. We then shall analyze the pollution data using traditional timeseries analysis such as timeseries trend and stationarity recognition, by moving average and the augmented Dickey-Fuller test (ADF test), Seasonal and Trend decomposition using Loess' algorithm (STL), and memory analysis using autocorrelation function (ACF and PACF).

Then, we shall use classical methods for prediction: Autoregressive Integrated Moving Average (ARIMA), and its seasonal form (SARIMA) and the Holt-Winter method. The DL methods developed are the following: the simple vanilla Recurrent Neural Network (RNN), Long Short-Term Memory (LSTM), its bi-directional form (Bi-LSTM), and its simplified form, the Gated Recuring Unit (GRU). To conclude, we shall compare between models and give remarks regarding event recognition and decision-making.

1.2. OBJECTIVES OF THE PROJECT

Here are the objectives structured according to the methodology proposed by CLERMONT, 2016.

1.2.1 Context [WHY]

As stated, this project applies data science to the domain of air pollution management. There has been great development in technological solutions to manage air pollution, which could be generally divided in two categories: preventive measures, as those that hinder pollutant emission (such as the catalytic filters for combustion engines), and protective measures, as those with seek to limit the exposure of humans to the pollution (such as ambient air-filtration systems for respiratory protection).

Data analysis is uniquely suited for another class of solutions, beyond engineering research solutions: public policies. Again, in the context of the usage of data analysis and models for pollution forecast, two operational purposes could be distinguished. First, preventive forecasting, to prepare authorities to implement policies for upcoming or ongoing pollution conditions (LAWRENCE; BHATHMANABHAN, 2026). Second, evaluative analysis to retroactively assess the effectiveness of an applied policy (FENG, *et al.*, 2024).

The advent of machine learning and deep learning has equipped researchers methods than outperform classical ones. These models have been extensively used for pollution monitoring, processing large data sets describing nonlinear, stochastic dynamic processes. To date, several models have been used so far, often integrating exogenous variables as features (e.g. wind speed, temperature, pressure), such as: Convolutional Neural Networks (CNN), LSTM, RNN, GRU and various hybrid architectures (LAWRENCE; BHATHMANABHAN, 2026).

Finally, recent policies in the city of Toulouse, located in the French region of Occitanie, notably its "Plan de Protection de l'Atmosphère 2025-2030", are now at work for public health protection and ensuring regulatory limit value for the pollutants. The plan highlights the dangers of NO_x, NO₂, coarse and fine particulate matter (PM₁₀ and PM_{2.5}), ozone (O₃), volatile organic compounds (VOC's) and ammonia (NH₃), which strongly relies on observations and predictions to accurately trigger administrative action (DE CREPY; DREAL, 2023).

1.2.2 Content [WHAT FOR]

The project shall begin with the description of the pollutant data, gathered from the GEOD'AIR database of the Zone Administrative de Surveillance (ZAS) code FR76ZAG01. The analyzed pollutants will be: NO₂, NO_x, O₃, PM₁₀, and PM_{2.5}, from March 21st, 2021, to March 21st, 2026, a five-year span, corresponding to the available data from the meteorological surveillance sites. Subsequently, an exploratory data analysis (EDA)

will be executed to identify stationarity, periodicity, trends, and autocorrelation in the datasets.

A following study will be conducted to evaluate the forecast of the classical methods and then, the DL models, which undergo optimization using Bayesian optimization methods. As atmospheric pollutants follow complex dynamics influenced by meteorological variables, The DL models incorporate those exogenous variables inputs, from the ERA5-Land database. The models will be evaluated according to error and compared between them, and the result of this project is going to be presented in terms of the best models to predict pollution.

Beyond immediate technical and formative aspects, this project allows foreseeable continuation. The findings are contextualized in the French framework of Responsabilité Sociale de l'Entreprise (RSE), leading to the conscience of the implications these models have in public policies and proactive polluting event recognition.

1.2.3 Key Performance Indicators [WHAT]

The main indicators in this study are the error related measures Root Mean Square Error (RMSE), Mean Absolute Error (MAE), and Mean Absolute Percentage Error (MAPE). The first one, RMSE reflects the accuracy of the prediction, disproportionately penalizing significant deviations from the expected value.

$$RMSE = \sqrt{\frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2} \quad (1.1)$$

Then, MAE calculates the arithmetic mean of absolute error values, providing a linear penalty for forecast errors and being highly interpretable as the average error in the exact same physical units.

$$MAE = \frac{1}{n} \sum_{i=1}^n |y_i - \hat{y}_i| \quad (1.2)$$

Finally, MAPE is a version of MAE which is the ratio of accuracy of the forecast, being scale-independent and therefore used as a cross-model comparison.

$$MAPE = 100 \frac{1}{n} \sum_{i=1}^n \left| \frac{y_i - \hat{y}_i}{y_i} \right| \quad (1.3)$$

1.2.4 Constraints: Budget [HOW MUCH] and deadline [WHEN]

As a computationally focused study utilizing open-source databases and existing university and student infrastructure, the Air-TS Forecast project required no preallocated

financial budget. The project was conducted from February 2026 to the final presentation on the 17th of June 2026.

Project milestones followed:

1. Phase 1: Initial meetings and project scope delimitation;
2. Phase 2 Pollution and environmental data gathering;
3. Phase 3: Baseline forecasting using classical models;
4. Phase 4: DL model analysis;
5. Phase 5: Hyperparameter tuning via Bayesian optimization;
6. Phase 6: Comprehensive cross-model final comparisons and improvements;
7. Phase 7: Compilation of final report and presentation.

1.2.5 Means [HOW]

- Databases: GEOD'AIR (for endogenous pollutant concentrations) and ERA5-Land (for exogenous meteorological variables).
- Software Stack: The computational pipeline is developed in Python, utilizing Jupyter Notebooks for interactive data exploration and reporting. The core environment relies on essential libraries including PyTorch (deep learning), scikit-learn (preprocessing and classical metrics), pandas (time-series data manipulation), and Optuna. The version control was made using GitHub and the repository containing the programs used for pollutant and environmental EDA and DL model deployment was published on: [tiagoross-ops/AirTS_Forecast](https://github.com/tiagoross-ops/AirTS_Forecast).
- Exploratory Data Analysis (EDA): The diagnostic phase includes time-series visualization, Power Spectral Density (PSD) analysis for extracting cyclical periodicities, stationarity verification via the Augmented Dickey-Fuller (ADF) test post-moving average smoothing, STL decomposition, and memory analysis using Autocorrelation and Partial Autocorrelation Functions (ACF and PACF).
- The exploratory analysis with regards to the exogenous meteorological data focuses on spatial aggregation and harmonic fitting to characterize cyclic environmental behaviors. To systematically process these variables, the following classes of metrics are established: spatial aggregation metrics (overall average, spatial average, zonal mean and meridional mean), time-dependent cyclical metrics (amplitude, phase shift, vertical shift, seasonal correlation)
- Forecasting Architectures: Baseline models consist of ARIMA, SARIMA, and Holt-Winters. Advanced architectures include RNNs, LSTMs, Bi-LSTMs, and

GRUs. Deep learning performance is continuously monitored via training and validation of loss curves to mitigate overfitting.

- Hyperparameter Tuning: Optimization of the neural network architectures is conducted using Bayesian optimization frameworks via the Optuna library.

1.3. PROJECT MANAGEMENT

Agile Methodology

The methodology illustrated in the Figure 1.2:a) follows an Agile approach based on iterative development. The process begins with an objective setting, where the goals and expected outcomes are defined. Once the objectives are established, a planning phase is carried out to organize the tasks, resources, and workflow required to achieve them.

The project then moves to task development, during which the planned activities are implemented. After completion, the results are presented and evaluated during the presentation of results stage. A decision point is then reached to determine whether corrections or improvements are required. If necessary, the process returns to the development stage, allowing modifications and refinements to be made. If the results are satisfactory, the project progresses to the next stage.

This iterative cycle promotes continuous improvement and flexibility, ensuring that each stage is validated before moving forward.

In this project, the Agile cycle was repeated three times, corresponding to the three main phases of the work:

1. Data collection and data exploration.
2. Classical and deep learning prediction models development.
3. Hyperparameter optimization and model evaluation.

Repeating the process for each phase allowed progressive refinement of the methodology and continuous improvement of the forecasting results.

Figure 1.2: b) illustrates the workflow adopted for the forecasting task. Two main data sources were used: air pollution data obtained from the GEO D'AIR platform and meteorological data extracted from the ERA5-LAND dataset.

The pollution data were used as input for both the classical statistical models and the deep learning models. The classical modeling branch included ARIMA, SARIMA, and

Holt-Winters, which rely on the temporal structure of the pollutant concentration series to generate forecasts.

For the deep learning branch, the pollution data were combined with meteorological variables from ERA5-LAND in order to provide additional information about atmospheric conditions influencing pollutant concentrations. Several neural network architectures were investigated, including RNN, LSTM, Bi-LSTM, and GRU.

The outputs generated by both modeling approaches were then compared and evaluated to produce the final data predictions. This workflow allowed the assessment of both traditional time-series forecasting techniques and data-driven deep learning approaches using the same study framework.

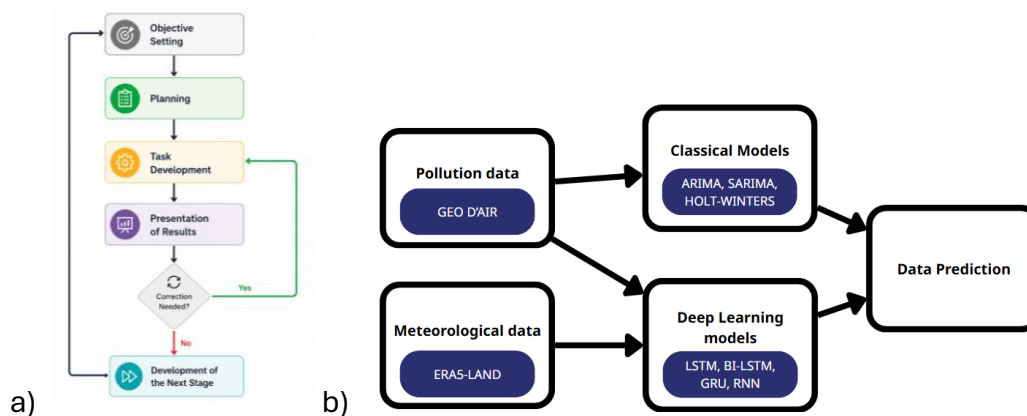


Figure 1.2: a) Agile method used in this project and (b) the workflow of model architectures deployment.

1.4. LEGAL AND ETHICAL CONSIDERATIONS

1.4.1 Stakeholders

To ensure clear accountability and project governance, the primary stakeholders of the Air-TS Forecast project are defined as follows:

- Development Team: Lucas Reinoso Urabayen and Tiago Toloczko Ross.
- Academic Supervisor: Dr. Raymond Houé Ngouna.
- Institution: UTTOP (Université de Technologie de Tarbes Occitanie Pyrénées).
- Potential End Users and Beneficiaries: Local authorities (such as the municipality of Toulouse), environmental monitoring agencies, and the public.

1.4.2 Environmental and Social Responsibility

This project actively contributes to environmental awareness and public health prevention by analyzing atmospheric data to identify underlying patterns. Specifically, it seeks to correlate environmental conditions and pollution peaks with localized temporal anomalies, such as large-scale festivals or sporting events. By integrating meteorological data (temperature, humidity, atmospheric pressure, and dew point) with targeted air pollution indicators (gas concentrations and particulate matter), the system generates predictive insights valuable to both citizens and public administrators.

From an environmental perspective, the project promotes:

- A rigorous, data-driven approach to environmental monitoring.
- A deeper understanding of urban pollution dynamics and dispersion.
- Analytical support for sustainable urban planning and decision-making.

From a social perspective, the predictive capabilities of the models aim to:

- Empower vulnerable populations to minimize exposure during anticipated pollution peaks.
- Elevate public awareness regarding the direct impacts of air quality.
- Support municipal authorities in deploying preventive public health actions ahead of critical pollution events.

Ultimately, the project emphasizes the responsible use of environmental data, aiming to provide transparent, accessible, and actionable information for societal benefit.

1.4.3 Data protection in the context of France's RGPD

The Air-TS Forecast project exclusively utilizes publicly available environmental datasets sourced from the Copernicus Climate Change Service (ERA5-Land reanalysis dataset) and Geod'Air (air quality measurements for the Toulouse agglomeration). The aggregated data comprises meteorological indicators and atmospheric pollution measurements.

Because the project strictly processes environmental observations and public event schedules, it involves no Personally Identifiable Information (PII) or sensitive individual data. Consequently, within the framework of the European General Data Protection Regulation (GDPR)—known in France as the *Règlement Général sur la*



Protection des Données (RGPD)—the project inherently complies with the highest standards of data privacy. It strictly adheres to the following principles:

- Exclusive utilization of open-access, anonymized datasets.
- Total transparency regarding data provenance and methodology.
- Complete absence of individual tracking, surveillance, or demographic profiling.

The foundational objective is to advance environmental monitoring and public health prevention while operating fully within the ethical boundaries of EU data protection regulations and the civic goals of Toulouse's PPA.

2. THEORETICAL BACKGROUND

2.1 TIMESERIES ANALYSIS – THEORY AND REMARKS

Time-series analysis is a well-established discipline applied across numerous domains, including finance, economics, medicine, and geospatial sciences, to study the behavior of temporally evolving variables. Typically, these time-based observations are discrete, acquired primarily through periodic sensor measurements (PRENAT, 2012).

Atmospheric pollutants such as NO₂, NO_x, O₃, PM_{2.5}, and PM₁₀ exhibit complex temporal behaviors. Their concentrations are influenced by recurrent human activities, meteorological conditions, seasonal effects, and stochastic fluctuations. As a consequence, pollution measurements are naturally represented as a time series, where observations are collected sequentially over time.

Notation is as follows: let $\mathbf{X}$ be a random real process indexed by the set $\mathbf{T}$ of timesteps, where $\mathbf{X} = \{\mathbf{x}_t\}$, t being the timesteps in $\mathbf{T}$, and the elements $\mathbf{x}_t$ are the measures vectors collected at each timestep, with values in the state space $E = \mathbb{R}^k$. The process is said to be univariate if $k = 1$, that is, if the vectors $\mathbf{x}_t$ are one-dimensional (scalars) and chronological if the set $\mathbf{T}$ maps the Natural numbers, $\mathbf{T} = \mathbf{N}$ (PRENAT, 2012).

For modeling and forecasting purposes, a chronological time series is typically formulated as the sum of a deterministic mean function and a zero-mean stochastic process. These stochastic processes often exhibit inherent periodicities that can be identified through PSD analysis, a technique that visualizes the dominant frequencies of the series as distinct peaks within a periodogram. For a given time series $\mathbf{X}_t$, the Discrete Fourier Transform (DFT), which maps the time-domain signal into the frequency domain to isolate these spectral components, is defined in Equation 2.1 (PRENAT, 2012).

$$I_n(\omega_j) = n^{-1} \sum_{t=1}^n |X_t e^{-i\omega_j t}|^2, \quad i = \sqrt{-1}, \quad \omega = -\pi + j \frac{2\pi}{n}, \quad j \in \mathbb{N} \quad (2.1)$$

2.1.1 Stationarity

Before applying time series forecasting models, it is important to study whether the PM_{2.5} series is stationary. A stationary time series is one whose statistical properties remain relatively stable over time. In simple terms, this means that the mathematical hope, variance, and general behavior of the series do not change too strongly across the observation period, expressed as $E(\mathbf{x}_t) = E(\mathbf{x}_t) = \text{constant}$, as stated in Equation 2.2. Also, in a stationary series, covariance $E[\mathbf{X}_{t_2} - E(\mathbf{X})][\mathbf{X}_{t_1} - E(\mathbf{X})]$ is only dependent on intervals $\tau = t_2 - t_1$, being written as a single-variable function (Equation 2.3) (PRENAT, 2012).

$$\frac{dE(X)}{dt} = 0 \rightarrow E(X_t) = E \quad (2.2)$$

$$\gamma_x(\tau) = E[X_t - E][X_{t+\tau} - E] \quad (2.3)$$

Stationarity is especially important for classical models such as ARIMA and SARIMA, because these models assume that the structure of the series is stable enough to be learned and forecasted. If a series is not stationary, transformations such as differencing or smoothing may be required before modeling. A technique to be used in this study is the numerical differentiation (Equation 2.4) of the dataset to extract the trending component of the series for ARIMA and SARIMA models. For example, the raw dataset could have a non-zero first derivative for mathematical hope, thus not being stationary. The dataset passes through differentiation and now it is the first derivative which is analyzed. Now, the first derivative for mathematical hope is more likely to be zero, but one could repeat n times numerical differentiation on the dataset until achieve the zero-valued derivative for mathematical hope on the n^{th} order differentiated series for analysis.

$$\frac{df}{dx} \approx \frac{f(x_i + h) - f(x_i)}{h}, \quad h = x_i - x_{i-1} \quad (2.4)$$

To evaluate stationarity, the Augmented Dickey-Fuller test, also called the ADF test, was applied. This statistical test checks whether a time series has a unit root. A unit root indicates that the series is non-stationary, meaning that shocks or changes can have a persistent effect over time. The ADF test is interpreted using the p-value. The p-value indicates whether there is enough statistical evidence to reject the hypothesis that the series is non-stationary. The hypotheses are:

- H_0 : the unit root is present in an autoregressive (AR) model;
- H_1 : the series is stationary

A common decision rule is:

- $p\text{-value} < 0.05 \rightarrow \text{reject } H_0 \rightarrow \text{the series is considered stationary}$
- $p\text{-value} \geq 0.05 \rightarrow \text{fail to reject } H_0 \rightarrow \text{the series is considered non-stationary}$

The simple AR model is given by Equation 2.5, where y_t is the variable of interest; t is the time index, ρ a coefficient and equals 1 if the series has a root, and u_t is the error term, modelled as white noise. A regression model (Equation 2.6) and the augmented version of the test (Equation 2.7) (ADF) for stationarity are then applied to the dataset, with the intuition being that if the series y is stationary, then it has a tendency to return to a constant (WIKIPEDIA, 2025; WIKIPEDIA, 2025);

$$y_t = \rho y_{t-1} + u_t \quad (2.5)$$

$$\Delta y_t = (\rho - 1) y_{t-1} + u_t = \delta y_{t-1} + u_t \quad (2.6)$$

$$\Delta y_t = \alpha + \beta t + \gamma y_{t-1} + \delta \Delta y_{t-1} + \dots + \delta_{p-1} \Delta y_{t-p+1} + \varepsilon_t \quad (2.7)$$

To reveal trends and set the boundaries for stationarity, a moving average method was used on the raw timeseries data.

2.1.2 Seasonal-Trend Decomposition

Following stationarity analysis, a time series Y can be decomposed into three distinct components: trend, seasonality, and residuals, as expressed in Equation 2.8. Cleveland et al. (1990) proposed a robust decomposition procedure based on locally weighted scatterplot smoothing (Loess), formally known as Seasonal and Trend decomposition using Loess (STL). This algorithmic approach utilizes two iterative loops: an inner loop consisting of six steps to iteratively update the seasonal and trend components, and an outer loop designed to compute robustness weights that mitigate the influence of transient outliers. The original authors demonstrated the efficacy of this method by applying it to long-term atmospheric carbon dioxide (CO_2) concentrations (referenced in Figure 1.3), thereby directly validating its applicability to the pollutant concentration forecasting undertaken in this project.

$$Y = T + S + R \quad (2.8)$$

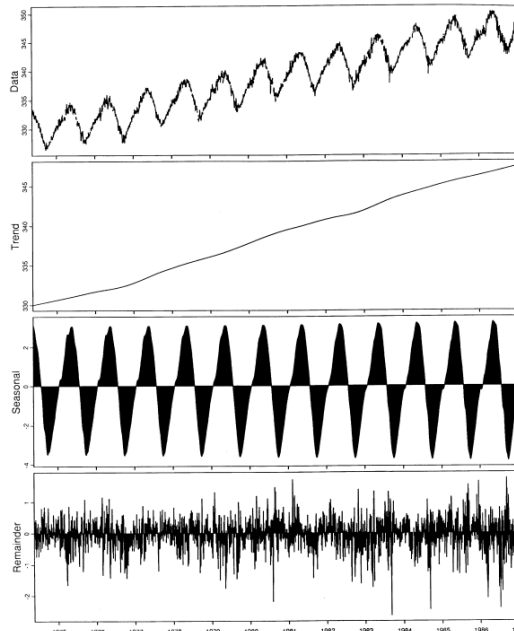


Figure 2.1: Decomposition Plot of Daily Carbon Dioxide concentration data in ppm. Source: CLEVELAND; et al., 1990.

Consequently, STL offers a robust yet straightforward framework highly suited for Exploratory Data Analysis (EDA). It is readily accessible through open-source Python libraries and has been successfully adapted for various forecasting applications (CLEVELAND; *et al.*, 1990). Because the selection of algorithmic parameters significantly impacts overall decomposition performance, the tuning process for this study was directly informed by the analytical techniques discussed previously.

Given the highly dynamic nature of atmospheric data, it is important to align the mathematical assumptions of the STL algorithm with the physical realities of pollutant dispersion. While STL is fundamentally an additive decomposition model, air quality time series frequently exhibit multiplicative behavior, necessitating variance-stabilizing transformations prior to analysis.

Beyond mapping these predictable atmospheric cycles, the primary value of the STL method for Exploratory Data Analysis lies in its isolation of the residual component. In urban environmental engineering, these extracted residuals are not merely mathematical error; rather, they serve as direct physical indicators of highly stochastic anthropogenic anomalies. Because extreme pollution spikes are predominantly driven by irregular, non-linear human events, such as sudden traffic congestion or isolated industrial emissions, the residual extraction successfully decouples these unpredictable impacts from the natural diurnal rhythm. By isolating these stochastic emission events, the STL framework provides a deeply clarified dataset, enhancing the feature extraction and predictive accuracy of the subsequent multivariate deep learning models.

2.1.3 Autocorrelation Function

Finally, stochastic processes must also be evaluated in terms of their autocorrelation.

This metric quantifies the degree to which two observations at different time steps, separated by a specific time lag, are correlated with one another. The autocorrelation function (ACF), provided in Equation 2.9, is derived from the autocovariance function (Equation 2.10) by normalizing it against the variance of the time series. Furthermore, to identify the optimal lags for autoregressive (AR) processes and determine the appropriate parameters for the **ARIMA(p, d, q)** model, detailed in the subsequent sections, it is standard practice to utilize the partial autocorrelation function (PACF). As defined in the Durbin-Levinson Algorithm, in Equation 2.11, the PACF isolates the direct partial correlation of a stationary time series with its own lags by removing the linear influence of any intermediate time steps (WIKIPEDIA, 2026; WIKIPEDIA, 2026).

$$R_{xx}(t_1, t_2) = E[X_{t_1}, \bar{X}_{t_2}] \quad (2.9)$$

$$K_{xx}(t_1, t_2) = E[X_{t_1}, \bar{X}_{t_2}] - \mu_{t_1} \mu_{t_2} \quad (2.10)$$

$$\phi_{n,n} = \frac{\rho(n) - \sum_{k=1}^{n-1} \phi_{n-1,k} \rho(n-k)}{1 - \sum_{k=1}^{n-1} \phi_{n-1,k} \rho(k)}, \quad \rho(n) \text{ being the ACF} \quad (2.11)$$

2.2 CLASSICAL STATISTICAL MODELS

Before the emergence of modern Machine Learning and Deep Learning techniques, forecasting problems were primarily addressed through statistical models specifically designed to capture temporal dependencies and recurring patterns. These approaches remain highly relevant today because they provide interpretable results, require relatively low computational resources, and often serve as benchmark methods against which more complex models are evaluated.

Among the most widely adopted statistical forecasting methods are the Autoregressive Integrated Moving Average (ARIMA), its seasonal extension of Seasonal Autoregressive Integrated Moving Average (SARIMA), and the Holt-Winters Exponential Smoothing method. Each of these approaches' models temporal dynamics differently, allowing the identification of trends, seasonality, and autocorrelation structures within the data.

In this project, these classical statistical methods are employed as baseline forecasting models for pollutant concentration prediction. Their performance will subsequently be compared with Deep Learning architectures to evaluate the advantages and limitations of each approach when applied to atmospheric pollution forecasting.

2.2.1 Autoregressive Integrated Moving Average (ARIMA)

The Autoregressive Integrated Moving Average (ARIMA) model is one of the most widely used forecasting methods for time series analysis. Its objective is to predict future values based on the information contained in past observations and past forecasting errors. The model is particularly suitable for data exhibiting temporal dependencies, such as atmospheric pollutant concentrations.

An ARIMA model is represented as ARIMA (p,d,q), where the parameters (p), (d), and (q) describe different characteristics of the time series.

The Autoregressive Component (p):

The autoregressive component expresses the degree to which the current value of the series is influenced by previous observations. The parameter (p) represents the number of past observations used to estimate the current value.

In practical terms, a pollutant concentration measured at a given time is often correlated with concentrations observed during previous hours or days. This phenomenon is known as a temporal memory. The autoregressive term allows the model to exploit this dependency by incorporating past values into the prediction process.

A larger value of (p) implies that the model considers a longer historical influence, while a smaller value focuses only on the most recent observations.

The Integrated Component (d):

The integrated component corresponds to the number of differencing operations applied to the original series to achieve stationarity. A stationary time series is characterized by statistical properties, such as mean and variance, that remain approximately constant over time. Many forecasting techniques, including ARIMA, require stationarity because the relationships learned from historical data are assumed to remain valid in the future.

When a time series exhibits a trend or long-term evolution, differencing is applied by subtracting consecutive observations. The parameter (d) therefore represents the degree of transformation required to remove trends and stabilize the series.

For example:

- $(d = 0)$: the series is already stationary;
- $(d = 1)$: the first difference is applied;
- $(d = 2)$: a second differencing operation is required.

The appropriate value of (d) is generally determined through stationarity tests such as the Augmented Dickey-Fuller (ADF) test.

The Moving Average Component (q):

The moving average component models the influence of previous prediction errors on future forecasts. The parameter (q) specifies the number of past error terms incorporated into the model.

Unlike the autoregressive component, which relies on previous observations, the moving average component uses information about the discrepancies between observed and predicted values. This mechanism allows the model to compensate for systematic forecasting errors and improve prediction accuracy.

A larger value of (q) indicates that the model considers a longer history of prediction errors, while a smaller value focuses only on recent forecasting deviations.

Together, the parameters (p), (d), and (q) enable ARIMA to capture temporal dependencies, remove non-stationary behavior, and account for residual forecasting errors, making it a powerful baseline method for environmental time series forecasting.

2.2.2 Seasonal Autoregressive Integrated Moving Average (SARIMA)

While the ARIMA model can capture temporal dependencies between consecutive observations, many real-world phenomena exhibit repetitive patterns occurring at regular intervals. Air pollution is a typical example, as pollutant concentrations are influenced by recurring human activities and meteorological cycles. Daily traffic peaks, weekly work schedules, and seasonal weather variations often generate periodic behaviors in atmospheric pollutant concentrations.

To account for these recurring patterns, the Seasonal Autoregressive Integrated Moving Average (SARIMA) model extends the classical ARIMA framework by introducing a seasonal component. In addition to considering the influence of recent observations, SARIMA incorporates information from previous seasonal cycles.

For example, when forecasting the concentration of nitrogen dioxide (NO_2) at a given hour, the model may not only use the observations recorded during the previous hours but also consider the concentration observed at the same hour on the previous day or during the previous week. This additional seasonal memory allows the model to better reproduce periodic behaviors frequently observed in environmental data.

A SARIMA model is generally denoted as SARIMA (P,D,Q) s , where the parameters (P,D,Q) describe the seasonal autoregressive, differencing, and moving average components, while s represents the seasonal period. The value of s depends on the frequency of the data and the dominant periodicity identified during exploratory analysis.

Because atmospheric pollution is strongly affected by recurring human and environmental processes, SARIMA is often considered one of the most effective classical forecasting methods for air quality prediction.

2.2.3 Holt-Winters Exponential Smoothing

The Holt-Winters method, also known as Triple Exponential Smoothing, is a forecasting technique specifically designed for time series that simultaneously exhibits a trend and a seasonal pattern.

Unlike ARIMA and SARIMA, which rely on autoregressive relationships and past forecasting errors, Holt-Winters decomposes the time series into three fundamental components: the level, the trend, and the seasonality.

The level represents the average magnitude around which the observations fluctuate. The trend describes the long-term evolution of the series, indicating whether pollutant concentrations tend to increase or decrease over time. Finally, the seasonal component captures repetitive variations occurring at regular intervals, such as daily or weekly pollution cycles.

The forecasting process continuously updates these three components as new observations become available. Consequently, the model can adapt to gradual changes in pollutant concentrations while preserving the recurring seasonal behavior present in the data.

2.3 DEEP LEARNING MODELS

In the early days of computer science, Alan Turing, regarded as the father of this field, had already established a test to verify whether a computing machine could be deemed “intelligent”, in an attempt to answer the question: “can machines think?” (TURING, 1950). Later in the same decade, Claude Shannon, John McCarthy, Nathaniel Rochester and Marvin Minsky, tackled the question in the Dartmouth Summer Research Project on Artificial Intelligence of 1956, and coined the term “Artificial Intelligence” (A.I.) (BRAFFORD, *in* MARQUIS; PAPINI; PRADE, 2020). By that time, research studies about the animal brain and the role of neurons were advancing. Frank Rosenblatt, a psychologist, proposed the perceptron architecture, in which the algorithm would function similarly to the biological brain, firing neurons to learn (ROSENBLATT, 1958; LE CUN, 2019).

Perceptron units are the artificial neurons used in ML models, each of them computing a weighted sum $\mathbf{z}$ of its inputs (Equation 2.12, which is added up to a bias $\mathbf{b}$ to compose the internal state of the perceptron $\mathbf{s}$. When the internal state exceeds a threshold value, the neuron fires and outputs the weighted sum to the next layer. Since the first perceptron using logical activation functions were limited to hyperplane n -dimensional linear solutions, other neuron variants were developed, such as the sigmoid threshold unit of steepness $\mathbf{l}$ (usually equal to 1) (Equation X), or the hyperbolic tangent function (Equation 2.13) (LE CUN, 2019).

a)

$$s = z + b, \quad z = \sum_{i=1}^n W_i x_i \quad (2.12)$$

$$y = \frac{1}{(1 - e^{-ls})} \quad (2.13)$$

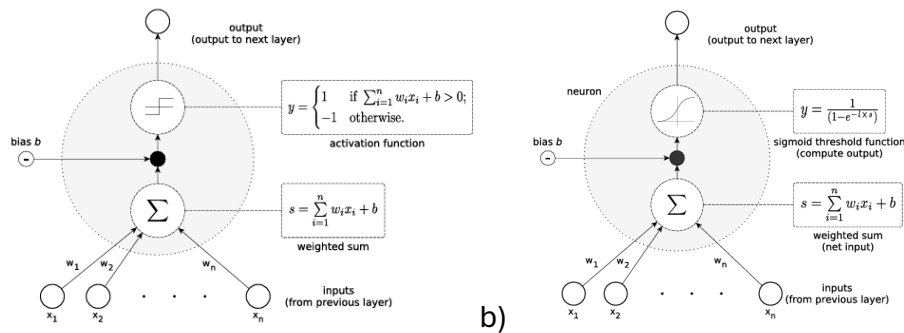


Figure 2.2: a) Logical activation perceptron and (b) sigmoid activation artificial neuron. Source: STAUNDEMEYER; MORRIS, 2019.

After a two-decade winter between the 70s and the 80s due to reduced funding, A.I. research experienced a plethora of new mathematical foundations that boosted confidence in the field. One of the most promising was the backpropagation method, proposed by the French electronics engineer Yann le Cun and the team of the American psychologist David E. Rumelhart (LE CUN, 1988; RUMELHART; HINTON; WILLIAMS, 1986). Furthermore, in the dawn of the XXI century, computing capacity greatly improved, allowing for massive information input and processing through these artificial neural networks (ANN). As landmarks, we could cite the Deep Convolutional Networks, pioneered by Alex Krizhevsky and coworkers in 2012 (KRIZHEVSKY; SUTSKEVER; HINTON, 2012), and the attention mechanism for generative A.I. which inaugurated the transformer architecture (VASWANI, et al., 2017) and gave rise to the recent generative A.I. boom¹.

These models are, in their essence, connectionist, and rely on statistical weight optimization. Alternatives to the probabilistic ANN were also developed in parallel, relying on symbolic and logical structures to produce outputs. This subfield, known as knowledge engineering (*ingénierie des connaissances*, in French), accounts for software (expert systems) in which a base of facts or rules is established and logical operations could be made to achieve a solution for a given query. Those systems have the advantage of being “explainable”, which means that the solution, or answer, possesses a traceable chain of reasoning. These logical approaches consisting of a rules database are currently seen in the field of Knowledge Management (KM) in industries, such as the method of case-based reasoning (CBS) (ALTHOFF; WEBER, 2007; BERJARANO, et al., 2014²).

¹ We’ve been introduced to advances in Information and Communication Technology Trends by Dr. Cedric Béler, professor at the École Nationale d’Ingénieurs de Tarbes (ENIT) with whom we’ve developed a project in parallel with this one on the 8th semester of the ENIT engineer diploma.

² We had the opportunity to have Dr. Laurent GENESTE, from the ENIT, coworker of Mr. BERJARANO, as professor in the topic of Knowledge and Decision Engineering, at the beginning of the same semester of this project.

Nowadays, the market is filled with commercial progress in the domain of generative A.I. by numerous corporations deploying their models. Amidst all fascination, A.I. techniques remain a powerful toolbox for solving complex research problems, having been applied in several domains, such as predictive maintenance (KELEKO, *et al*, 2022); medical research (NOÉ; DE FABRITIIS; CLEMENTI, 2020); among others. A.I. is truly a major achievement for all humankind, the converging point between technology, mathematics, and computer sciences. Nevertheless, it also encompasses moral dilemmas and questions, such as job market issues, concerns regarding data centers, such as vulnerable population targeting and global warming contribution which should be addressed by specialists and governmental entities.

2.3.1 Deployment of ANNs for study

Artificial Neural Networks (ANNs) are, on their basis, inspired by the behavior of simple animals, whose nervous system adapts to the environmental inputs by synaptic plasticity, which correspond to the mathematical formulation of weight modification. Those architectures are then meant to solve prediction problems as the classical models presented before, in mathematical terms, finding a function $f(\mathbf{x})$ which predicts the output values $\mathbf{Y}$ given input values $\mathbf{X}$, which is achieved by error minimization algorithms (LE CUN, 2019).

The optimization algorithms are based on a cost function L , which is usually the Mean Squared Error (MSE) described in Equation 2.14, where p is the number of samples, $y(i)$ the true observed value and $\hat{y}(i)$ is the predicted value. Optimization algorithms are then employed to minimize the cost function, most notably the stochastic gradient descent (SGD), proposed by Robbins and Monro in 1951 (LE CUN, 2019). In practice, these functions receive, as input, a representative subset of the data, called a batch, to calculate the minimum of the function, and the size of these subsets is a hyperparameter called “**batch-size**”, chosen for optimization in this project.

$$L = \frac{1}{p} \cdot \sum_{i=0}^{p-1} (y(i) - \hat{y}(i))^2 \quad (2.14)$$

By defining a specific **learning rate**, e , the SGD method iteratively updates the parameters of the prediction model (such as polynomial coefficients for a polynomial model, or the neuron weights in an ANN) to fit for each subsequent x_i until convergence upon a local minimum for the cost function. The update rule for the gradient descent for a given weight w is given in Equation 2.15, where C is a generic cost function (LE CUN, 2019).

$$w_i = w_{i-1} - e \cdot \frac{\partial C(X(i), Y(i), w)}{\partial w} \quad (2.15)$$

Le Cun, 2019, establishes that all ML deployment must begin with these four steps: (1) Collect a learning set; (2) propose a predictive model that defined by a set of tunable parameters $\mathbf{W}$, such that $\mathbf{Y} = \mathbf{f}(\mathbf{W}, \mathbf{X})$; (3) propose a cost function $L(\mathbf{W})$ to evaluate the model's performance; (4) apply an algorithm to find the values of $\mathbf{W}$ which minimize $L(\mathbf{W})$.

In this project, as stated above, the training set, or the output dataset $\mathbf{Y}$ is the forecast of pollutant concentration between March 21, 2025, and March 21, 2026, and the input dataset is the concentration values between March 21, 2021, and March 21, 2026, later with the addition of exogenous variables. This last dataset will be subject to a set of predictive models (RNN, LSTM, Bi-LSTM, GRU) to find the best one among them. The cost function, for model evaluation, is the MSE function, and gradient descent is stochastic enhanced by the Adam (Adaptive Moment Estimation) algorithm, rather than the standard SGD. Adam accelerates convergence by computing adaptive learning rates for each parameter, using estimations of first and second moments of the gradient, and initial bias correction parameters (KINGMA; BA, 2015). These aspects are summarized in Figure X

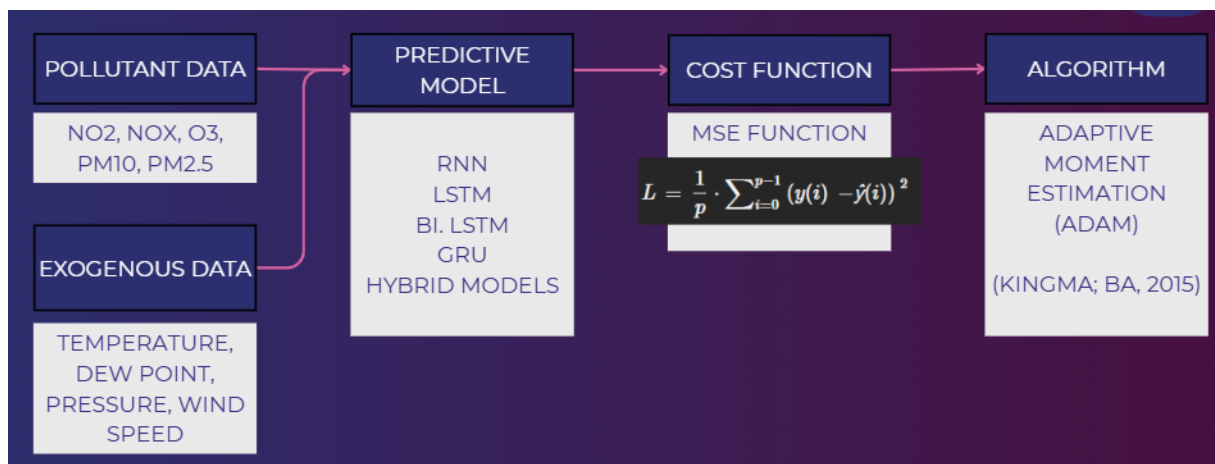


Figure 2.3: Techniques used in this project, based on the definition sequence by Le Cun, 2019.

The DL architectures for ANN are mainly networks of interconnected neurons disposed in multiple sequential layers, alternating linear and non-linear layers. In linear layers, inputs pass through a weighted sum to calculate the output, whereas in the non-linear ones, the output is given by passing the input through a nonlinear function, called the activation function. This alternating architecture constitutes a universal estimator, as some theorems show that any function could be approximated using alternating linear and non-linear stacking provided that they have sufficient connections (LECUN, 2019).

Most generative AI applications use convolutive versions of ANN for natural language processing and image recognition and generation, called convolutional neural networks (CNN). Another model would be feedforward neural networks (FFNN). In those feedforward multilayered networks, neurons that are not connected directly to the environment are called hidden neurons and compose the **hidden layers**, that will be subject to optimization. Those layers compose the overall architecture of multilayer, forward-connected networks which are the basis of the used models in this project, exemplified in Figure X, and with the sigmoid activator, can act as universal estimators (STAUNDEMEYER; MORRIS, 2019).

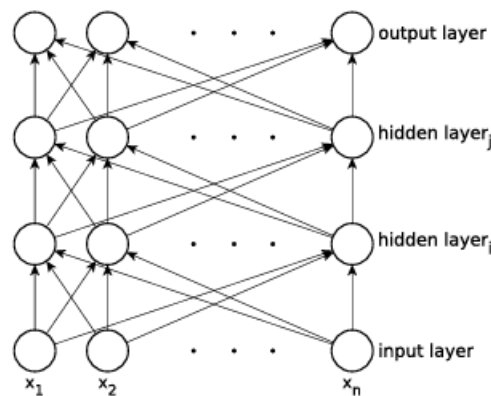


Figure 2.4: A multilayer feed-forward neural network with one input layer, two hidden layers, and an output layer. Using neurons with sigmoid threshold functions, these neural networks are able to express non-linear decision surfaces. Source: STAUNDEMEYER; MORRIS,

2.3.2 Recurrent Neural Networks

Recurrent neural networks (RNN) used in this study are an advanced version of the FFNNs with circular self-feedback connections, allowing RNNs to build a memory, which is ideal for timeseries analysis

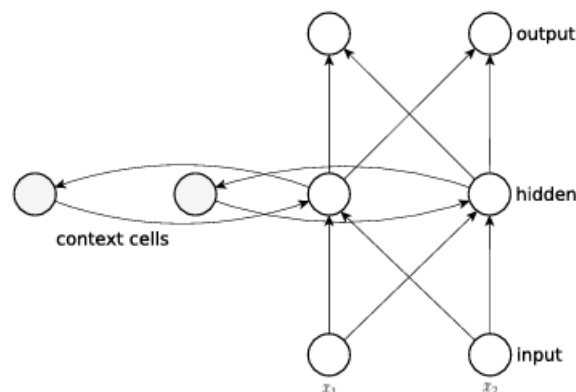


Figure 2.5: An Elman neural network, an example of connected RNN with context cells. Source: STAUNDEMEYER; MORRIS, 2019.

We finally arrive at the learning algorithm used to adapt the weights of the RNN in order to predict the results, which in this study is backpropagation through time (BPTT). This technique is an improved version from the algorithm by Le Cun, allowing for iterative weight update by applying the same logic found in the chain rule, for each layer and adapted for the iterative nature of RNN. Let e_o be the output error given by the difference between the desired output, d_o , and the network output, y_o . The overall error of the network for the simple backpropagation (BP) algorithm is given by equation 2.16, and the update formula by Equation 2.17. BPTT assumes that, for each finite time step, called an epoch, there is FFNN with an identical behavior to the RNN, as shown in Figure Y (STAUNDEMEYER; MORRIS, 2019).

$$E = \frac{1}{2} \sum e_o^2 \quad (2.16)$$

$$\Delta W_{[u,v]} = -\eta \frac{\partial E}{\partial W_{[u,v]}} \quad (2.17)$$

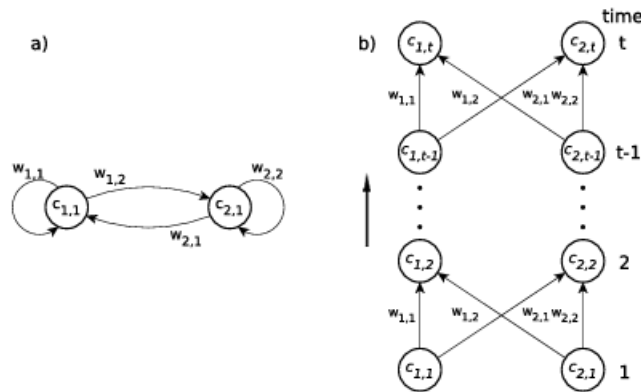


Figure 2.6: a) A simple fully recurrent neural network with a two-neuron layer. The same network unfolded over time with a separate layer for each time step is shown in (b). The latter representation is a feed-forward neural network. Source: STAUNDEMEYER; MORRIS,

The simplest version of the RNN deployed is called vanilla RNN. Let the enter signal be $\mathbf{x} = x_1 + x_2 + \dots + x_n$, and the output $\mathbf{o} = o_1 + o_2 + \dots + o_n$, defined by $h_0 = (0, \dots, 0)$ and o_i , with $h_i = m(\theta, x_i, h_{i-1})$. The recurrent neural network is an ANN with parameters θ that receives as inputs x and h , and provides as outputs o and h , with the self-feedback loop. The model is then trained exactly as stated before, using self-feedback neurons with the update function (STAUNDEMEYER; MORRIS, 2019).

2.3.3 LSM and its variants

Due to the mathematical formulation of the backpropagation algorithm, can't go beyond 5 - 10 timesteps, after which the backpropagation error blows-up or vanishes, this last being the problem of vanishing gradient and leading to enormous amounts of time in training. Solving the error in time, we obtain Equation 2.18. If the module of the term, inside the product is bigger than one, the equation grows exponentially, and if it is smaller than one, the equation decreases exponentially, causing the error to vanish. (HOCHREITER; SCHMIDHUBER, 1997; STAUNDEMEYER; MORRIS, 2019).

$$\frac{\partial \vartheta_v(t')}{\partial \vartheta_u(t')} = \sum_{u_{t'} \in U} \dots \sum_{u_{t-1} \in U} \left(\prod_{\tau=t'+1}^t f'_{u_\tau} \left(z_{u_\tau}(t - \tau + t') \right) W_{[u_\tau, u_{t-1}]} \right) \quad (2.18)$$

Moreover, the following relationship shows that the global error vanishes if even a local error vanishes (STAUNDEMEYER; MORRIS, 2019).

$$\sum_{o \in O} \frac{\partial \vartheta_v(t')}{\partial \vartheta_u(t')}$$

One solution to this problem is the application of Long Short-Term Memory (LSTM) Neural Networks, proposed by HOCHREITER and SCHIDHUBER, 1997, which use constant error carousels (CECs) to be able to bridge minimal time lags for over 10^3 time steps. This is achieved by multiplicative gate units which act as memory handlers inside a more complex cell, shown in Figure X. Each information flow is subject to an equation, and these equations are listed below, where sigma denotes the sigmoidal function.

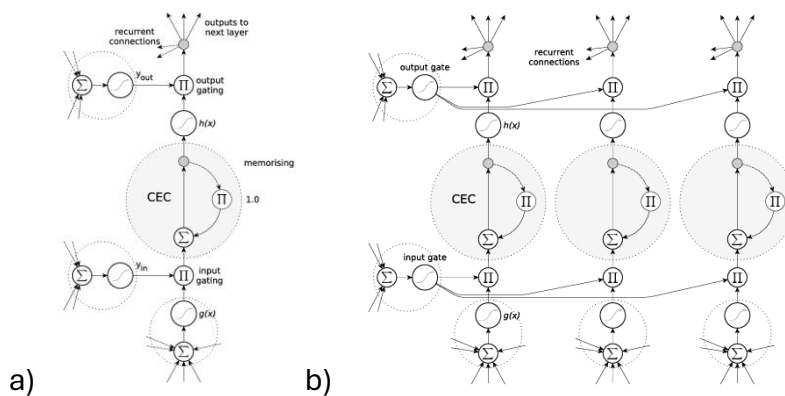


Figure 2.7: a) A standard LSTM memory block. The block contains (at least) one cell with a recurrent self-connection (CEC) and weight of '1'. The state of the cell is denoted as s_c . Read and write access is regulated by the input gate, y_{in} , and the output gate, y_o . Source: STAUNDEMEYER; MORRIS, 2019.

Equations (TEYTAUD, 2025):

$$\text{Cell input activation function: } g = \tanh(W_{ch} h_{t-1} + W_{cx} x_t) \quad (2.19)$$

$$\text{Input update gate activation: } y_{in} = \sigma(W_{ih} h_{t-1} + W_{ix} x_t) \quad (2.20)$$

$$\text{Forget gate activation: } f_t = \sigma(W_{fh} h_{t-1} + W_{fx} x_t) \quad (2.21)$$

$$\text{Cell state (not included in some architectures): } s_c = i_t \odot c'_t + f_t \odot c_{t-1} \quad (2.22)$$

$$\text{Output gate activation function: } y_{out} = \sigma(W_{oh} h_{t-1} + W_{ox} x_t) \quad (2.23)$$

$$\text{LSTM output (hidden state): } y_{mc}(\tau + 1) = o_t \odot \tanh(c_t) \quad (2.24)$$

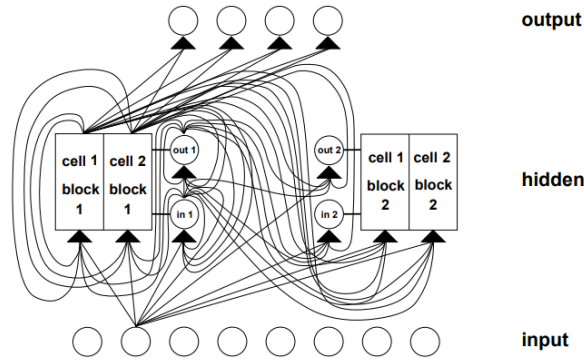


Figure 2.8: Example of a full net with 8 input units, 2 memory cell blocks of size 2 and four output units. Source: HOCHREITER; SCHMIDHUBER.

Memory blocks from LSTM-RNNs are very efficient in memory storage from limited amounts of data, due to information control through input and output gates. Those gates prevent irrelevant information from entering and synergize with forget gates that reset the state of different cells inside the block. On the other hand, LSTM topology is fixed a priori, and its memory does not adapt on the training process, which can become a limitation for applications needing more memory or a lack of optimization for situations needing less memory (STAUNDEMEYER; MORRIS, 2019).

A variant of LSTMs is their bidirectional form (Bi-LSTM). While conventional RNNs analyze only past data, Bi-LSTMs allow for both the future and the past of a given point in the context of LSTM. Also, this architecture implements full error gradient calculation, being more fitted to the standard BPTT usage (STAUNDEMEYER; MORRIS, 2019).

Another possible architecture would be the Gated Recurrent Unit (GRU) proposed by CHO, *et al.*, 2014, under the name of RNN Encoder-Decoder. This unit adaptatively remembers and forgets information but is simpler than the standard LSTM as it lacks the context gate. Its simplicity is found to be more powerful than the standard LSTM for most tasks (STAUNDEMEYER; MORRIS, 2019).

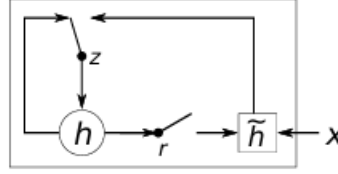


Figure 2.9: An illustration of the hidden activation function proposed by Cho et al., 2014. The update gate z selects whether the hidden state is to be updated with a new hidden state $\tilde{h}$. The reset gate r decides whether the previous hidden state is ignored (CHO);

$$\text{Reset gate: } r_j = \sigma([U_r h_{t-1}]_j + [W_r x]_j) \quad (2.25)$$

$$\text{Update gate: } z_j = \sigma([U_z h_{t-1}]_j + [W_z x]_j) \quad (2.26)$$

$$\text{Actual activation of the unit: } h_j^{(t)} = z_j h_j^{(t-1)} + (1 - z_j) h_j^{(t)} \quad (2.27)$$

$$\text{New hidden state: } \tilde{h}_j^{(t)} = \phi([U(r \odot h_{t-1})]_j + [W x]_j) \quad (2.28)$$

Finally, we also implemented a hybrid architecture of vanilla RNN with LSTM for information flux optimization. This architecture is expected to use the vanilla RNN simplicity for fast data input and the LSTM memory blocks to store memory and deal with the vanishing gradient.

2.3.4 Optimization

Bayesian optimization is a probabilistic, model-based strategy designed to efficiently find the global minimum of complex, computationally expensive objective functions. Unlike exhaustive grid searches or random sampling, it constructs a surrogate probability model of the objective function, dynamically balancing the exploration of new, untested parameter configurations with the exploitation of known optimal regions. It is well known in other fields such as materials science.

Within the context of this research, Bayesian optimization is leveraged to systematically navigate the high-dimensional hyperparameter space of the recurrent deep learning architectures. By intelligently driving the selection of structural variables, such as sequence length, hidden dimensions, and learning rates, this methodology maximizes computational efficiency, and, for this study, the goal was set to minimize the Mean Absolute Percentage Error (MAPE) of the pollutant concentration forecasts.

2.4 POLLUTANTS THEORETICAL CYCLES AND SOURCES

Table 2.1: Damages and injuries caused by several pollutants. Source: SCHRAUFNAGEL; *et al.*, 2019.

TABLE 1] How Different Types of Air Pollution Damage Tissue

Pollutant	Injury Determinants	Tissue Affected
Sulfur dioxide	Highly soluble	Upper airway and skin damage
Nitrogen dioxide	Less soluble (nitrogen dioxide and ozone are irritating)	Deeper lung penetration
Ozone		Bronchial and bronchiolar injury
Carbon monoxide		Carbon monoxide: tissue hypoxia
Particulate matter (PM ₁₀ , PM _{2.5} , PM _{0.1})	Size, structure, and composition determine toxicity	Large particles: mucous membranes, upper airways Small particles: bronchioles and alveoli Ultrafine particles: systemic tissue reactions

Forecasting pollutant concentrations presents a significant challenge in time-series analysis. As previously established, atmospheric dynamics are often better represented by multiplicative models due to the highly stochastic nature of pollutant evolution, a complexity that standard additive classical models frequently fail to fully capture. A rigorous understanding of these underlying environmental cycles is essential for identifying inherent modeling limitations, and it will serve as the foundation for evaluating the predictive performance of each architecture in the subsequent sections.

2.1.1 Nitrogen dioxide (NO₂) and NO_x

Under standard atmospheric conditions, elemental nitrogen and oxygen do not spontaneously react. The formation of nitrogen oxides (NO_x) primarily occurs during high-temperature combustion processes. While trace amounts are naturally generated by bacterial respiration, volcanic activity, and lightning discharges, the most prominent anthropogenic source is the internal combustion engine.

Because combustion-driven technologies are ubiquitous in modern society, ambient NO₂ exposure predominantly results from motor vehicle emissions, while indoor exposure is largely driven by butane heaters, kerosene heaters, and gas stoves. Notably, indoor exposure levels can be up to three times higher in households utilizing gas cooking appliances. Upon inhalation, NO₂ initiates oxidative stress within the respiratory tract, leading to bronchoconstriction, airway hyperresponsiveness, and an impaired pulmonary host defense mechanism (U.S. EPA, 2016).

Nox encompasses the total concentration of NO and NO₂, following the fast conversion cycle between these two species in the stratosphere and troposphere. These concentrations are linked with the ozone cycle, and sunlight input determines the ratio of NO to NO₂. In fact, ozone concentration depends on the ratio of these two species and

requires a third body M to mediate molecular oxygen collision (SEINFELD; PANDIS, 2016, *apud* WIKIPEDIA).



$$\frac{[NO_2]}{[NO]} = \frac{k_3 [O_3]}{j_{NO_2}} \text{ (Leighton relationship)} \quad (2.32)$$

2.4.1 Ozone

Unlike primary pollutants that are emitted directly from specific sources, ground-level ozone (O_3) is a secondary atmospheric pollutant. Its formation is fundamentally driven by complex photochemical interactions between nitrogen oxides (NO_2) and volatile organic compounds (VOCs). Under the influence of ultraviolet (UV) radiation from sunlight, these precursor molecules, which predominantly originate from vehicular exhaust, industrial emissions, and chemical solvents, undergo a series of catalytic reactions that result in the rapid accumulation of O_3 . Because this photolysis process is highly dependent on both solar radiation and ambient temperature, tropospheric ozone concentrations exhibit a pronounced diurnal and seasonal periodicity, typically reaching their maximum peak during the mid-afternoon hours of the warmer summer months.

Due to its highly reactive molecular structure, prolonged exposure to ambient O_3 poses severe risks to human health, acting as a potent respiratory oxidant. Upon inhalation, ozone reacts aggressively with the epithelial lining fluid of the respiratory tract, inducing severe oxidative stress and subsequent inflammatory responses. This cellular damage causes a range of adverse pulmonary outcomes, including the exacerbation of asthma, measurable reductions in lung capacity, and an increased susceptibility to respiratory infections. Furthermore, because elevated ozone concentrations frequently coincide with extreme heat events, this compounding environmental stress disproportionately impacts vulnerable populations, making O_3 mitigation a critical priority for modern urban air quality management.

2.4.2 Fine and ultrafine particulate matter

Particulate matter (PM) constitutes a highly complex, heterogeneous mixture of solid particles and liquid droplets suspended in the atmosphere. Unlike purely gaseous pollutants, these aerosols are primarily categorized by their aerodynamic diameter, with

PM₁₀ (coarse particles around 10 µm) and PM_{2.5} (fine particles around 2.5 µm) serving as the primary metrics for environmental monitoring. These particulates originate through two distinct pathways: primary emissions, which are mechanically or thermally released directly into the air via fossil fuel combustion, vehicular brake and tire wear, and construction dust, and secondary formations.

Secondary aerosols are generated through the complex atmospheric nucleation and condensation of precursor gases, including sulfur dioxide SO₂, NO_x, and ammonia NH₃. Because these particles possess physical mass, their atmospheric residence time and spatial dispersion are heavily governed by meteorological variables such as wind advection and precipitation. Consequently, PM concentrations are particularly susceptible to extreme localized accumulation during winter thermal inversions, when stagnant atmospheric conditions trap the particles close to the surface.

The toxicological severity of particulate matter is fundamentally dictated by its aerodynamic size, which determines its penetration depth within the human body. While PM₁₀ is largely deposited within the upper respiratory tract, where it induces localized irritation and exacerbates preexisting conditions such as asthma and chronic obstructive pulmonary disease (COPD), the finer PM_{2.5} fraction poses a significantly more profound physiological threat.

Due to their microscopic dimensions, PM_{2.5} particles easily pass through the body's primary mucociliary defenses, penetrating deep into the alveolar regions of the lungs. Once lodged in the alveoli, these fine particulates, often carrying toxic trace metals and polycyclic aromatic hydrocarbons (PAHs), can cross the alveolar-capillary membrane and enter the systemic bloodstream. Chronic exposure to PM_{2.5} is known to trigger widespread systemic inflammation and oxidative stress, directly contributing to accelerated atherosclerosis, elevated risks of ischemic heart disease, stroke, and a measurable increase in premature global mortality (SCHRAUFNAGEL, 2019).

3 DATA EXPLORATION AND BASIC STATISTICS

3.1 Pollution data description

3.1.1 Data gathering methods

The pollution data used in this project were obtained from the Geod'Air public API, which provides official air quality measurements in France. The study focused on the monitoring zone identified by the code FR76ZAG01. This code corresponds to the Occitanie region in southern France, more specifically the urban area around Toulouse.

The selected pollutants were NO_2 , NO_x , O_3 , PM_{10} , and $\text{PM}_{2.5}$, as they are commonly used indicators of urban air quality and traffic-related pollution. The available pollution data covered the period from 21 March 2021 to 21 March 2026.

The original Geod'Air files contain measurements with a high temporal resolution, mainly at an hourly scale. This hourly sampling is useful because air pollution can vary strongly during the day, for example due to traffic peaks, meteorological conditions, and daily human activity patterns. For the exploratory analysis, the data could therefore be studied at different temporal scales, such as hourly, daily, weekly, or monthly.

For the forecasting models, the raw timestamped measurements were converted into a regular time series. The data were indexed by a timestamp and then resampled to a daily frequency using the mean concentration of each pollutant for each day. This means that each daily value represents the average pollution level observed over that day. Missing values were handled using linear interpolation, which allowed the time series to remain continuous before being used by the neural network models.

In addition to pollution data, environmental variables were collected from the Copernicus Climate Data Store, using the ERA5-Land daily statistics dataset. These variables included temperature, dew point temperature, soil temperature, wind components, surface pressure, and albedo. The environmental data were also processed at a daily scale so they could be merged with the pollution dataset.

Throughout the report, the main analysis focuses on $\text{PM}_{2.5}$ as a representative example, since this pollutant is particularly relevant for urban air quality and public health due to its ability to penetrate deeply into the respiratory system. The other pollutants, namely NO_2 , NO_x , O_3 , and PM_{10} , were also processed and analyzed using the same methodology, but their detailed results are presented in the appendix.

3.1.2 Time series plots

These figures show the hourly evolution of PM2.5 concentrations between 2021 and 2026, as well as different temporal transformations of the series.

Order 0 corresponds to the original series, meaning the PM2.5 concentrations observed directly over time. It is a raw signal, without transformation. Order 1 corresponds to the first temporal differentiation. It measures the variation between two consecutive observations: $PM2.5(t) - PM2.5(t-1)$, divided by the timestep value, therefore making it possible to observe rapid increases and decreases in concentration. This transformation is often used to reduce trend effects and make the series more stationary.

Order 2 corresponds to the second temporal difference. It measures the variation of the variation: $[PM2.5(t) - PM2.5(t-1)] - [PM2.5(t-1) - PM2.5(t-2)]$. It highlights even sharper changes in the dynamics of the series. This type of transformation can be useful for classical time series models, such as ARIMA or SARIMA, which generally perform better when the data are stationary.

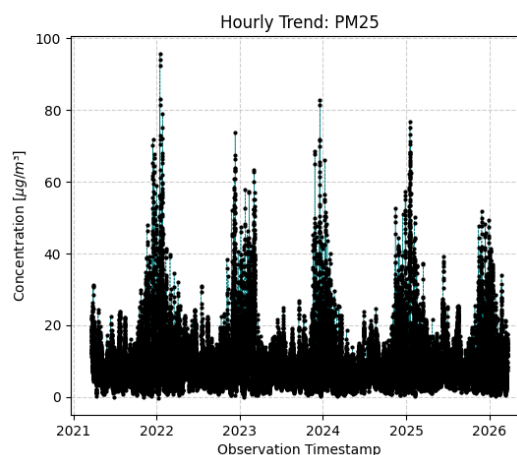


Figure 3.1: Hourly trend plot of PM2.5.

Figure 3.1 shows that PM2.5 concentrations generally remain low to moderate, but with several intense pollution episodes. These peaks are particularly visible around the winter periods, especially in 2022, 2023, 2024, 2025, and 2026. This suggests an important seasonal dynamic, probably linked to meteorological conditions, atmospheric stability, and the accumulation of fine particles.

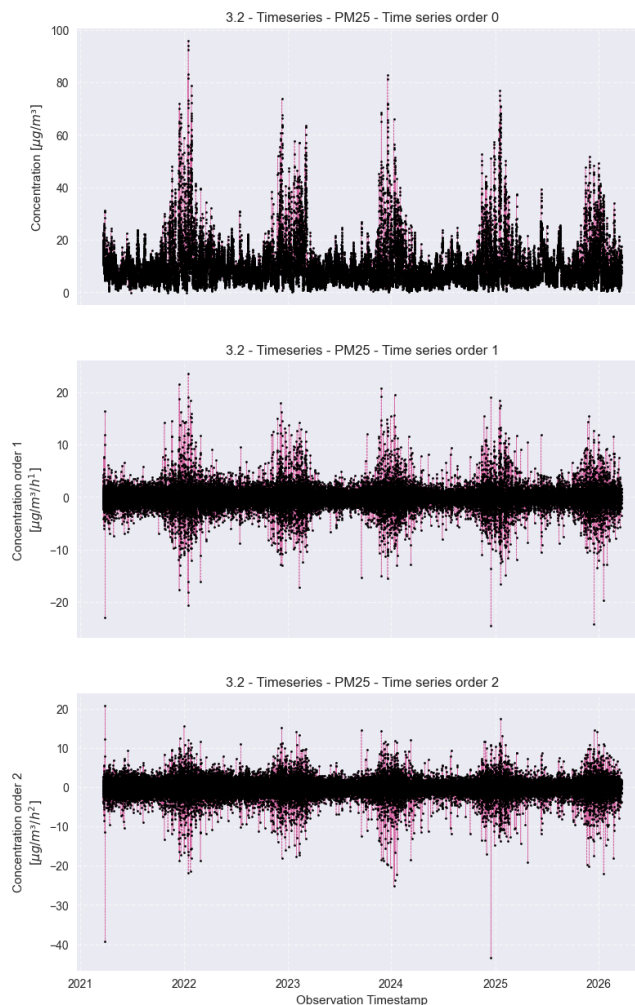


Figure 3.2: Differentiated plots of PM2.5 timeseries.

In figure 3.2, the order 0 series reproduce the overall behavior of the observed concentration. The order 1 series highlights rapid hour-to-hour variations: positive peaks indicate sudden increases, while negative values indicate rapid decreases. Finally, the order 2 series further emphasizes breaks and abrupt changes, but it also becomes noisier. Overall, these transformations help to better understand the temporal structure of the data and prepare the series for statistical modeling.

3.1.3 Power Spectral Density analysis

The PSD analysis highlights several spectral peaks for the three pollutants. The most pronounced peaks occur at short periods, mainly around 12 h and 24 h, indicating a strong daily component in pollutant concentrations.

NO₂ exhibits the highest spectral amplitudes, particularly at 12 h, reflecting a more pronounced periodic variability than that observed for PM10 and PM2.5. These

particulate pollutants also display periodic components at the same time scales, although with lower amplitudes.

Additional peaks are observed at longer periods, notably around 4382 h and 8764 h. These periods correspond approximately to six months and one year, respectively, which may reflect seasonal variations in pollutant concentrations.

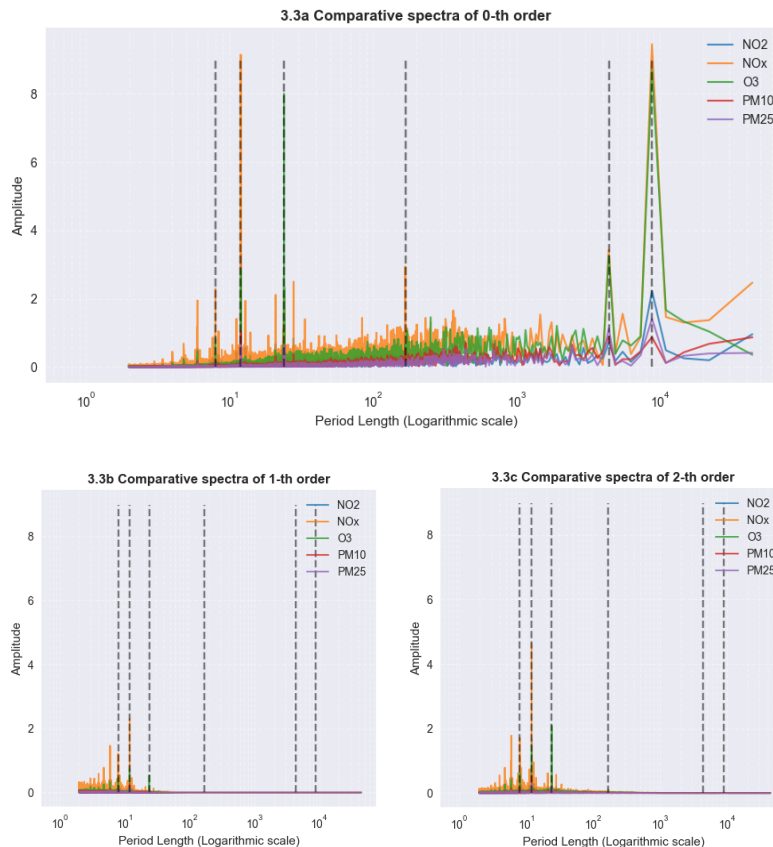


Figure 3.3: (a) Representative periodograms for three selected pollutants, highlighting their dominant spectral peaks. The dashed lines represent 8h, 12h, 24h, half-a-year and a year periods. (b, c) Periodograms of the first- and second-order differenced time series

3.1.4 Moving average, trend and stationarity

In the plots below, the grey curve represents the $PM_{2.5}$ concentration data at the selected temporal scale. The orange curve represents the rolling trend, calculated as a moving average over a specific time window. This rolling trend smooths short-term fluctuations and helps visualize the general evolution of the pollutant concentration.

The results of the Augmented Dickey-Fuller (ADF) test indicate that the time series at hourly, daily, and weekly aggregation levels exhibit stationarity, successfully rejecting the null hypothesis with p-values below the standard 0.05 significance threshold. This confirms that, at these finer temporal resolutions, the $PM_{2.5}$ series lacks a statistically

significant unit root or persistent stochastic trend. Conversely, the monthly, semiannual, and annual aggregations yield p-values exceeding 0.05, thereby failing to reject the null hypothesis of non-stationarity. However, it is crucial to note that these extended temporal windows result in significantly reduced sample sizes. This data sparsity severely diminishes the statistical power and reliability of the ADF test, particularly at the annual scale.

Ultimately, this multi-scale analysis demonstrates that the manifestation of stationarity is highly dependent on the chosen temporal resolution. The daily aggregation level emerges as the optimal resolution for predictive modeling: it retains a sufficient sample size for robust algorithmic training, effectively attenuates high-frequency hourly noise, and accurately preserves the critical seasonal and temporal periodicities required for accurate forecasting.

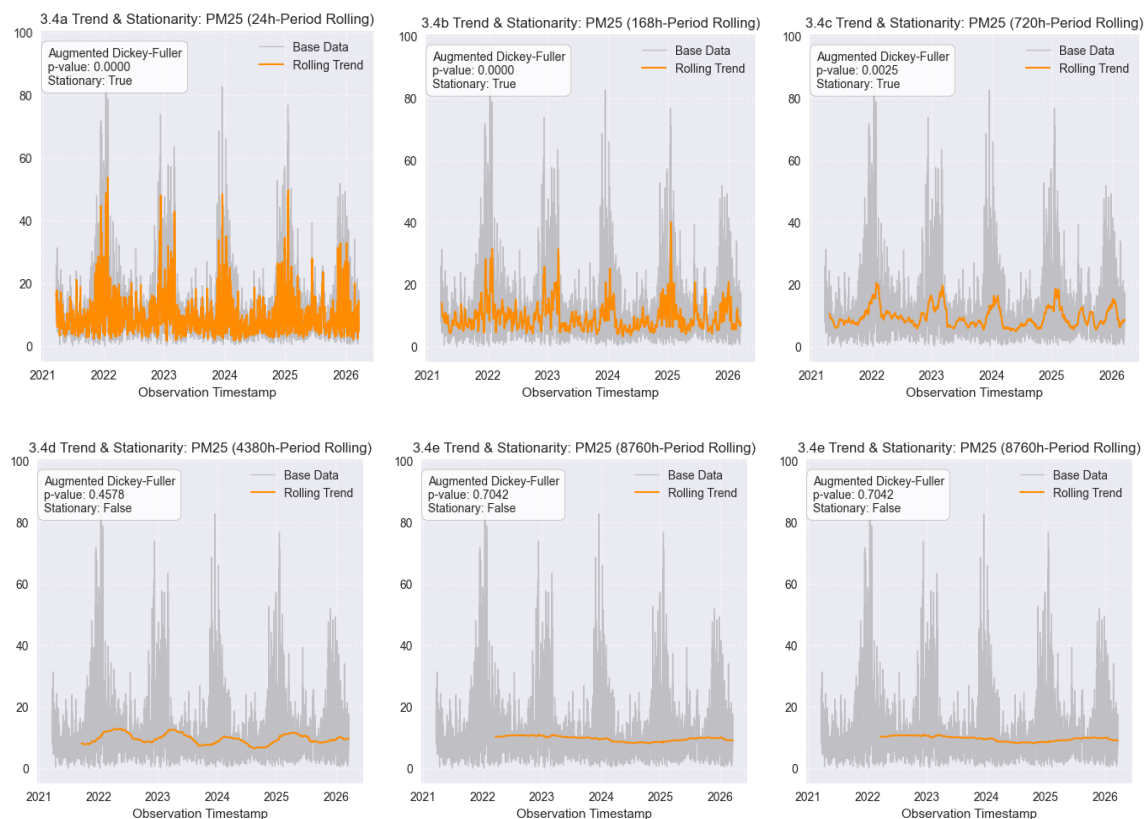


Figure 3.4: Evaluation of underlying trends and local stationarity in the PM2.5 time series utilizing rolling mean analysis.

To facilitate standardized comparisons across the various pollutants, the data was aggregated at both daily and 180-day (semi-annual) temporal resolutions. This dual-scale

approach revealed that all evaluated pollutants exhibited highly consistent behavioral dynamics over time, as shown in Figure 3.5.

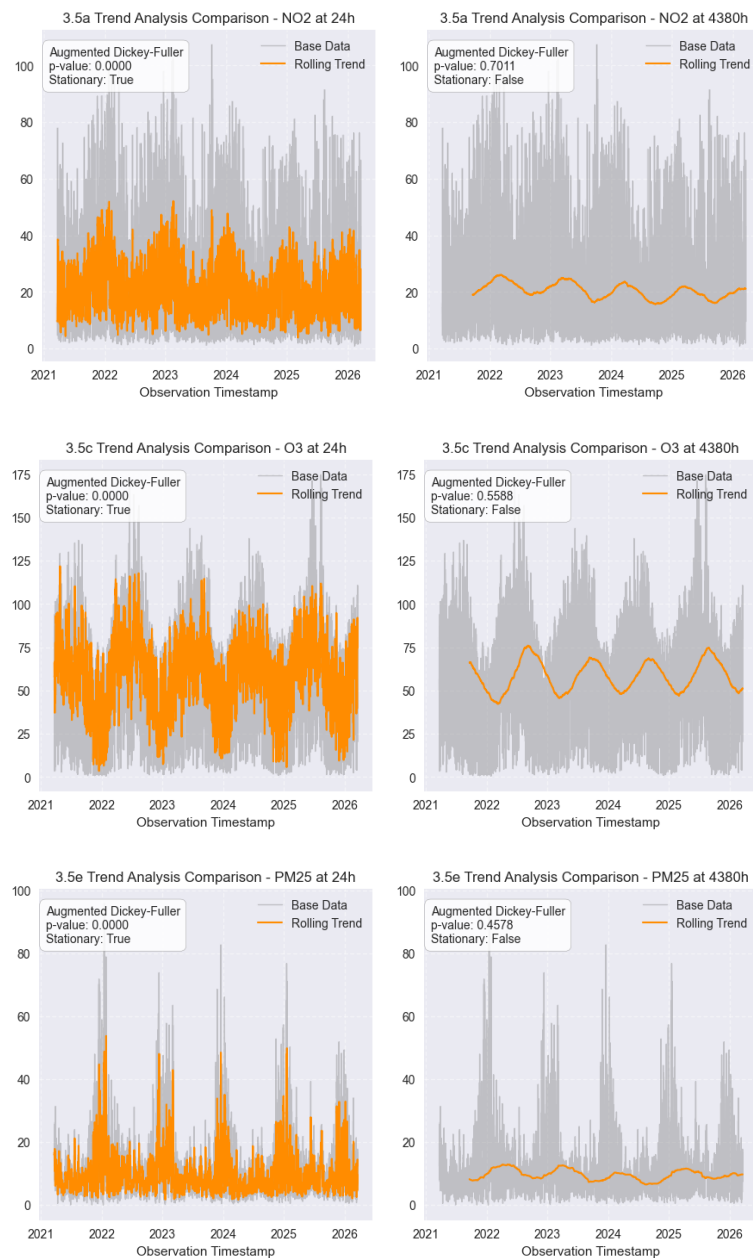


Figure 3.5: Evaluation of underlying trends and local stationarity utilizing rolling mean analysis for a representative pollutant from each chemical class, comparing 24-hour and 4,380-hour (semi-annual) temporal windows.

To explicitly demonstrate the efficacy of mathematical differencing in eliminating underlying trends, Figure 3.6 contrasts the raw and transformed data across the two

established temporal windows (24-hour and 4,380-hour) for $PM_{2.5}$. While the rolling mean of the raw extended window reveals a distinct non-stationary drift, reflective of the gradual decline in overall pollutant concentrations, applying a first-order difference effectively flattens this trajectory. By stabilizing the mean around zero, this transformation successfully isolates the stochastic fluctuations from the deterministic baseline. Consequently, this visual comparison highlights how the differencing operation neutralizes long-term macroeconomic or policy-driven shifts, thereby satisfying the strict stationarity requirements necessary for subsequent autoregressive predictive modeling.

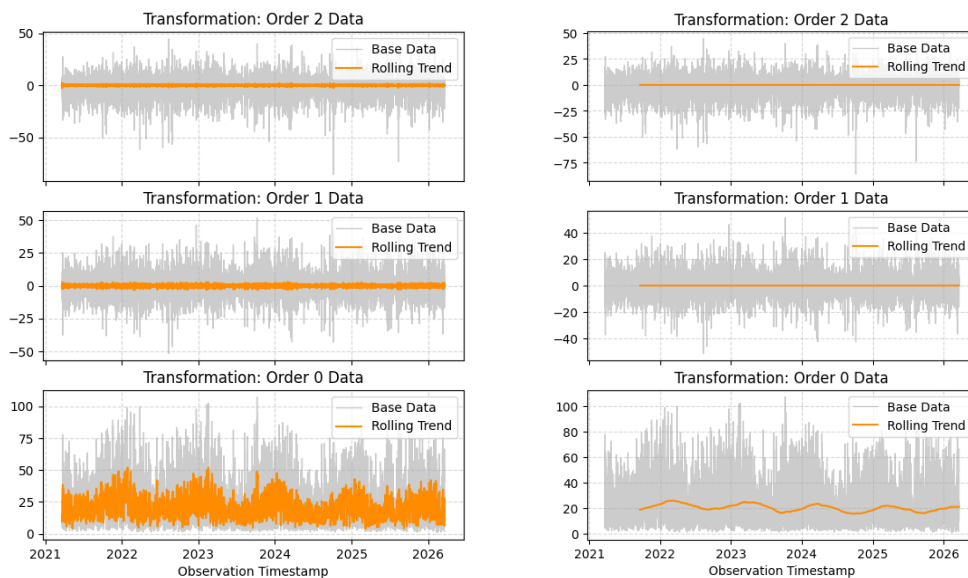


Figure 3.6: Comparison between transformed data for 24 hour- and 4380 hour-rolling average window.

3.1.5 STL Decomposition

To better understand the temporal structure of the $PM_{2.5}$ series, a Seasonal-Trend decomposition using Loess (STL) was performed. The decomposition separates the observed signal into three components: trend, seasonality, and residuals, allowing the identification of the dominant sources of variability.

The results show that the trend component remains relatively weak throughout the study period, with a low strength index, indicating the absence of a strong long-term increasing or decreasing pattern in $PM_{2.5}$ concentrations. The seasonal component is more pronounced, suggesting the presence of recurring temporal patterns, although its contribution remains moderate compared with the total variability of the series.

Most of the variability is contained within the residual component, which exhibits large fluctuations and several extreme values. This indicates that a significant proportion of $PM_{2.5}$ variability is driven by short-term events and irregular factors that cannot be fully explained by trend or seasonal effects alone.

The comparison between the standard and robust STL decompositions shows that the robust approach reduces the influence of outliers while preserving the main trend and seasonal structures. Overall, the decomposition confirms that $PM_{2.5}$ concentrations are characterized by relatively weak long-term trends, moderate seasonality, and strong irregular variability.

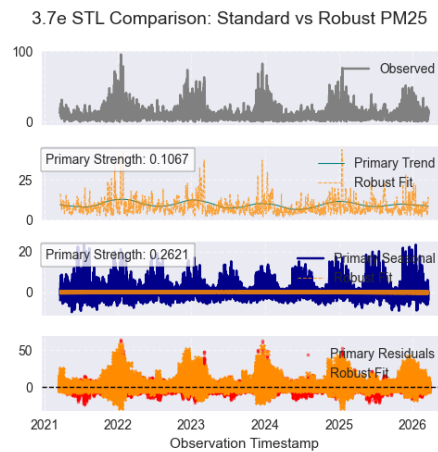


Figure 3.7: STL Decomposition for PM2.5.

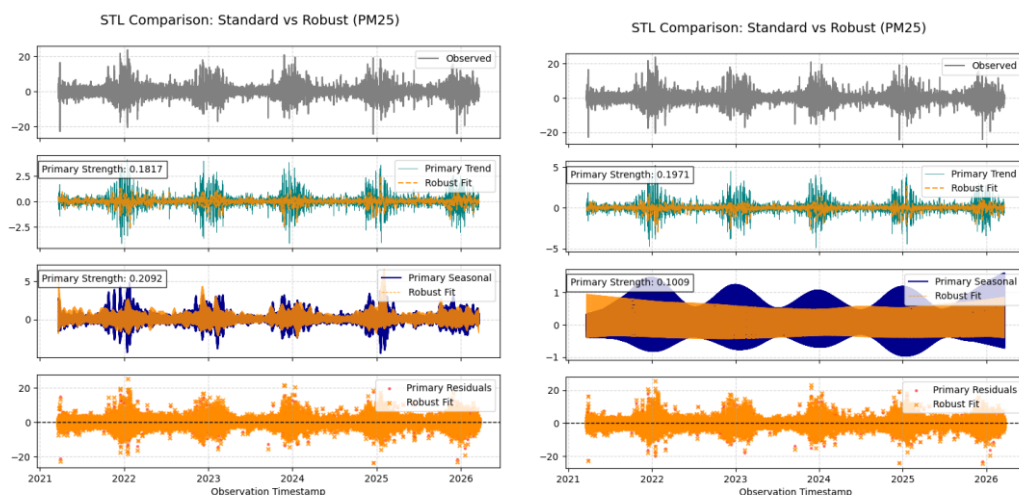


Figure 3.8: STL Decomposition Comparison for first- and second-order differentiation for PM2.5.

3.1.6 Memory analysis

The Autocorrelation Function (ACF) and Partial Autocorrelation Function (PACF) were analyzed to evaluate the temporal dependence structure of the $PM_{2.5}$ time series and to support the selection of forecasting models.

The ACF shows a slow decay of the correlation values across increasing lags, indicating that PM_{25} concentrations are strongly influenced by previous observations and exhibit significant temporal persistence. The oscillatory pattern observed at regular intervals also suggests the presence of a seasonal component in the series.

The PACF presents a dominant spike at the first lag, followed by substantially smaller values for subsequent lags. This behavior indicates that the strongest direct influence comes from the most recent observations, while the effect of more distant lags is largely explained through intermediate values.

Overall, the ACF and PACF analyses confirm that the PM_{25} series is not purely random and contains both temporal dependence and seasonal structures. These characteristics justify the use of time-series forecasting approaches such as ARIMA, SARIMA, Holt-Winters, and deep learning models.

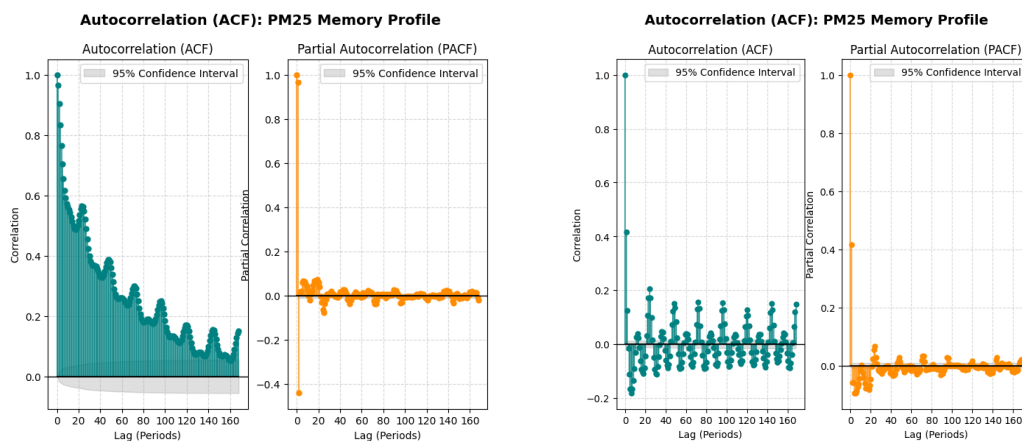


Figure 3.9: ACF and PACF for raw $PM_{2.5}$ timeseries and its derivative.

3.1 ENVIRONMENTAL DATA DESCRIPTION

3.2.1 Data Gathering Methods

The environmental data was collected from COPERNICUS' "ERA5-Land post-processed daily statistics from 1950 to present" database, which allows users to download data in .nc files, well spread among environmental data analysis. To better use the data for analysis and visualization, a procedure based on ETL pipelines was adopted,

since Copernicus transfers between 70 and 90TB of data daily, making it a challenge to extract a high amount of data.

The extraction phase was developed using ERA5's API integrated into python code: for each month, the extraction was made such that each variable data was stored into a file, which were themselves stored in a folder for the corresponding month and then placed in a folder meant to store all months' folders. Each month presented, as variables, the ones depicted in Table X, and each variable presented three "dimensions": latitude, longitude, and valid time. The chosen latitude and longitude range were from 43,7° to 43,5° and from 1,4° to 1,5°, a region comprising only Toulouse area (Figure 3.9).



Figure 3.10: Toulouse territory comprised between the bounds of the collected data, shown in Google Earth.

Table 3.1: Exogenous variables, abbreviations and their units

Variable	Abbreviation	Unit
2m dewpoint temperature	D2M	K
2m temperature	T2M	K
Skin temperature	SKT	K
Soil temperature level 1	SLT1	K
Soil temperature level 2	SLT2	K
Soil temperature level 3	SLT3	K
Soil temperature level 4	SLT4	K
10m u component of wind	U10	m/s
10m v component of wind	V10	m/s
Surface pressure	SP	Pa
Forecast albedo	FAL	[0,00 – 1,00]

After extraction, the data passed by a transformation in order to condense all variables from one month into a single database, a dictionary containing the variable abbreviated name as key and the data as values. From this step on, the pipeline was branched into two modes of data storage: the first, adapted to data visualization and basic statistics, converting the variables into 3-dimensional arrays (referred as “3D data”), and the second, adapted to Machine Learning Techniques, which uses a raveling function to return flat tabular data frames (referred as “tab data”).

Those monthly databases were then loaded into Hierarchical Data Format (HDF5) files², stored in a dedicated folder, to preserve the dictionary structure and facilitate both scaling and modularity. All those steps were iterated over a period ranging from March 21st, 2021, to March 21st, 2026.

For this section, we shall focus on data exploration and visualization, using the loaded 3D data. This process followed a retrieval logic which read a single file, applied a transformation from 4-dimensional to n-dimensional space, $n = 1, 2, 3$, depending on the desired analysis, and outputted a dictionary structure with the variable names as keys and a data frame as value. For the entire period analysis, those steps were iterated, and the output was a dictionary with variable names as keys and data frame lists as values. Those output dictionaries were then used for preliminary data analysis.

The first analysis, as carried out with pollution data, was a monthly overall average to detect major periodic tendencies. Following this analysis, the latitude and longitude span were divided into smaller squares according to a granularity parameter set to $0,1^\circ$, which were called regions, totaling 4 of them. For each region, a timeseries graph was generated with the average regional values to inspect different tendencies per geographical area. Lastly, to identify geographical trends, a 3-dimensional plot was generated plotting the overall period average as a function of latitude and longitude.

3.1.2 Data exploration methods

- **Spatial Aggregation Metrics**
- **Grand Mean:** The overall spatial and temporal average of the environmental variable across the entire dataset.
- **Spatial Average:** The aggregated average value computed across specific geographic grid cells.
- **Zonal Mean:** The spatial average computed strictly along specific latitudes.

- **Meridional Mean:** The spatial average computed strictly along specific longitudes.

To model the cyclical nature of the meteorological data, a sinusoidal function of the form $y(t) = A \sin(\omega t + \phi) + C$ is fitted to the time series, yielding the following descriptive parameters:

- **Amplitude (A):** The peak magnitude of the seasonal oscillation.
- **Phase Shift (ϕ):** The temporal displacement of the seasonal cycle, indicating exactly when peak values occur within the calendar year.
- **Vertical Shift / Mean (C):** The baseline average level around which the seasonal curve oscillates.
- **Seasonal Correlation:** The statistical correlation percentage between the observed time series and the fitted sinusoidal model, serving as an indicator of how strictly the variable adheres to a periodic cycle.

Some of the plot examples are shown below in Figure 3.10. The rest is shown in the appendix.

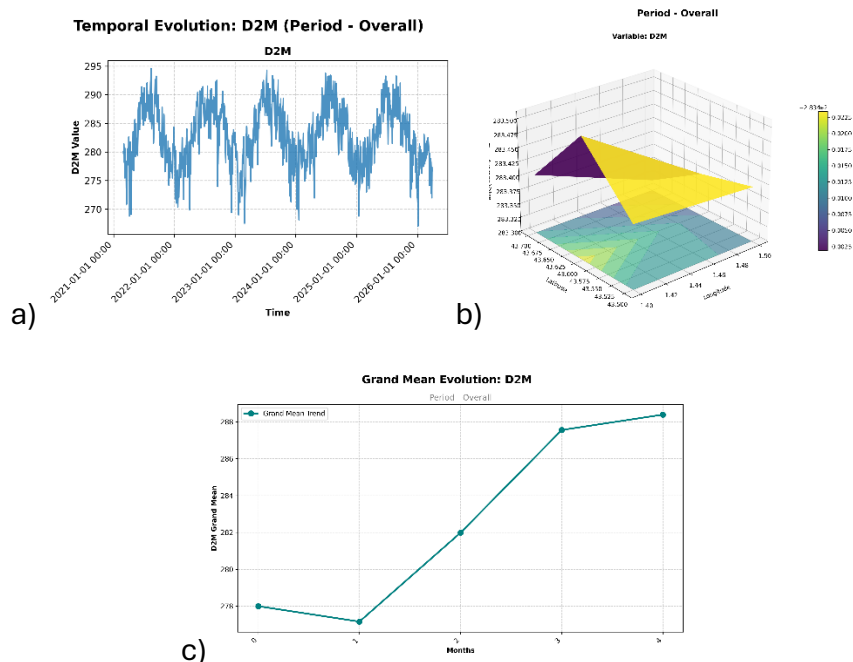


Figure 3.11: a) Timeseries evolution of dewpoint over two meters in Toulouse from 2021 to 2026. b) average spatial distribution of the temperature over the concerned period; c) anual mean.

4. TRADITIONAL STATISTICAL METHODS

In this project, the analysis and modelling were carried out for several atmospheric pollutants, including PM_{2.5}, PM₁₀, NO₂, NO_x, and O₃. However, in order to provide a more detailed and readable analysis in the main body of the report, this section focuses only on PM_{2.5}. The results obtained for the other pollutants are presented in the appendix.

PM_{2.5} refers to fine particulate matter with a diameter less than or equal to 2.5 micrometers. Due to their small size, these particles can penetrate deeply into the respiratory system, making PM_{2.5} a particularly important pollutant in air quality studies.

The objective of this section is to analyze the PM_{2.5} time series using classical statistical forecasting models. Three approaches are considered: the ARIMA model, the SARIMA model, and the Holt-Winters method. These models make it possible to study the temporal structure of the series, including trend, seasonality, and dependence between successive observations.

The ARIMA model represents the general dynamics of the series based on its past values and previous errors. The SARIMA model is a seasonal extension of ARIMA and allows regular cycles that may appear in pollution data to be taken into account. Finally, the Holt-Winters method is based on exponential smoothing and incorporates the level, trend, and seasonality of the series.

This statistical modelling step provides an initial benchmark before comparison with more complex models. It establishes interpretable reference models for PM_{2.5} forecasting, while maintaining a methodology that can also be applied to the other pollutants studied.

For the statistical forecasting experiments, the PM_{2.5} time series was divided chronologically into two parts. The first 80% of the observations were used as the training set, while the remaining 20% were reserved for prediction and evaluation. This approach respects the temporal order of the data and avoids using future information during model training. In this case, the training period extends from March 2021 to March 2025, and the test period covers March 2025 to March 2026. In the plots, only the final part of the training period is displayed in order to improve readability, but the models were trained using the complete 80% training dataset.

4.1 ARIMA METHOD

The ARIMA forecast captures the average level of the PM2.5 series but remains relatively stable over the prediction period. It does not fully reproduce short-term peaks, which suggests that the model has limitations when dealing with sudden pollution variations.

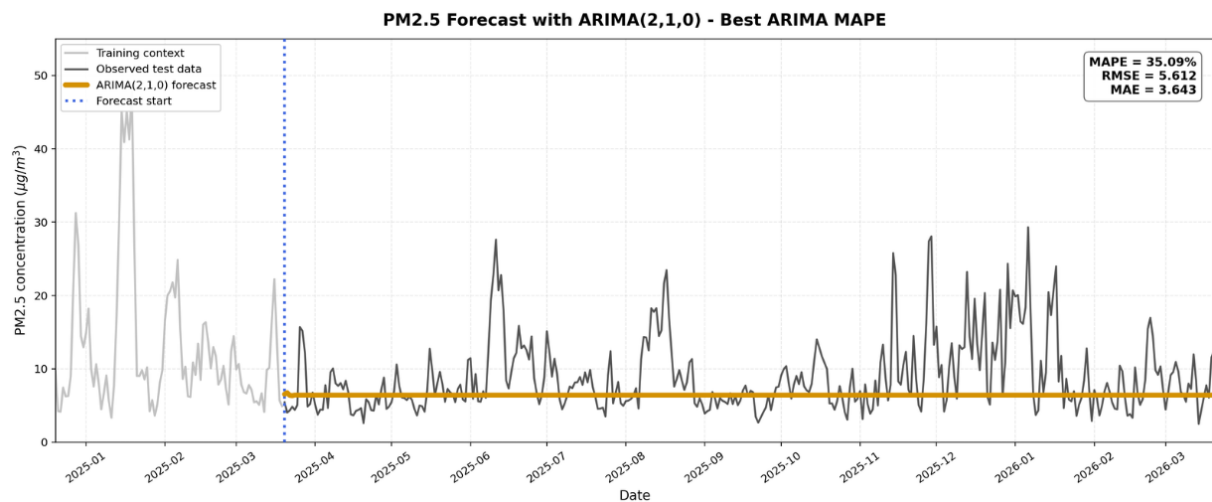


Figure 4.1: Best ARIMA model (in orange) applied to forecasting the pollutant concentration series (in black)

The ARIMA (2,1,0) model achieved the best performance among the tested models, with a MAPE of 35.09%, an RMSE of 5.612, and an MAE of 3.643. However, the forecasts remain nearly constant around the mean value of the series and fail to reproduce the large fluctuations observed in PM2.5 concentrations. These results indicate that the model captures the overall trend of the signal but has limitations in representing concentration peaks and short-term variability.

4.2 SARIMA METHOD

The SARIMA forecast follows the seasonal evolution of the PM2.5 series more closely than ARIMA.

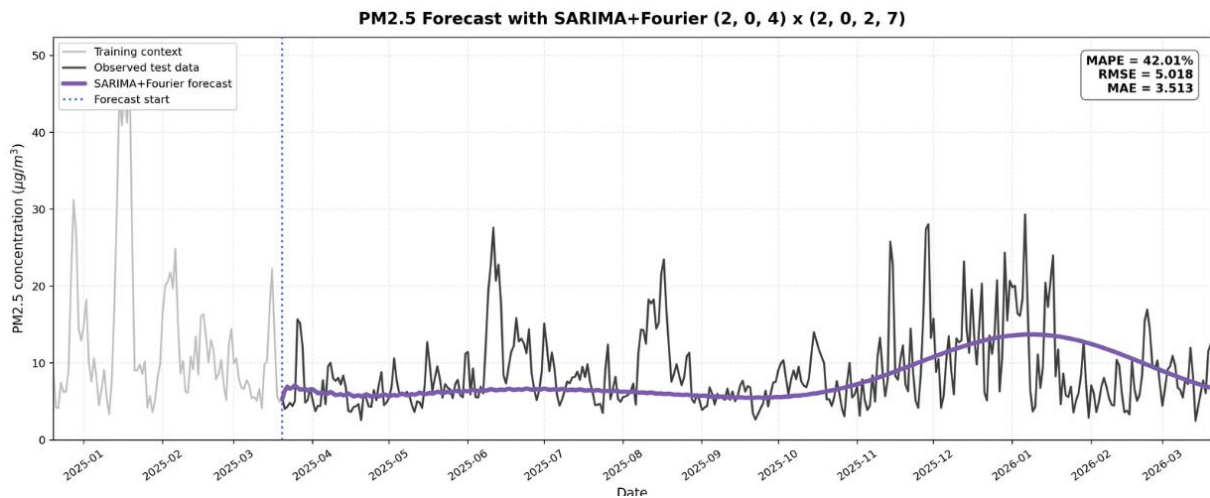


Figure 4.2: Best SARIMA model (in orange) applied to forecasting the pollutant concentration series (in black)

The SARIMA + Fourier $(2,0,4) \times (2,0,2,7)$ model can represent a seasonal component of the series, as shown by the smooth evolution of the forecasts over time. The model achieved a MAPE of 42.01%, an RMSE of 5.018, and an MAE of 3.513. Although the RMSE and MAE values are slightly lower than those obtained with the ARIMA model, the MAPE remains higher. The model captures the overall trend and seasonal variations of the series but fails to reproduce the concentration peaks observed in the real data. Therefore, it provides a better representation of the underlying dynamics of the series, while short-term fluctuations remain difficult to predict.

4.3 HOLD WINTER'S METHOD

The Holt-Winters forecast shows a smoother decreasing trend over the prediction period. Although the model captures the general direction of the series, it tends to underestimate several observed peaks, which increases the forecasting error.

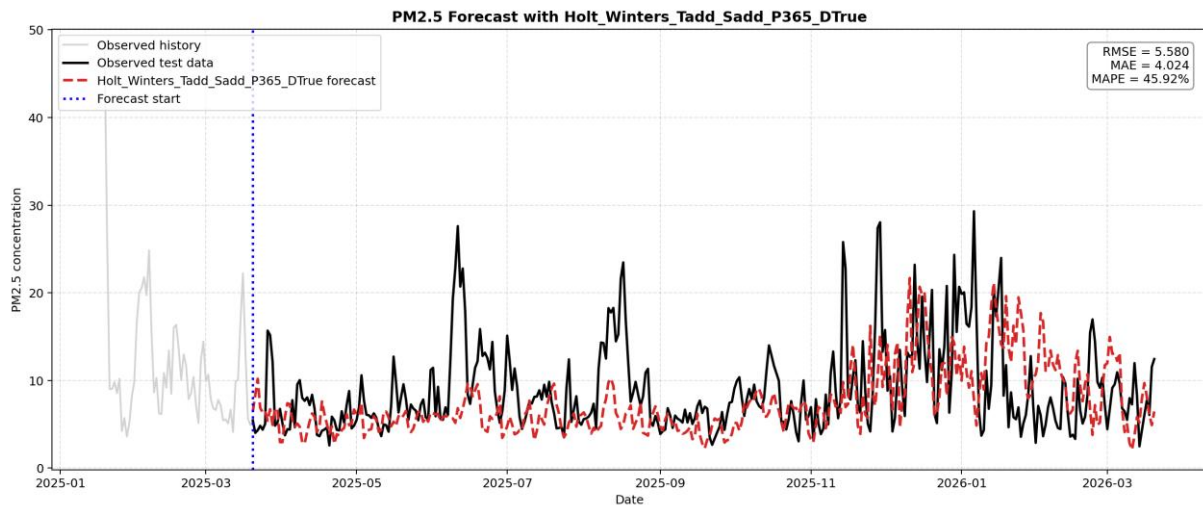


Figure 4.3: Best Holt-Winters model (in orange) applied to forecasting the pollutant concentration series (in black)

The Holt-Winters model (additive trend, additive seasonality, damped trend) achieved a MAPE of 45.92%, an RMSE of 5.580, and an MAE of 4.024. The forecasts follow the general evolution of the PM2.5 concentrations and capture part of the seasonal variability present in the series. However, the model tends to smooth the signal and is unable to accurately reproduce the magnitude and timing of the observed concentration peaks. Although it reflects the overall behavior of the series, its predictive performance remains limited for short-term fluctuations and extreme pollution events.

4.3 RESULTS COMPARISON

In this comparison, the optimization criterion was MAPE, which is why ARIMA appears as the best-performing model in relative percentage terms, with a MAPE of 35.09%. However, this does not necessarily mean that ARIMA is the best model overall. When looking at other metrics such as RMSE and MAE, models like SARIMA+Fourier show lower absolute errors, meaning that their predictions can be closer to the observed values in real magnitude. Therefore, ARIMA can be considered the best model for the specific objective of minimizing MAPE, but the final model selection should also consider RMSE, MAE, and the visual behavior of the forecasts.

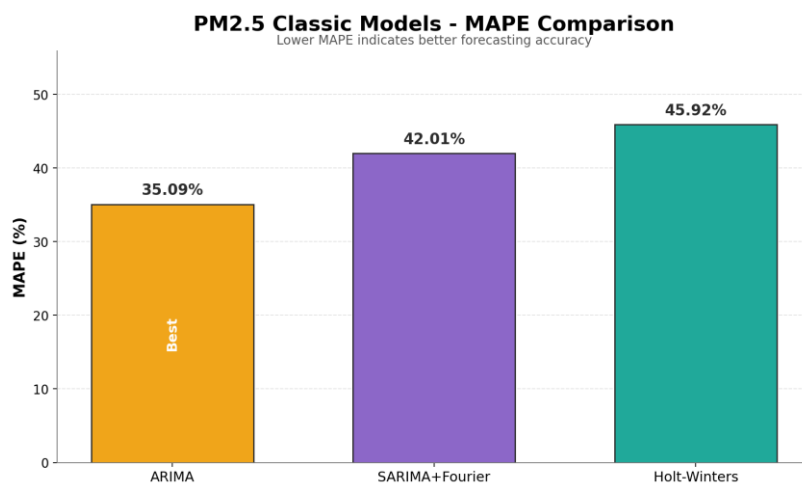


Figure 4.4: Comparison between classical models

5. DEEP LEARNING METHODS

The deep learning framework developed for this study was structured into three sequential phases: (1) baseline hyperparameter optimization; (2) the integration of these optimized configurations into the univariate base models for each target pollutant; and (3) the incorporation of exogenous data as predictive features, followed by a secondary hyperparameter tuning phase specifically tailored to these multivariate architectures.

All computational experiments were executed locally utilizing an Intel Core i7 (13th Generation) central processing unit. Operating within this hardware parameters necessitated pragmatic engineering trade-offs regarding resource allocation and the management of extended training times, which frequently spanned several hours per architecture. Balancing these strict computational constraints against the requirement for high predictive accuracy underscores the core applied-engineering nature of this project, ensuring that the resulting models are not only mathematically robust but also practically deployable within standard hardware environments.

5.1 HYPERPARAMETERS BAYESIAN OPTIMISATION

Following preliminary tests—based on initial hyperparameter guidelines provided by the project supervisor to validate the algorithms and the overall project architecture—Bayesian optimization was conducted using the Python library Optuna to minimize the Mean Absolute Percentage Error (MAPE). The initial phase was exploratory, executing over 200 trials across a subset of pollutants and models. This allowed us to analyze the convergence behavior of the trials and evaluate the outputs generated by the Optuna framework.

Observing that most optimization curves converged within 100 trials, subsequent optimizations were capped at 100 trials per model for each pollutant, including those incorporating exogenous variables.

Standard practice often involves defining a small number of discrete values per parameter to encourage the optimizer to explore diverse configurations. However, during the initial trials, we observed that the optimizer frequently evaluated duplicate parameter sets rather than effectively exploring the space. To resolve this, we defined a more comprehensive and granular search space for the parameters.

The chosen hyperparameter search space is detailed in the table below. Because some configurations were highly time-consuming to train, Optuna's Hyperband pruner

was implemented to early-stop unpromising trials and optimize computational resources.

Table 5.1: Choice of hyperparameters, their computational meaning in the context of RNNs and levels chosen

Hyperparameter	Meaning	Levels
Number of layers	The total number of hidden layers within the network architecture.	{2, 4, 6}
Hidden Dimensions	The number of hidden units (or neurons) present within each hidden layer.	{64, 128, 256}
Batch size	The number of training samples processed in a single forward and backward pass before updating the model's internal weights.	{16, 64, 128}
Look-back Window	The number of historical time steps provided to the network as input to capture temporal dependencies.	{15, 30, 45}
Prediction horizon	The number of future timesteps in the model is configured to forecast in a single output sequence.	{7, 21, 28}
Learning Rate	The step size determines the magnitude of weight updates during the optimization process.	{0,001; 0,01}

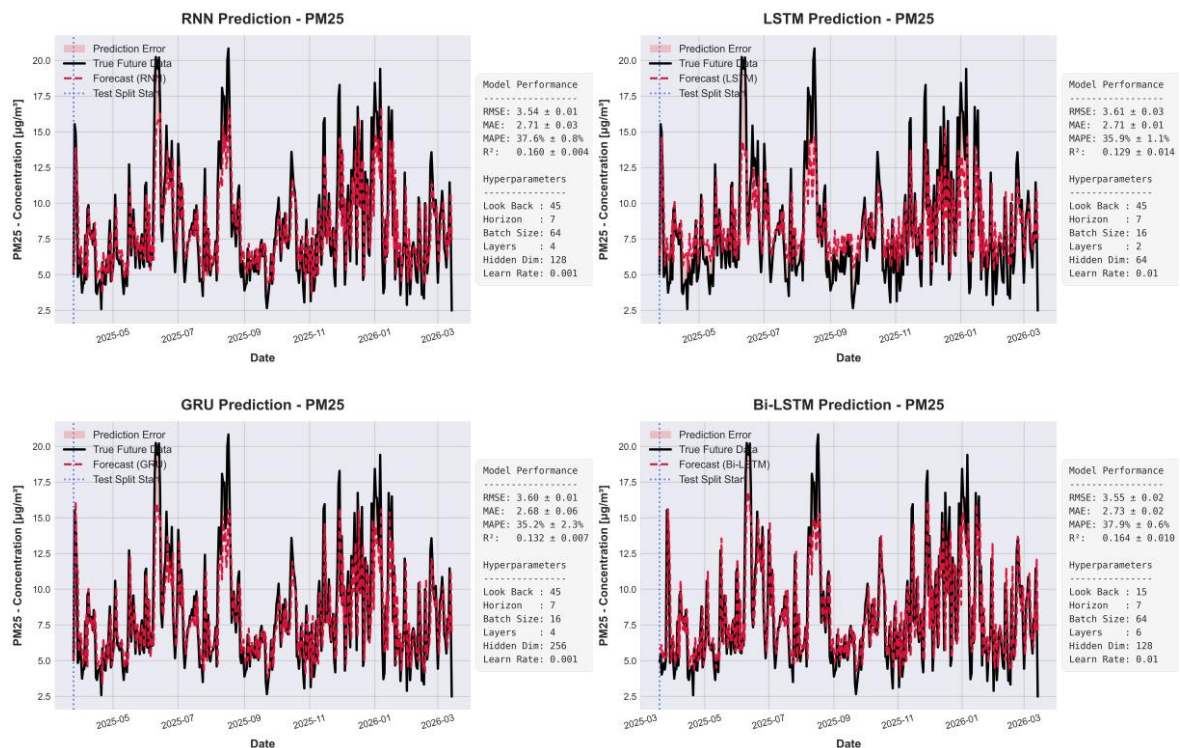
Optuna allows to make analysis based on the optimization studies, such as the hyperparameter relevance for each, the convergence curves and an attempt to map graphically the maxima and the minima of the loss function as a function of the hyperparameters.

Optuna provides several built-in visualization tools to analyze the results of the optimization studies. These features include evaluating the relative importance of each hyperparameter, plotting the optimization history (convergence curves), and graphically mapping the loss landscape to identify the objective function's minima across the hyperparameter space.

5.2 UNIVARIATE PREDICTION OF POLLUTANT TREND

To ensure robustness, the deep learning models were evaluated over three independent runs. The resulting performance metrics were averaged, and the standard deviations were recorded for comparative analysis. Overall, all recurrent architectures demonstrated strong predictive capabilities, with their forecast curves closely tracking the ground truth data in the test set. Deep Learning models also were compared in terms of R^2 to assess the best fitting for each pollutant.

Using $PM_{2.5}$ as a representative example, we observed that the LSTM and GRU models outperformed the alternative architectures in terms of MAPE. This superior performance is likely due to their optimal balance of long-term memory retention and architectural simplicity. Because the vanilla RNN struggles to capture long-term dependencies, primarily due to the vanishing gradient problem, it is prone to temporal shifts when predicting pollution spikes, which heavily penalizes the MAPE score. Conversely, the Bi-LSTM model appeared overly complex for this specific univariate application. Furthermore, the hybrid RNN-LSTM architecture performed comparably to the standalone LSTM and GRU models, reinforcing the necessity of gated memory mechanisms for effective sequential data handling.



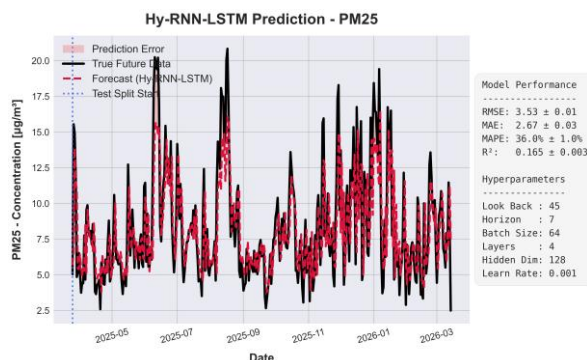


Figure 5.1: PM2.5 concentration forecasts across all evaluated models. The vertical blue dashed line delineates the start of the test dataset on March 21, 2025. The solid black line represents the observed ground truth data collected through March 21, 2026, while the dashed red line denotes the models' predictions. Optimal model performance is indicated by a high degree of overlap between the predicted and observed values. The discrepancy between the two curves, representing the residual prediction error, is highlighted by the light red shaded area.

Notably, the analysis revealed that the primary factor influencing model performance and error metrics is the specific pollutant under consideration, with distinct chemical families exhibiting highly correlated predictive trends. For instance, the predictive profiles for NO_2 and NO_x concentrations were remarkably consistent across all models. These profiles differed significantly from those of O_3 , $\text{PM}_{2.5}$, and PM_{10} , while the particulate matter variants ($\text{PM}_{2.5}$ and PM_{10}) demonstrated a similarly cohesive behavior with one another.

The elevated error metrics associated with particulate matter likely stem from its strong correlation with highly stochastic anthropogenic activity. Unlike nitrogen oxides or ozone, whose baseline concentrations are heavily regulated by predictable, large-scale biogeochemical processes (the nitrogen and oxygen cycles, respectively), urban particulate matter is predominantly driven by localized human emissions. Because these anthropogenic sources, such as traffic congestion or industrial activity, are highly non-linear and irregular, the resulting concentration peaks are inherently more difficult for the models to forecast accurately.

5.2 EFFECT OF ADDED ENVIRONMENTAL FEATURES

To incorporate environmental features into the pollutant forecasting models while mitigating unnecessary computational complexity, a Spearman's rank correlation analysis was conducted between the target pollutants and the collected meteorological variables. Subsequently, the top three most highly correlated features were selected and

integrated into each model to facilitate multivariate forecasting. The resulting Spearman correlation matrix is presented in Figure 5.X.



Figure 5.2: Pearson correlation matrix between pollutants and environmental features.

The correlation matrix confirms the previously observed similarities within pollutant families and reveals the distinct environmental dynamics driving their respective concentrations. For example, temperature and albedo exhibit a strong positive correlation with O_3 , but a strong negative correlation with NO_2 and NO_x . Conversely, atmospheric pressure demonstrates the exact opposite trend for both pollutants. This reflects the well-documented photochemical coupling between ozone and nitrogen oxides, whose formation and depletion cycles are inherently antagonistic.

For particulate matter ($PM_{2.5}$ and PM_{10}), wind speed emerges as the most significant variable. Because particulates have physical mass, unlike purely gaseous pollutants, their localized concentrations are heavily governed by atmospheric dispersion and advection dynamics. By capturing these underlying physical and chemical relationships, the multivariate models can leverage these exogenous variables to generate more accurate concentration forecasts. The $PM_{2.5}$ is taken again as a representative example in Figure X.

Analyzing the results on a per-pollutant basis, the models' predictive performance for the gaseous pollutants (NO_2 , NO_x , and O_3) actually deteriorated upon the inclusion of exogenous features across almost all architectures. This degradation likely stems from the fact that the collected meteorological data represents strictly surface-level conditions, which fail to capture the complex transport and mixing dynamics occurring in higher atmospheric boundary layers. Furthermore, the Bayesian optimization process

may have struggled with the increased dimensionality of these multivariate models, failing to converge effectively when optimizing strictly for a relative error metric such as MAPE. Finally, it is highly plausible that incorporating these environmental features introduced unnecessary complexity, or noise, into the modeling of these specific gaseous cycles, given that their natural photochemical periodicity is already well-established and highly predictable using solely historical univariate data.

Nevertheless, GRU and the RNN-LSTM hybrid had quite satisfactory results on O_3 , probably because of the simplicity of measure in this pollutant and the improved memory handling.

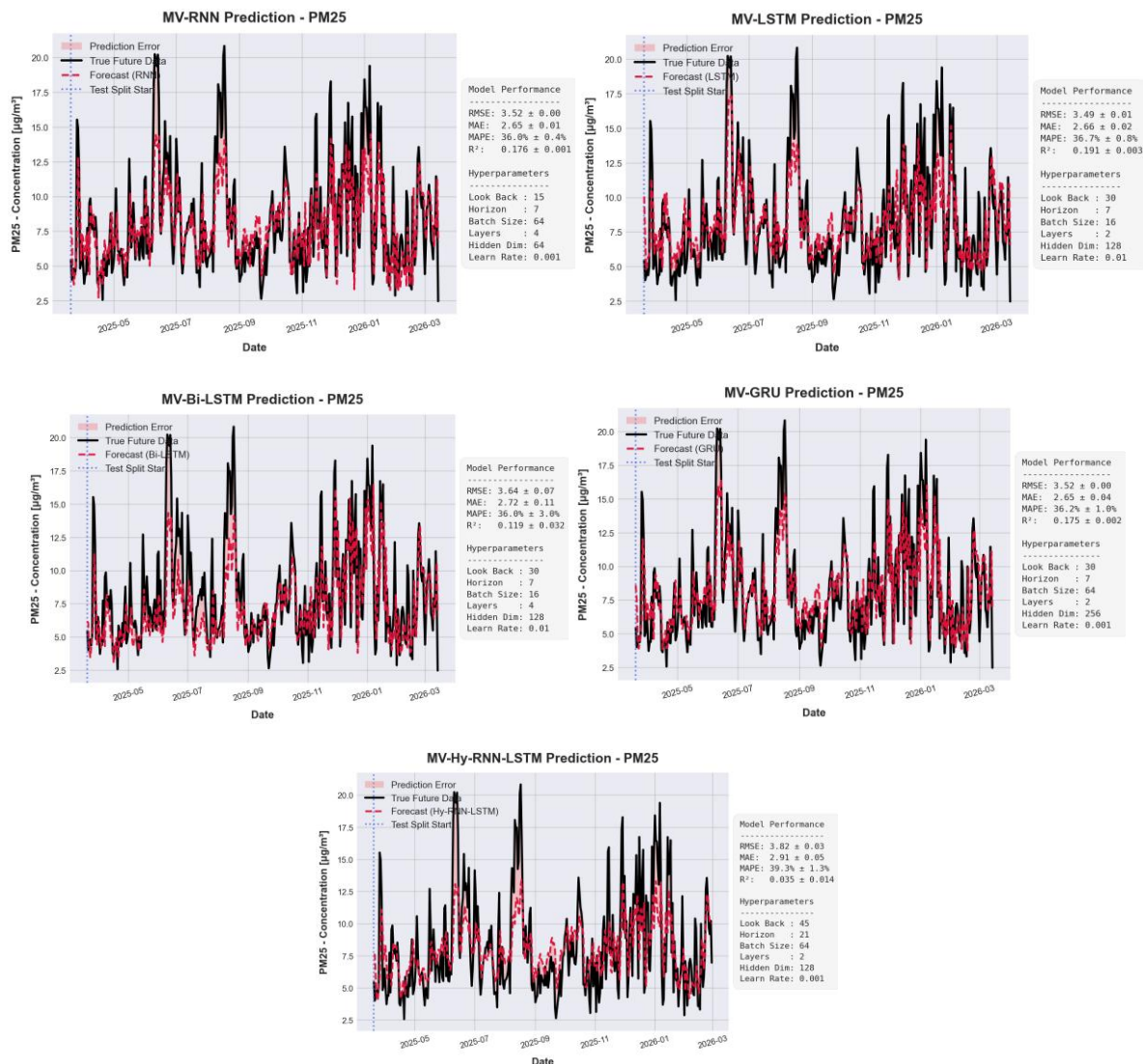


Figure 5.3: PM2.5 concentration forecasts across all multivariate evaluated models. The vertical blue dashed line delineates the start of the test dataset on March 21, 2025. The solid black line

represents the observed ground truth data collected through March 21, 2026, while the dashed red line denotes the models' predictions. Optimal model performance is indicated by a high degree of overlap between the predicted and observed values. The discrepancy between the two curves, representing the residual prediction error, is highlighted by the light red shaded area.

5.3 RESULTS COMPARISON

Figure 5.4 illustrates the comparative predictive performance of the evaluated deep learning architectures across the different chemical pollutant classes. Notably, O_3 demonstrates a significantly higher degree of predictability compared to the other evaluated pollutants, whereas the particulate matter fractions (PM_{10} and $PM_{2.5}$) exhibit the lowest forecasting accuracy. Overall, the models yielded Mean Absolute Percentage Error (MAPE) scores ranging from 20% to 40%, alongside Coefficient of Determination (R^2) values between 0.05 and 0.45. These performance metrics are highly consistent with the benchmark forecasting values reported in contemporary atmospheric modeling literature.

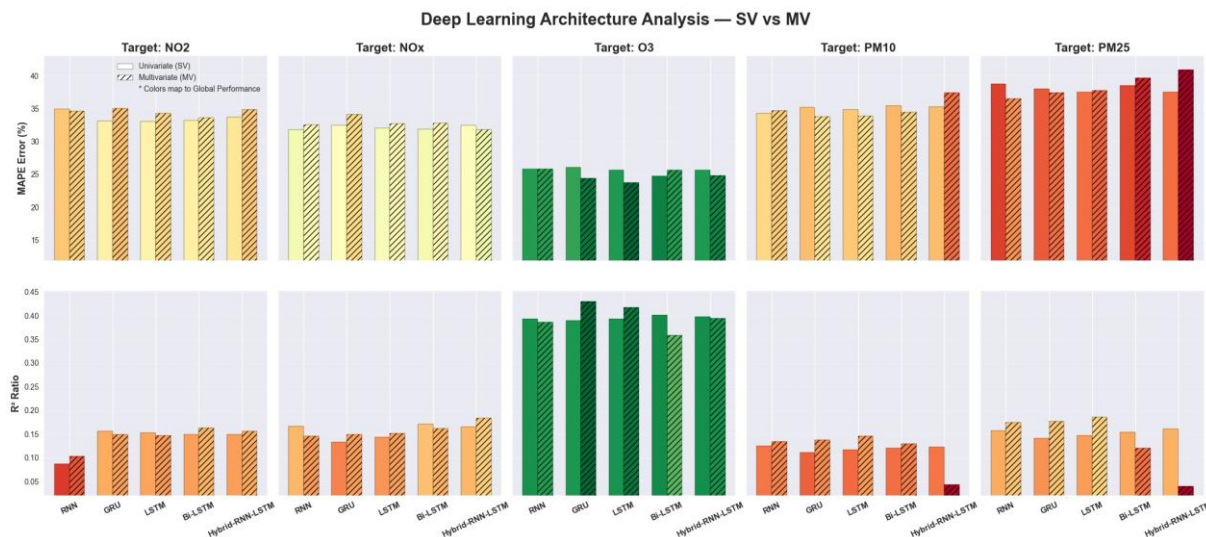


Figure 5.4: Comparative performance of the deep learning architectures, grouped by target pollutants. Solid bars denote univariate configurations, while hatched bars represent the multivariate models.

Figure 5.5 presents a comparative analysis of the univariate and multivariate forecasting configurations. The transition to a multivariate framework yielded highly architecture-dependent outcomes: both the GRU and LSTM models demonstrated a clear enhancement in predictive capability, evidenced by a reduction in forecasting error

and an increase in the Coefficient of Determination (R^2). In contrast, the standard RNN exhibited a stagnant performance profile, while the more sophisticated Bi-LSTM and hybrid architectures actually experienced a distinct degradation in accuracy. This counterintuitive performance drop can likely be attributed to the inherent structural complexity of these advanced networks. When multivariate inputs are introduced to already parameter-heavy architectures, the exponentially increased dimensionality can lead to overfitting and information bottlenecking, ultimately degrading the model's ability to generalize.

An alternative explanation lies within the limitations of the hyperparameter tuning phase conducted via the Optuna framework. It is highly probable that the vast, high-dimensional search spaces generated by these complex multivariate models overwhelmed the Bayesian optimization algorithm. This hypothesis is strongly supported by the excessively long computational times and delayed convergence observed during the optimization phase, suggesting that the algorithm may have prematurely settled on sub-optimal local minima rather than discovering the true global optimum.

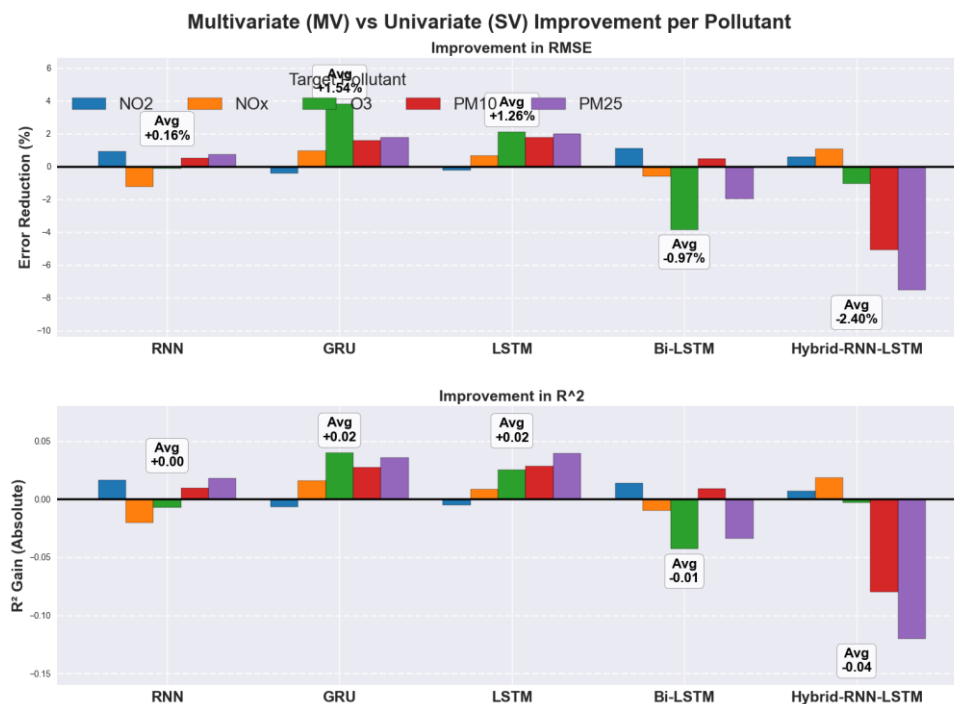


Figure 5.5: Comparative evaluation of univariate versus multivariate forecasting configurations, grouped by deep learning architecture. Performance enhancements are quantified by the relative reduction in forecasting error and the absolute gain in the Coefficient of Determination (R^2), where greater positive values indicate superior predictive accuracy.

Finally, Figure 5.6 presents a comprehensive cross-model comparison, evaluating the predictive efficacy of the univariate deep learning architectures against classical statistical baselines. Consistent with preceding analyses, the results unequivocally demonstrate the superior forecasting capability of the deep learning architectures. While there are isolated instances where the **ARIMA** model appears to yield marginally better quantitative error scores, this statistical advantage is largely artifactual. Standard global error metrics can occasionally misrepresent true predictive quality by disproportionately rewarding conservative, smoothed baseline predictions while penalizing the neural networks for slight temporal misalignments when attempting to forecast complex stochastic spikes. Consequently, these quantitative metrics must be contextualized alongside a rigorous visual inspection of the forecasted curves. As visual analysis corroborates, deep learning models are vastly superior at capturing the non-linear atmospheric variance, the high-frequency cyclicity, and the extreme emission events that classical models entirely fail to reproduce.

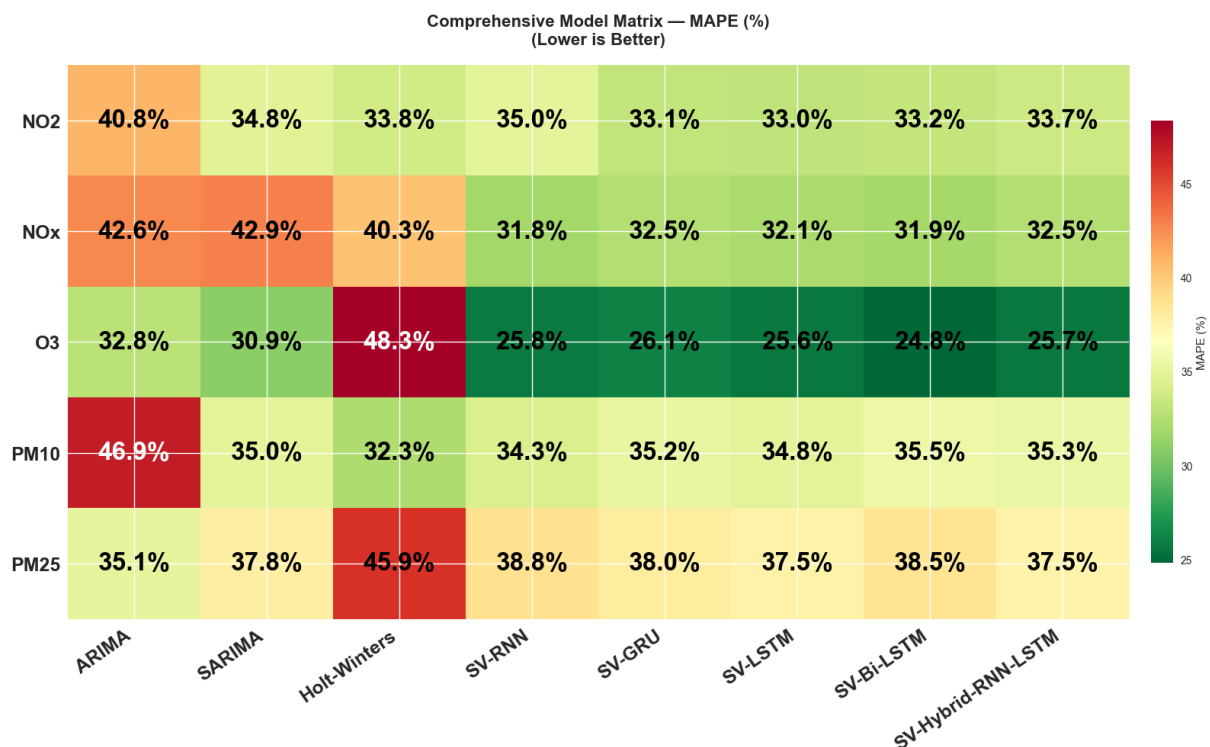


Figure 5.6: Heatmap matrix illustrating the comparative forecasting performance between the univariate deep learning architectures and the classical statistical models' performances across all evaluated pollutants. The color gradient quantifies the relative performance, red being the worse ones and green being the better ones.

CONCLUSION

This project served as a rigorous practical immersion into advanced time-series forecasting and deep learning methodologies. Beyond the core technical scope, it presented a multifaceted engineering challenge that necessitated the integration of both analytical and interpersonal competencies. The project team successfully navigated a complete data science workflow, from foundational statistical theory and complex algorithmic implementation to strict project, time, and computational resource management. Furthermore, the collaborative nature of the work fostered critical soft skills, notably the ability to effectively communicate highly technical concepts within a linguistically and culturally diverse environment. Ultimately, the pedagogical objectives of this 8th-semester project were fully realized. By successfully bridging the gap between theoretical atmospheric modeling and applied machine learning constraints, this endeavor has comprehensively equipped the team with the practical expertise required to tackle complex, industry-scale engineering challenges.

This report details the conclusive findings of the deployment project, which focused on developing decision-support models by leveraging predictive modeling for atmospheric pollution analysis. The results indicate that deep learning recurrent neural network (RNN) architectures consistently outperformed classical statistical baselines across all evaluated pollutant classes, and the integration of exogenous environmental features yielded distinct, architecture-dependent performance enhancements. Crucially, overall predictive accuracy was highly contingent upon the specific target pollutant, underscoring the unique, complex atmospheric dynamics and physical formation cycles inherent to each chemical species.

Overall, the GRU, LSTM, Bi-LSTM, and hybrid RNN-LSTM frameworks demonstrated the most robust forecasting capabilities across the majority of the pollutants. The average performance differentials between these top-tier models were marginal, largely falling within statistical error boundaries. Quantitatively, the models achieved Mean Absolute Percentage Error (MAPE) scores of 33–34% for NO_2 and NO_x , 26–28% for O_3 , 34–36% for PM_{10} , and 38–39% for $\text{PM}_{2.5}$.

Although these error values are marginally higher than the performance metrics reported in recent literature (Whu, Zu, & Wen, 2024), this variance must be contextualized by the inherent limitations of the training data. This project was conducted utilizing a constrained, six-year historical dataset originally compiled for general atmospheric monitoring, rather than being explicitly curated or pre-processed for machine learning

applications. In contrast, state-of-the-art deep learning benchmarks typically leverage highly optimized, multi-decade databases specifically engineered to maximize neural network convergence and predictive accuracy. Given these structural data limitations, the forecasting performance achieved by the architectures in this study remains highly robust and competitive.

While this project establishes a robust baseline for atmospheric time-series forecasting, several identified limitations present distinct opportunities for future research. Methodologically, subsequent iterations should incorporate a more rigorous fine-tuning of the Adam optimizer's momentum hyperparameters, alongside a refined, epoch-by-epoch analysis of the loss evolution to better characterize convergence dynamics and mitigate transient overfitting. From an architectural perspective, future work could explore the integration of Convolutional Neural Networks (CNNs) for spatiotemporal feature extraction, or the deployment of genetic algorithms for automated neural architecture search (NAS), also, other variants of LSTM and GRU which go beyond the traditional architectures such as Bi-GRU, and other hybrid architectures.

Furthermore, shifting the predictive paradigm toward event-driven forecasting, specifically targeting the identification and prediction of extreme pollution spikes, would significantly enhance the real-world utility of the system. Collectively, these advanced analytical vectors provide a comprehensive research roadmap, serving as highly suitable foundations for future End-of-Studies Projects (Projets de Fin d'Études - PFE) at ENIT focused on industrial deep learning deployment.

This preliminary study serves as a foundational exploration into the application of advanced time-series analysis for complex environmental challenges. Within the current scope, the primary focus remained on the successful deployment, algorithmic configuration, and preliminary performance evaluation of various neural network architectures. Future iterations of this research must incorporate a more nuanced analytical framework. Specifically, the evaluation criteria should expand beyond standard point-wise magnitude metrics, such as MAPE and the Coefficient of Determination (R^2), to include dynamic, time-dependent evaluation methodologies. Metrics such as Dynamic Time Warping (DTW) or directional accuracy would better capture temporal phase shifts, lag effects, and the precise timing of peak pollution events. Furthermore, conducting a rigorous cost-benefit analysis regarding computational overhead and training latency will be paramount.

Quantifying the exact trade-off between predictive accuracy and computational resource consumption is a critical prerequisite for selecting the optimal architecture for

each specific pollutant, ultimately facilitating the transition of these predictive models from an experimental framework into a scalable, production-ready product.

The final realization of this project would entail the transition from an experimental analytical framework into a fully integrated software application, specifically designed to assist governmental and municipal entities in proactively managing atmospheric pollution. Deploying such a system at scale necessitates a multidisciplinary approach, engaging a diverse network of institutional, technical, and civic partners. The proposed stakeholder mapping necessary for this product development phase is detailed in Figure C.1. Ultimately, this report underscores a fundamental engineering mandate: advanced computational methodologies must be leveraged for the public good. By equipping decision-makers with robust, data-driven forecasting tools, we can proactively address critical environmental challenges, mitigate public health risks, and tangibly improve urban quality of life.

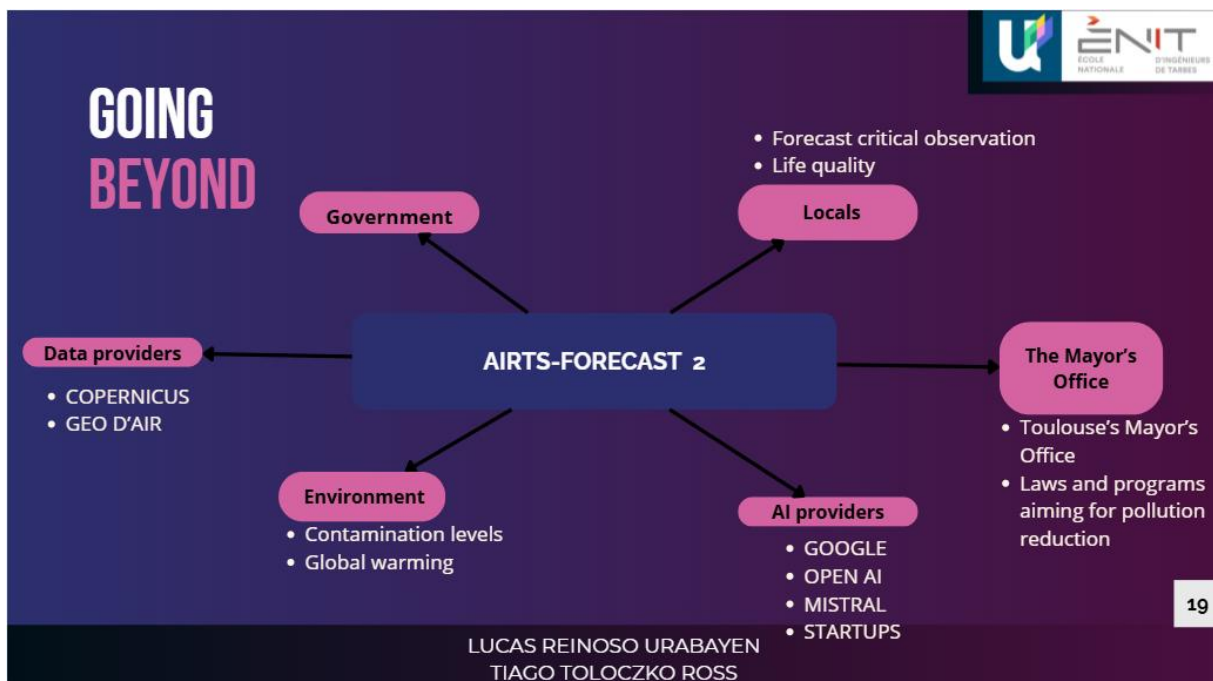


Figure 0.1: Foreseen stakeholders' chart for the future realization of this project as presented by the students, implicating various sectors of civil society.

REFERENCES

- AKIBA, T.; *et al.* Optuna: A Next-generation Hyperparameter Optimization Framework. In KDD. 2019.
- ALTHOFF, K.-D.; WEBER, R. O. Knowledge management in case-based reasoning. 2006. The Knowledge Engineering Review, vol. 20, 3, p. 305-310.
- BERJARANO, J. C. R.; *et al.* Case-based reasoning and system design: and integrated approach based on ontology and preference modeling. 2014. Artificial Intelligence for Engineering Design, Analysis and Manufacturing, vol. 28, p.49-69.
- BRAFFORD, P. Foreword: From Cybernetics to Artificial Intelligence; *in* MARQUIS, P.; PAPINI, O.; PRADÉ, H. A guided Tour of Artificial Intelligence Research: Volume III; Interfaces and Applications of AI. Springer Nature: Switzerland, 2020. P. xiii-xviii.
- CHO, K. Learning Phrase Representations using RNN Encoder-Decoder for Statistical Machine Translation. ArXiv, cs.CL, sep. 2014.
- CLERMONT, Ph. Gestion des Projets – Poly de cours. École Nationale des Ingénieurs de Tarbes: Tarbes, 2016
- GÓMEZ-GARCIA, M. A.; PITCHON, V.; KIENNEMANN, A. Pollution by nitrogen oxides: an approach to NO_x abatement by using sorbing catalytic materials. 2005. Environment International, vol. 31, p.445-467.
- HOCHREITER, S.; SCHUIDHUBER, J. Long Short-Term Memory. **Neural Computation**, v9, n 8, p. 1735-1780, 1997.
- KELEKO, A. T.; *et al.* Artificial Intelligence and real-time predictive maintenance in industry 4.0: a bibliometric analysis. 2022. AI and Ethics, 2, p.553-577.
- KINGMA, D. P.; LEI BA, J. Adam: A Method for Stochastic Optimization. 2015. ICLR.
- KRIZHEVSKY, A.; SUTSKEVER, I.; HINTON, G. E. imageNet Classification with Deep Convolutional Neural Networks. 2012.
- LE CUNN, Y. A Theoretical framework for Back-Propagation. 1988. Proceedings of the 1988 Connectionist Models Summer School, p. 21-28.
- LE CUNN, Y. Quand la machine apprend : La révolution des neurones artificiels et de l'apprentissage profond. Odile Jacob: Paris, 2019.
- NOÉ, F.; DE FABRITIIS, G.; CLEMENTI, C. Machine learning for protein folding and dynamics. 2020. Current Opinion in Structural Biology, vol. 60, p.77-84.

-
- PRENAT, M. **Séries temporelles**. Techniques de l'ingénieur, 2012.
- ROBBINS, H.; MONRO, S. A Stochastic Approximation Method. 1951. The Annals of Mathematical Statistics, vol. 22, n.3, p.400-407.
- ROOD, S.; *et al.* Recent advances in gasoline three-way catalyst formulation: A review. 2020. Proc. IMechE part D: J automobile Engineering. vol. 234, 4, p.936-949.
- ROSENBLATT, F. The Perceptron: A Probabilistic Model for Information Storage and Organization in the Brain. 1958. Psychological Review, vol 65., n. 4.
- RUMELHART, D. E.; HINTON, G. E.; WILLIAMS, R. J. Learning representations by back-propagating errors. 1986. Nature, vol. 323, n. 7.
- SCHRAUFNAGEL, D. E.; *et al.* Air Pollution and Noncommunicable Diseases. 2019. CHEST, vol. 155, 2, p.409-416.
- SEINFELD, J. H.; PANDIS, S. N. Atmospheric chemistry and physics : from air pollution to climate change (3rd ed.). Wiley: 2019. *Apud WIKIPEDIA*, NO₂. Last edited in feb. 2026. Available at: <https://en.wikipedia.org/wiki/NOx> .
- STAUDEMEYER, R. C.; MORRIS, E. R. Understanding LSTM: a tutorial into Long Short-Term Memory Recurrent Neural Networks. ArXiv, sep. 2019.
- TEYTAUD, O. **Le Deep Learning En Équations** : avec des équations et sans code. Cépaduès: Toulouse, 2025
- TRINH, T.-H.; *et al.* Filtration systems for particulate matter reduction in outdoor air: a review. 2025. Journal of Environmental Management, vol. 390.
- TURING, A. M. Computing Machinery and Intelligence. 1958. **Mind**, 49; p, 433-460.
- US ENVIRONMENTAL PROTECTION AGENCY. Integrated Science Assessment for Oxides of Nitrogen – Health Criteria. 2016. Available at: <https://www.federalregister.gov/documents/2016/01/28/2016-01548/integrated-science-assessment-for-oxides-of-nitrogen-health-criteria> .
- VASVWANI, A.; *et al.* Attention is All You Need. From the 31st Conference on Neural Information Processing Systems (NIPS 2017), Long Beach, CA, USA.
- WHU, X.; ZHU, J.; WEN, Q. Short-term prediction of PM_{2.5} concentration by hybrid neural network based on sequence decomposition. PLoS ONE, 19, 5, 2023.
- WIKIPEDIA**. Augmented Dickey-Fuller test. Last edited in dec. 2025. Available at: https://en.wikipedia.org/wiki/Augmented_Dickey%E2%80%93Fuller_test .



WIKIPEDIA. Autocorrelation. Last edited in may 2026. Available at:
<https://en.wikipedia.org/wiki/Autocorrelation> .

WIKIPEDIA. Dickey–Fuller test. Last edited in jun. 2025. Available at:
https://en.wikipedia.org/wiki/Dickey%E2%80%93Fuller_test .

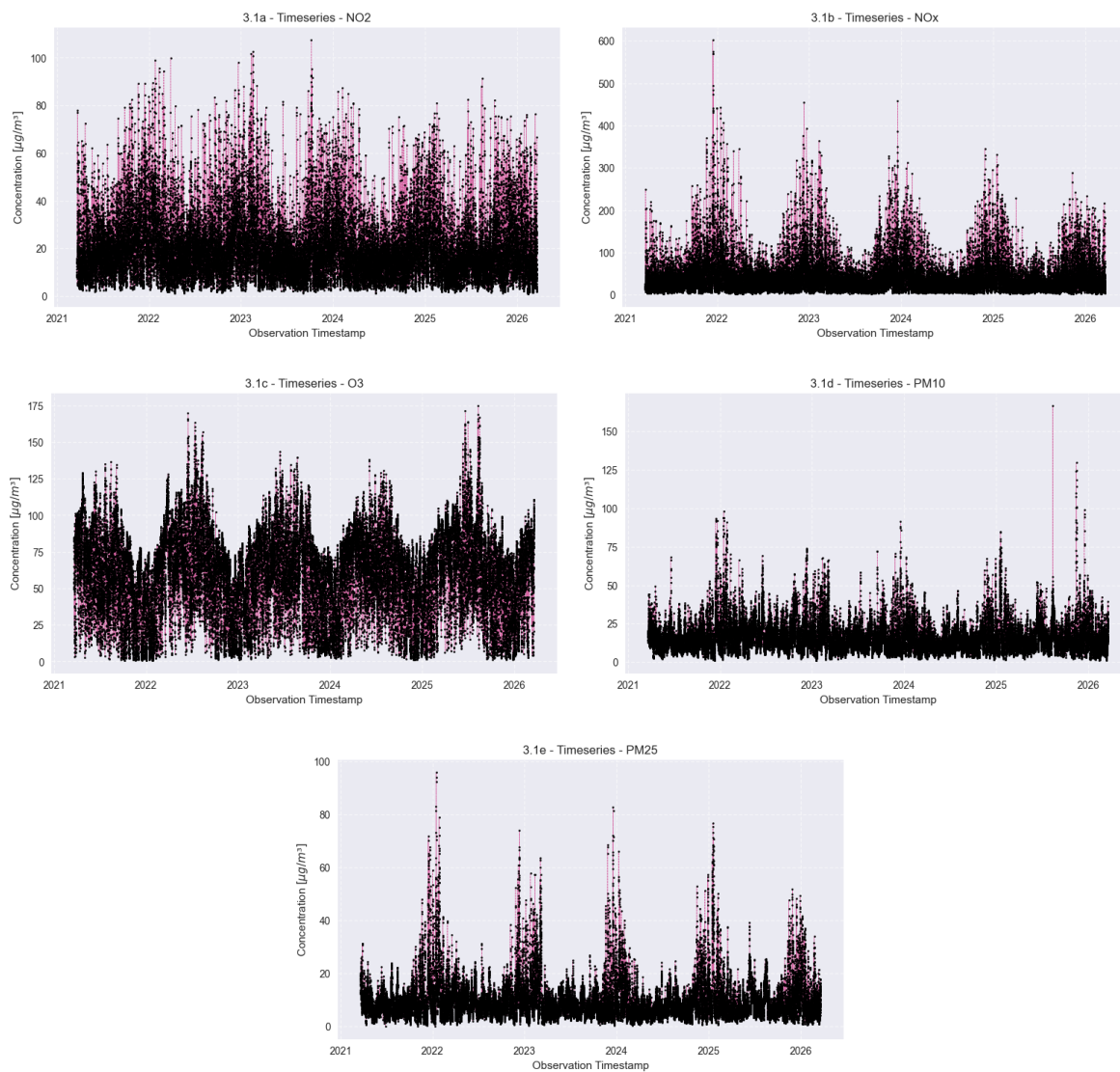
WIKIPEDIA. Parial autocorrelation function. Last edited in feb. 2026; Available at:
https://en.wikipedia.org/wiki/Partial_autocorrelation_function .

WORLD HEALTH ORGANIZATION [WHO]. Ambient (outdoor) air pollution. 2022. Available in:
[https://www.who.int/news-room/fact-sheets/detail/ambient-\(outdoor\)-air-quality-and-health](https://www.who.int/news-room/fact-sheets/detail/ambient-(outdoor)-air-quality-and-health) .

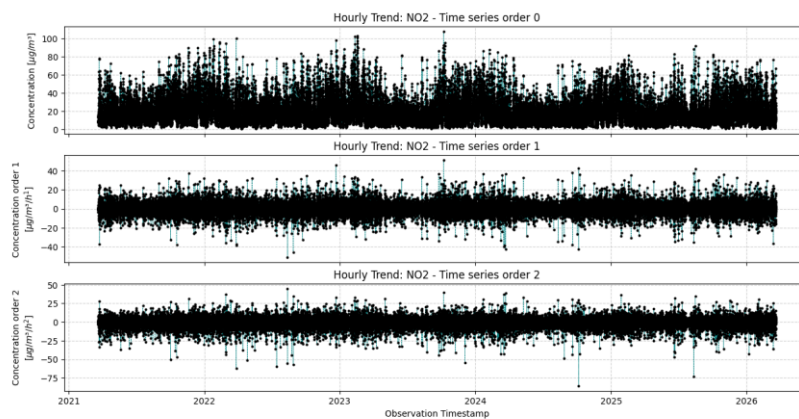
APPENDIX – OTHER PLOTS

1. OTHER POLLUTION EDA PLOTS

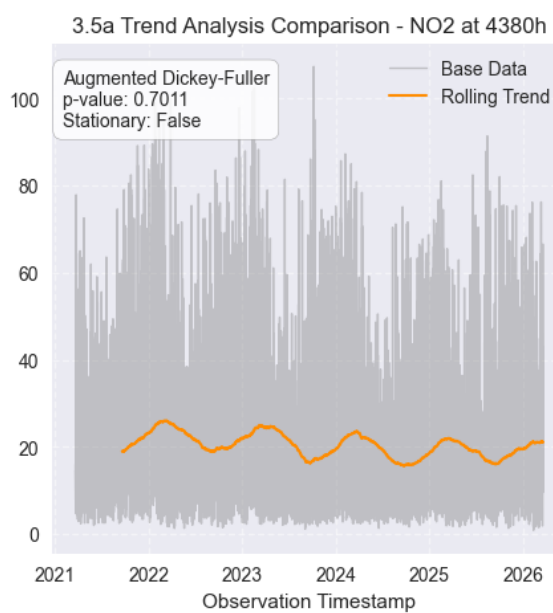
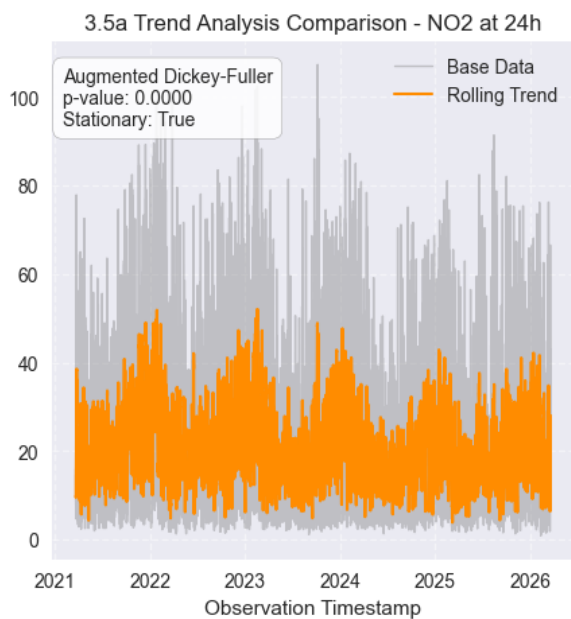
Hourly trend

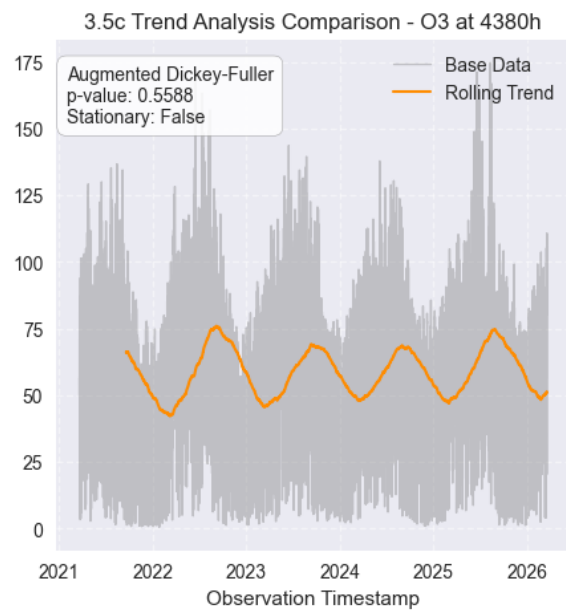
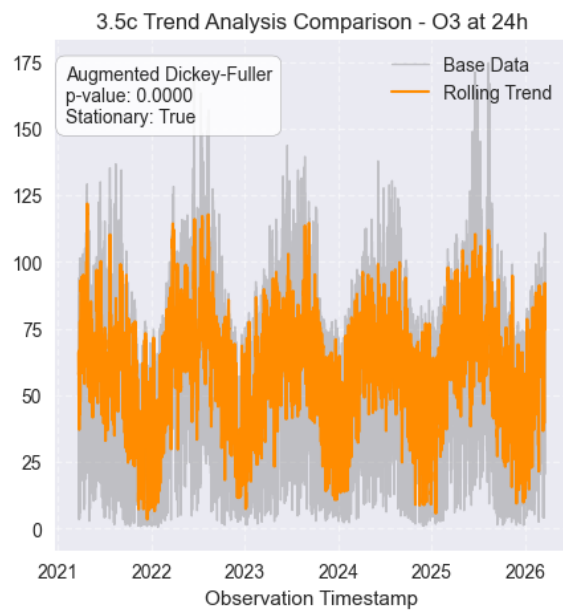
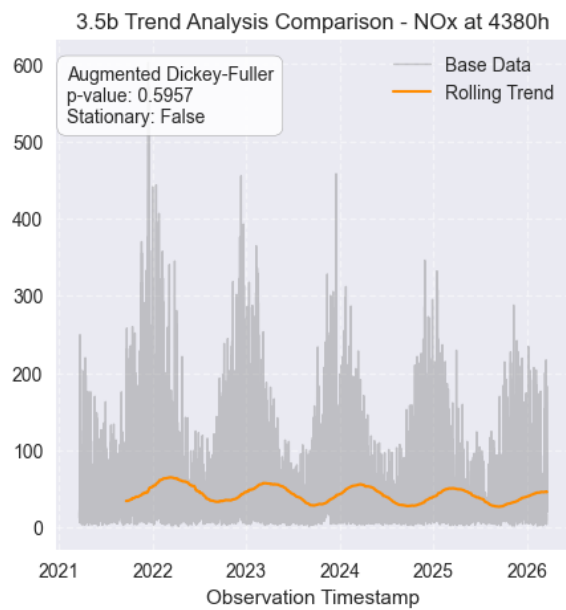
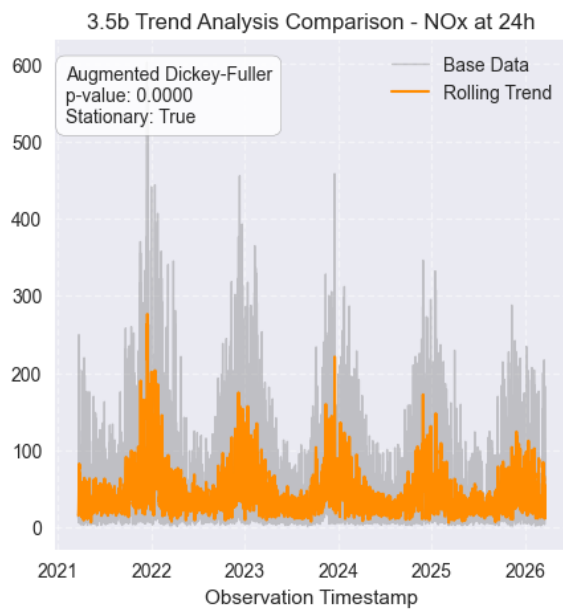


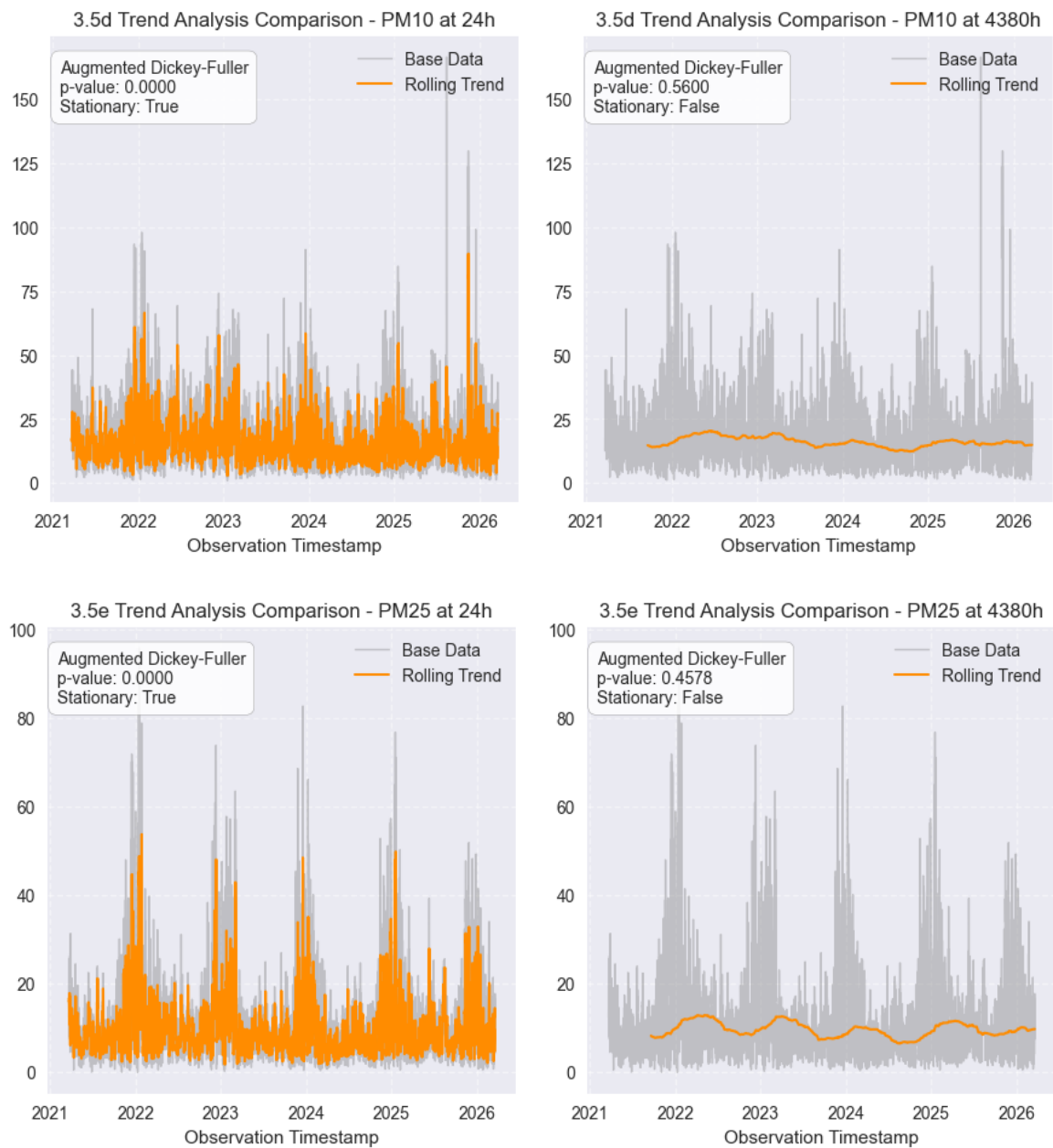
Hourly trend – Differentiation orders



Trend and stationarity comparison





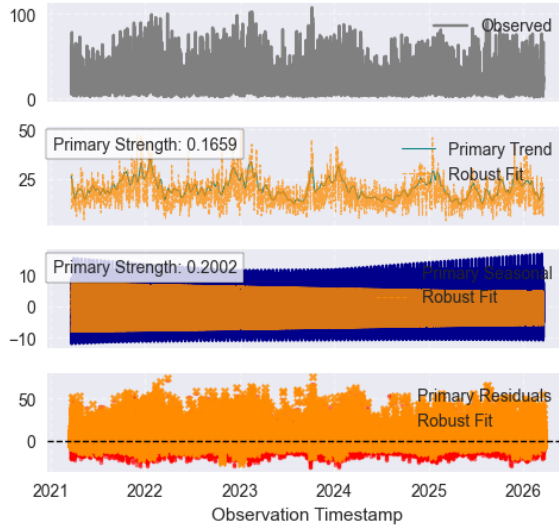


STL Decomposition

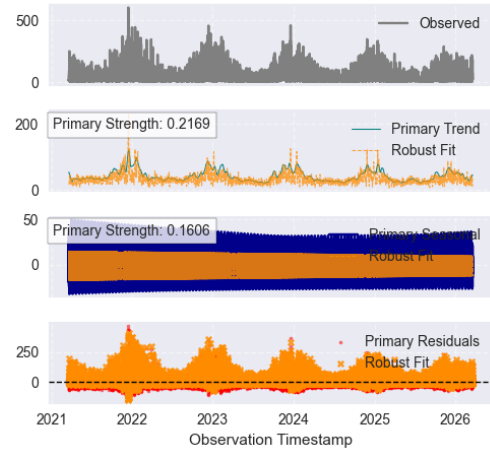
Period = 1 month

Seasonal = 6 months

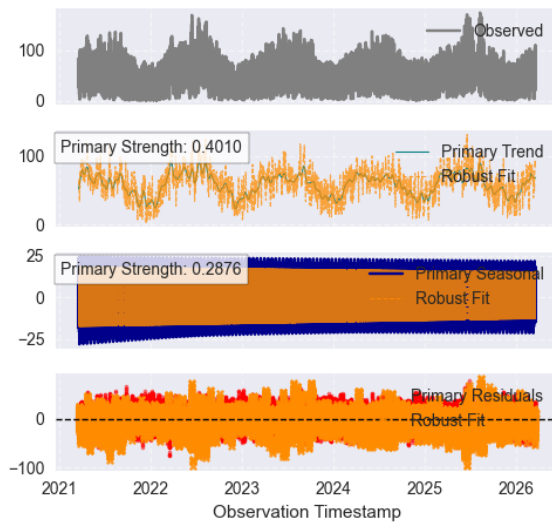
3.7a STL Comparison: Standard vs Robust NO₂



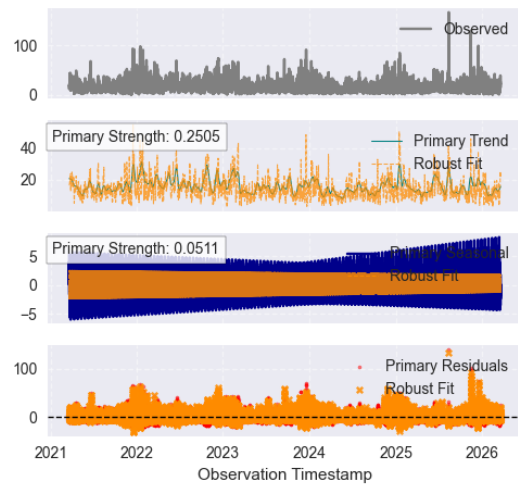
3.7b STL Comparison: Standard vs Robust NO_x



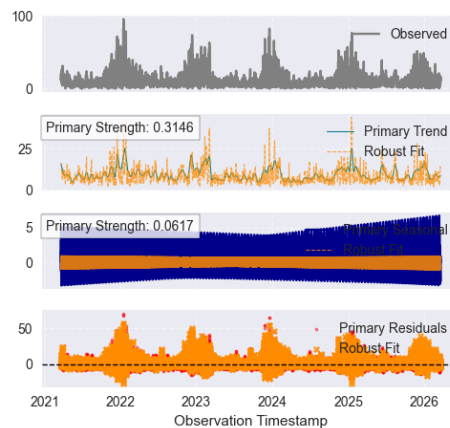
3.7c STL Comparison: Standard vs Robust O₃



3.7d STL Comparison: Standard vs Robust PM₁₀

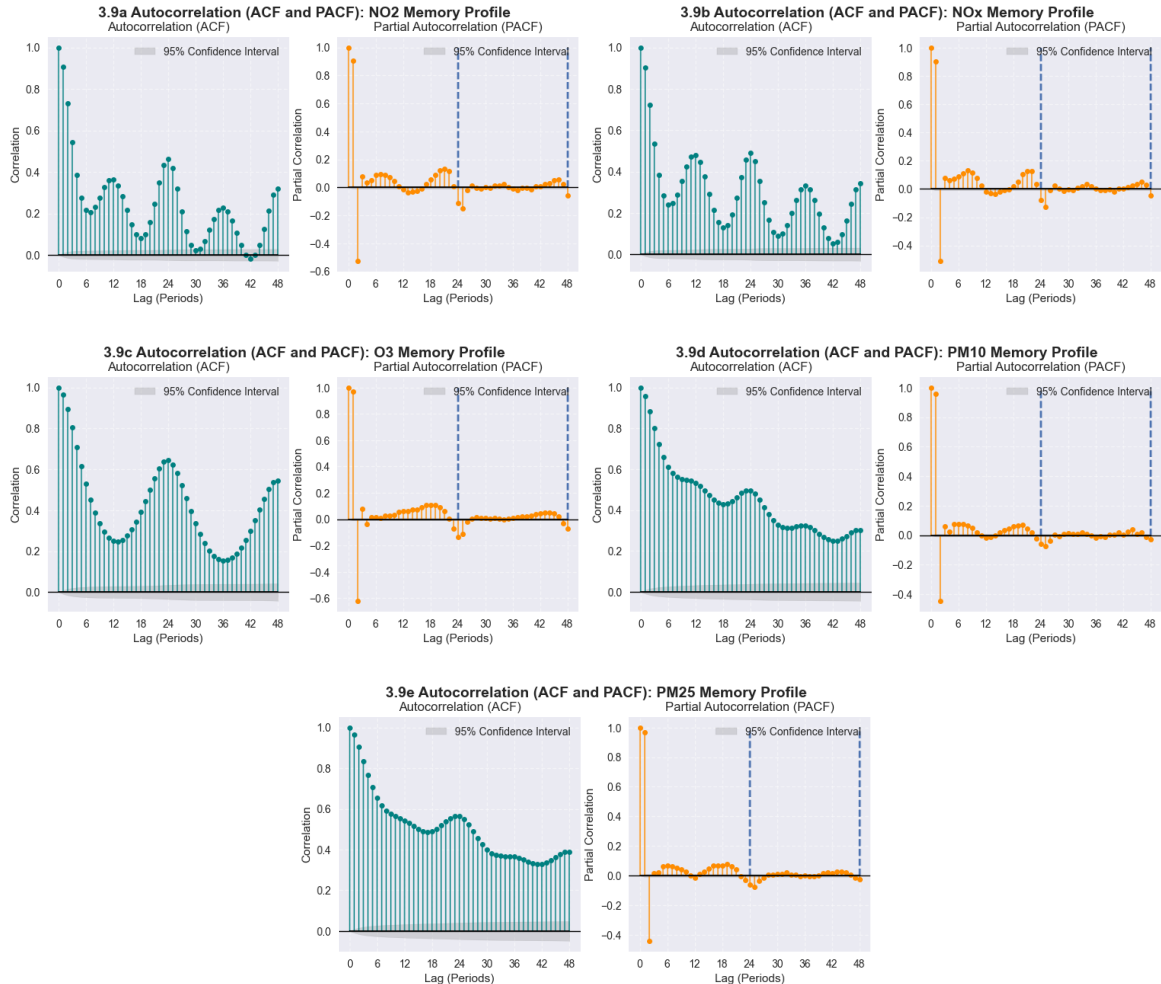


3.7e STL Comparison: Standard vs Robust PM₂₅

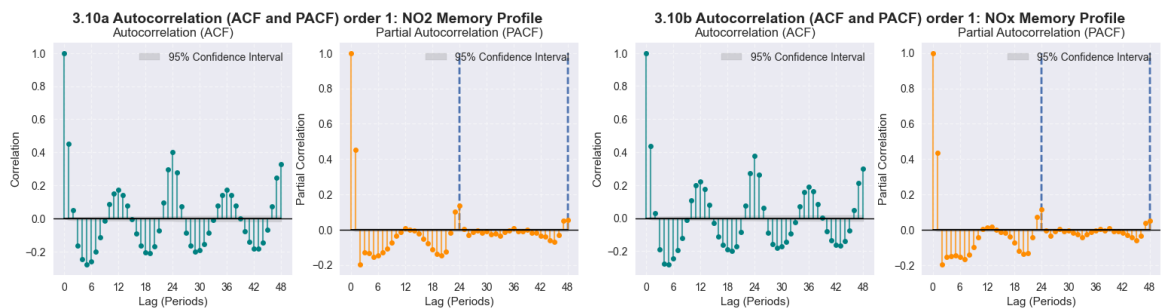


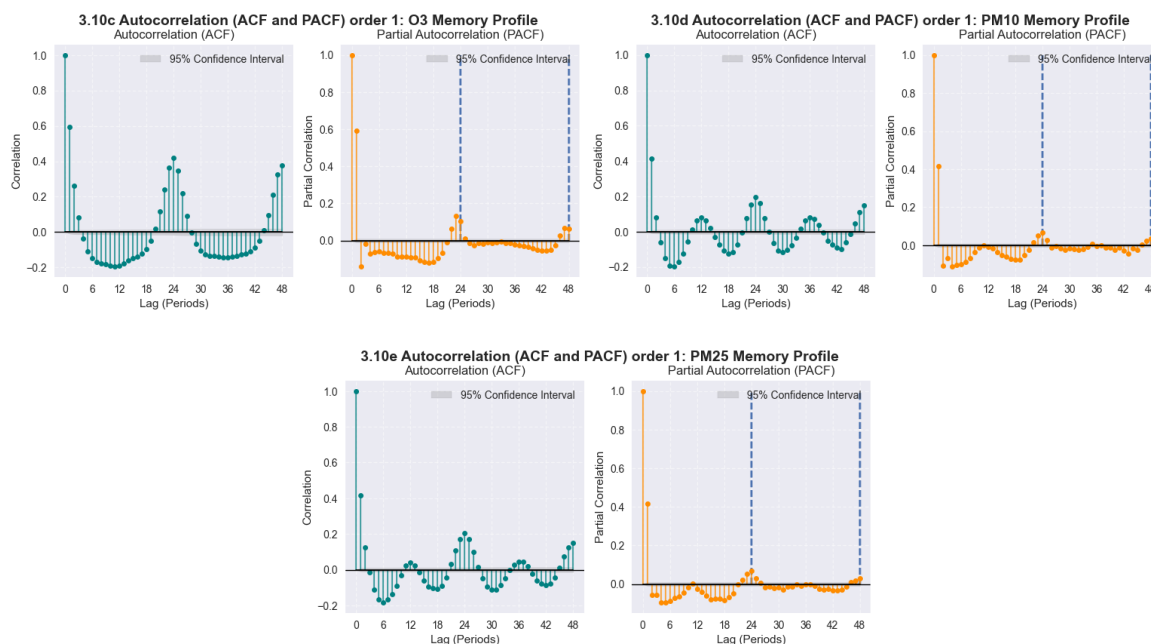
ACF and PACF plots

Order 0



First Order differential

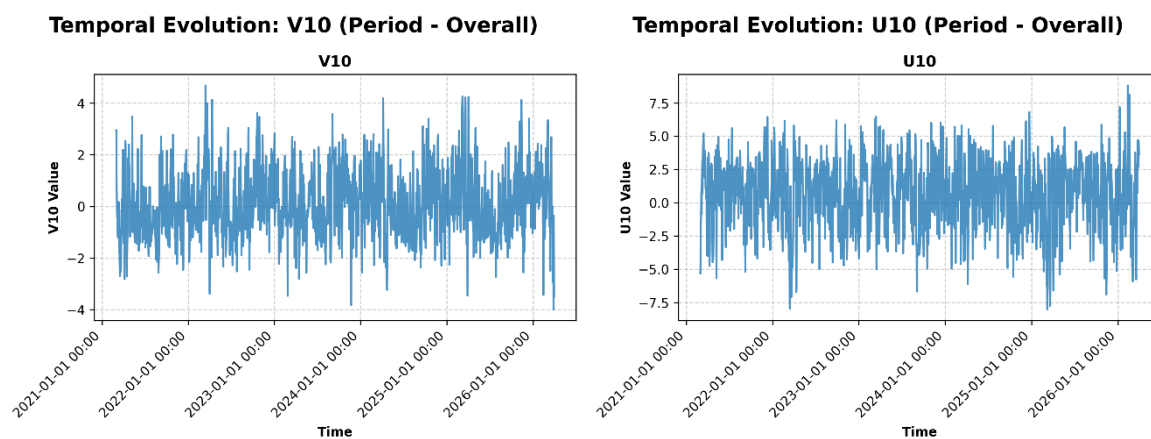




2. OTHER ENVIRONMENTAL EDA PLOTS

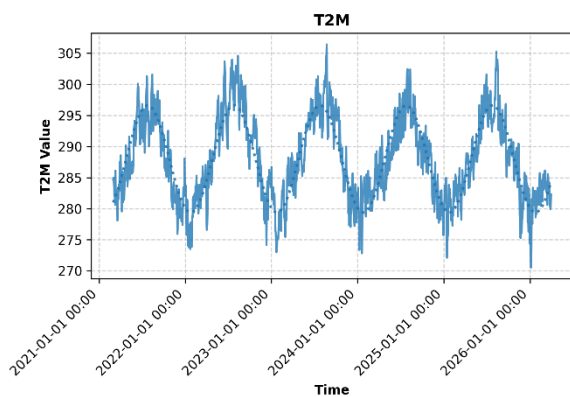
Exogenous variables time series plots

The blue dashed line is the cos fitting

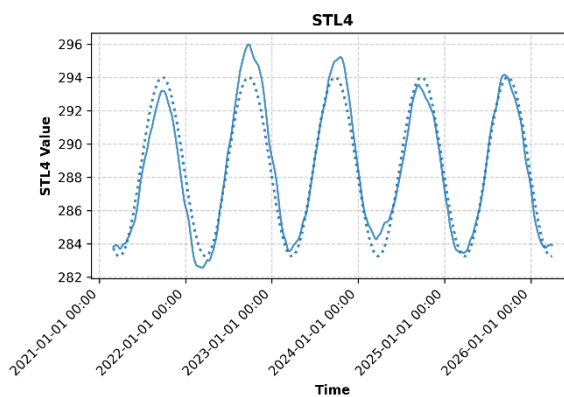




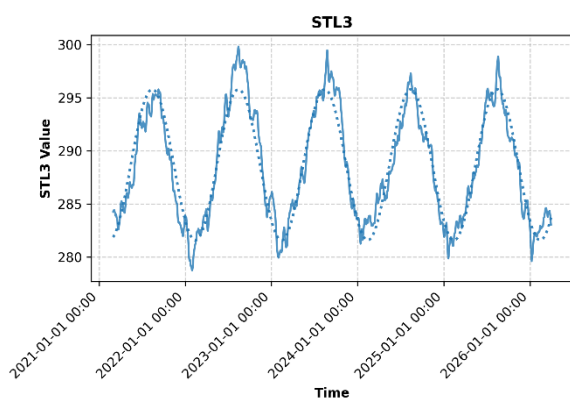
Temporal Evolution: T2M (Period - Overall)



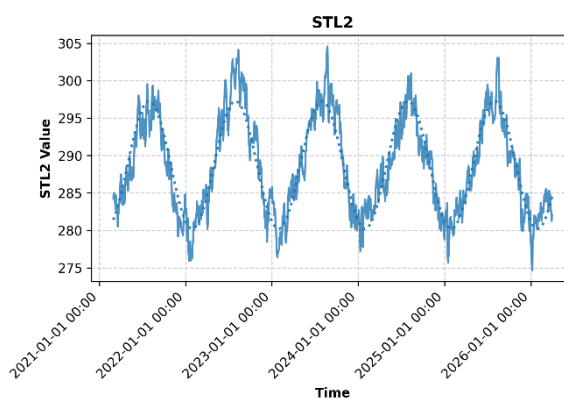
Temporal Evolution: STL4 (Period - Overall)



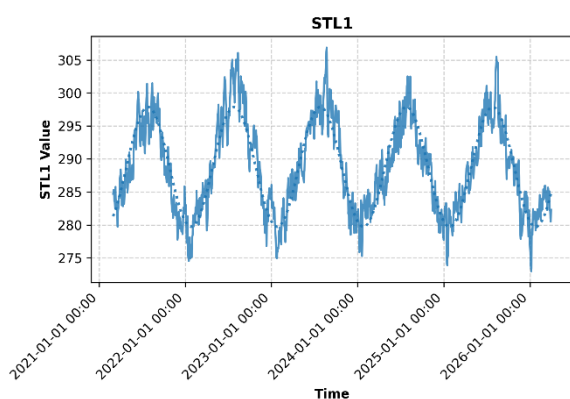
Temporal Evolution: STL3 (Period - Overall)



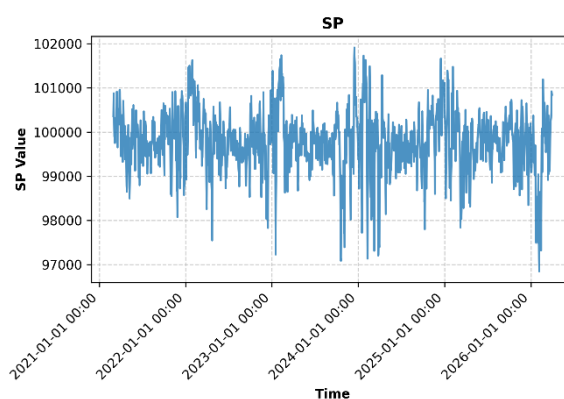
Temporal Evolution: STL2 (Period - Overall)



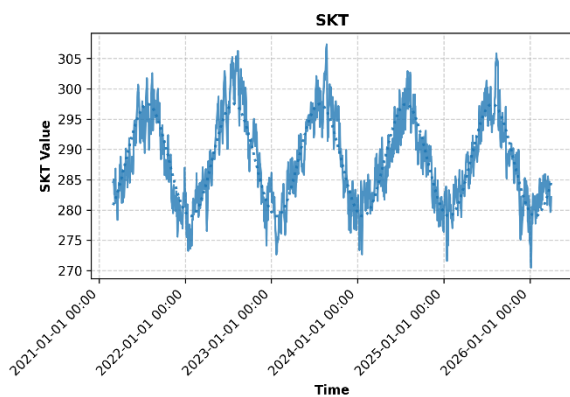
Temporal Evolution: STL1 (Period - Overall)



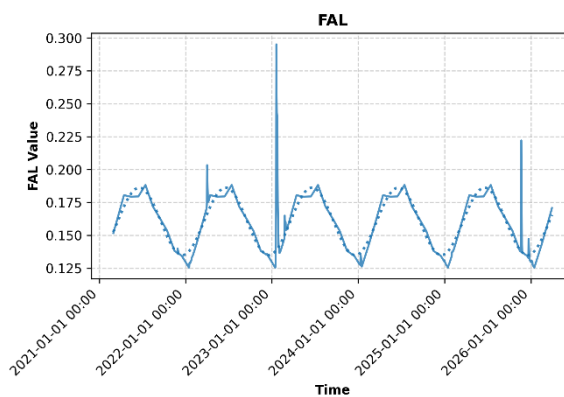
Temporal Evolution: SP (Period - Overall)



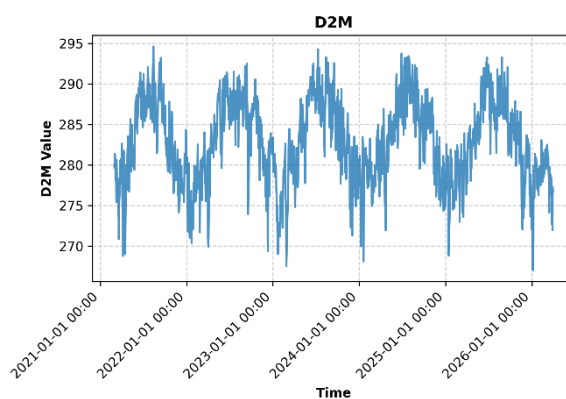
Temporal Evolution: SKT (Period - Overall)



Temporal Evolution: FAL (Period - Overall)



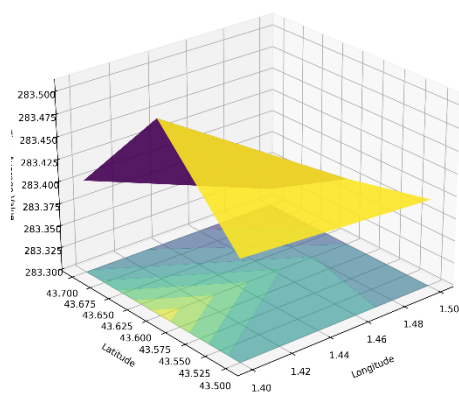
Temporal Evolution: D2M (Period - Overall)



Spatial average plots

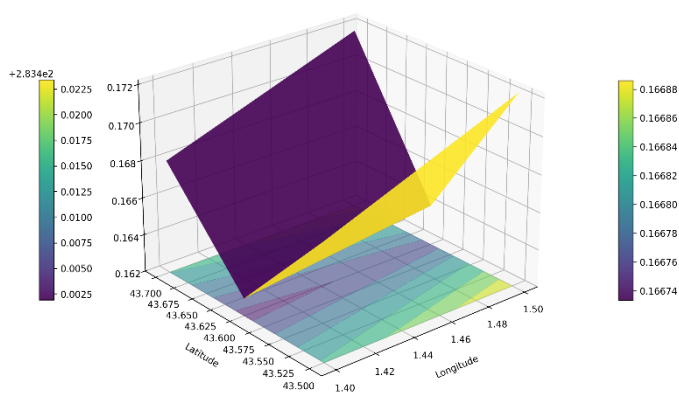
Period - Overall

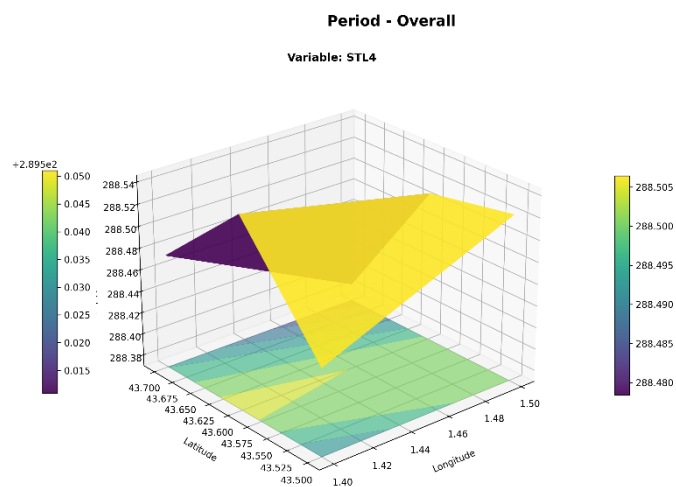
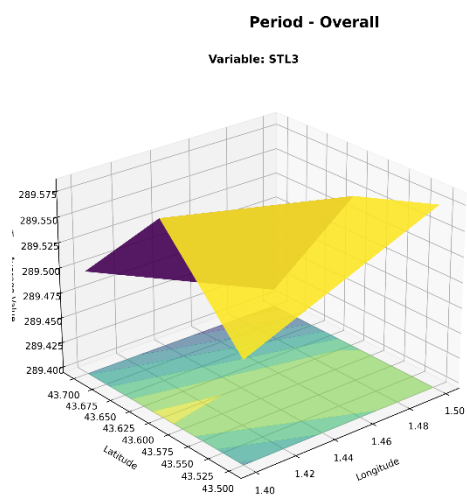
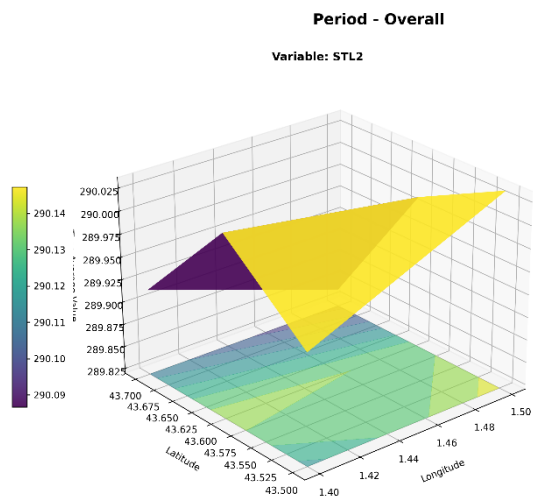
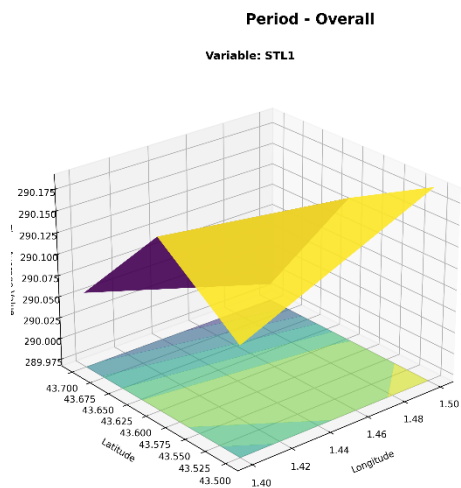
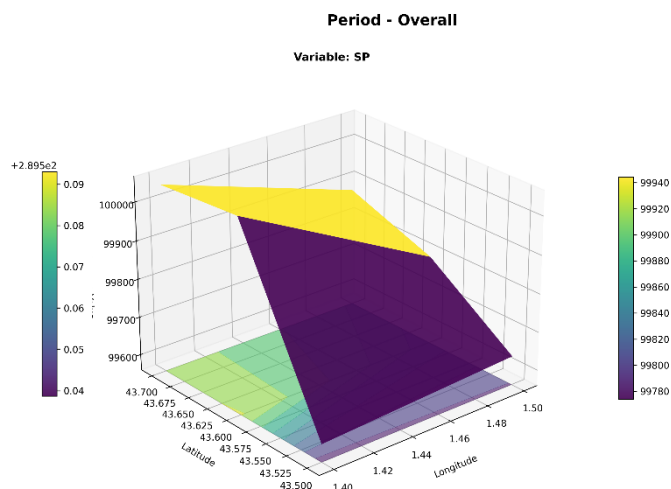
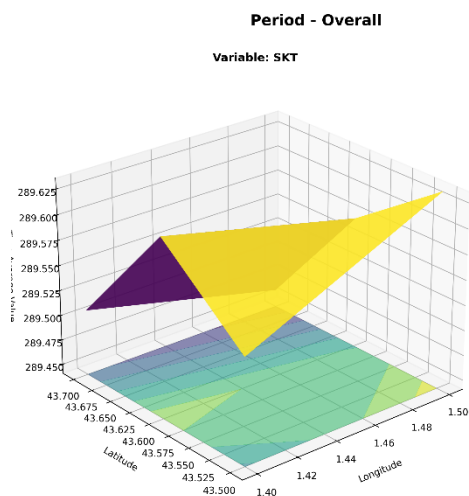
Variable: D2M

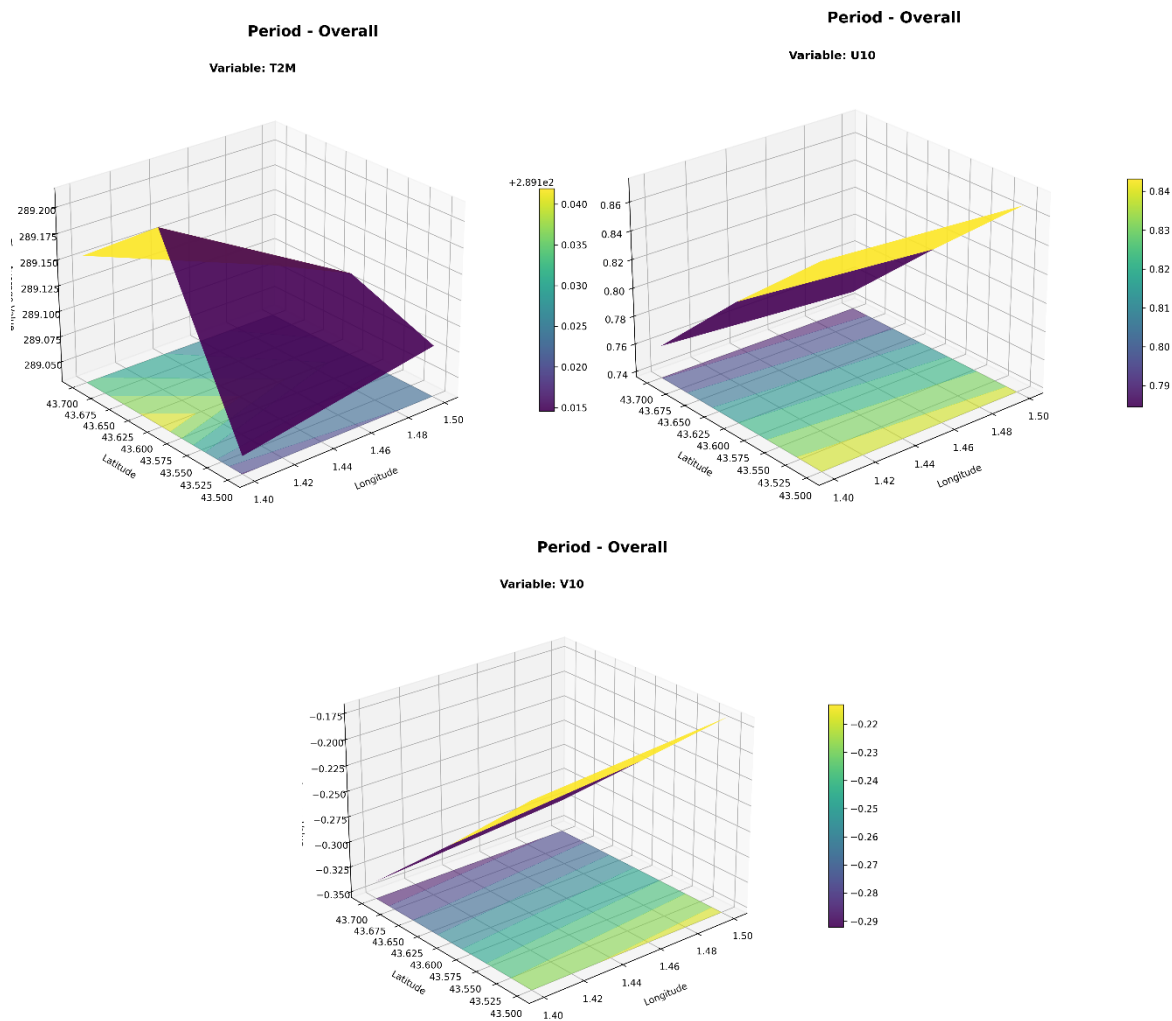


Period - Overall

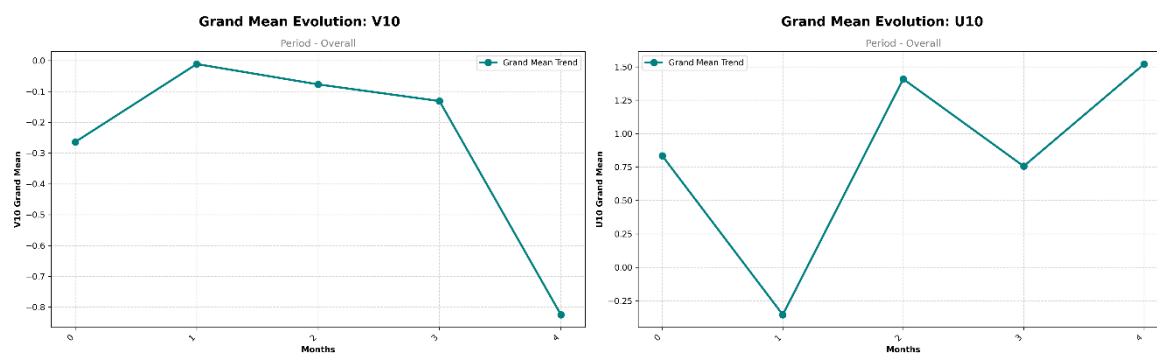
Variable: FAL

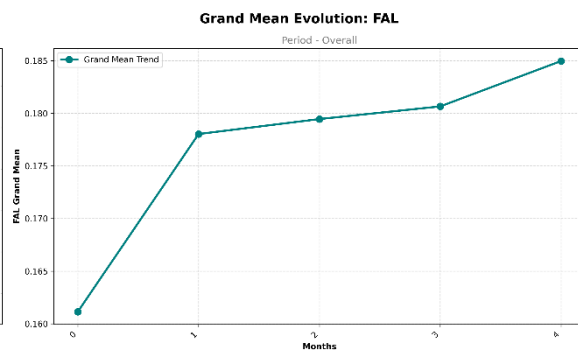
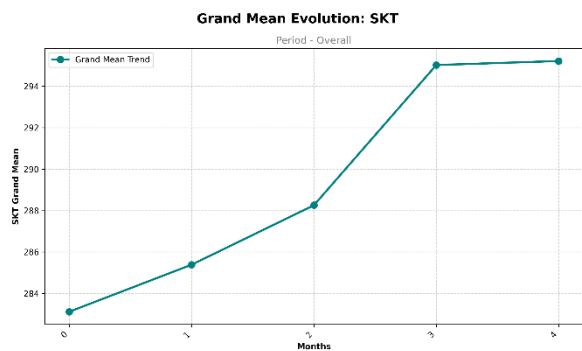
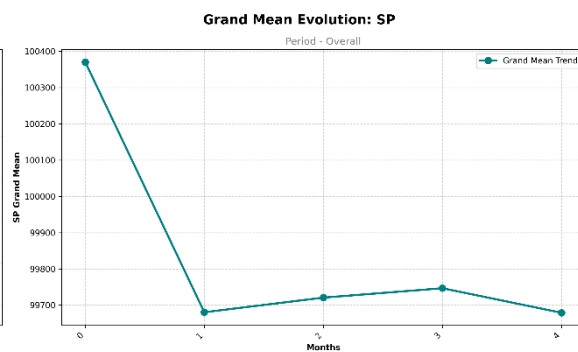
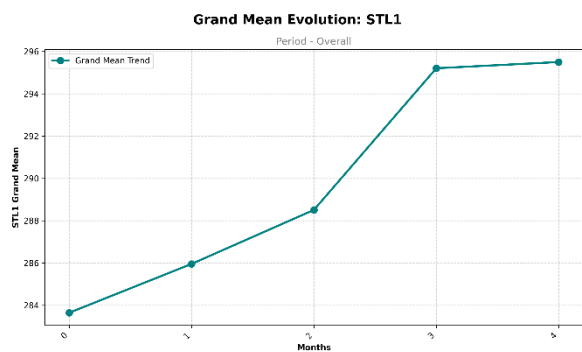
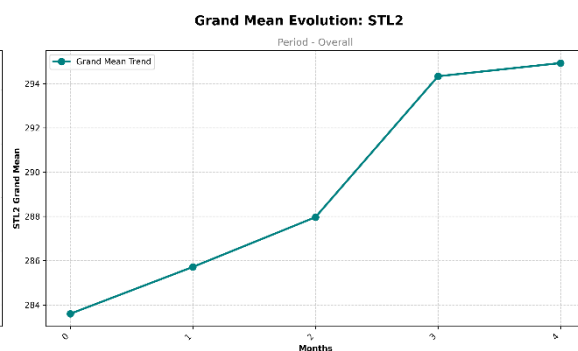
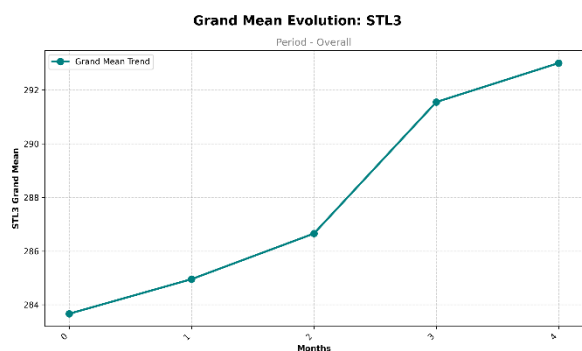
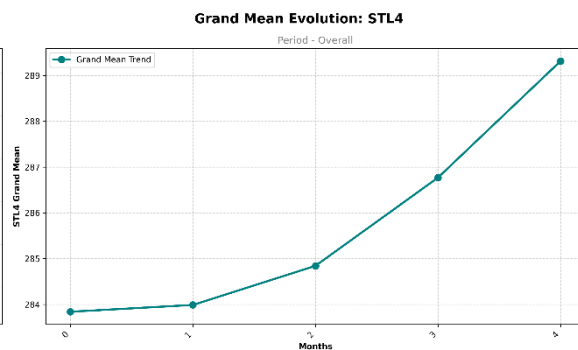
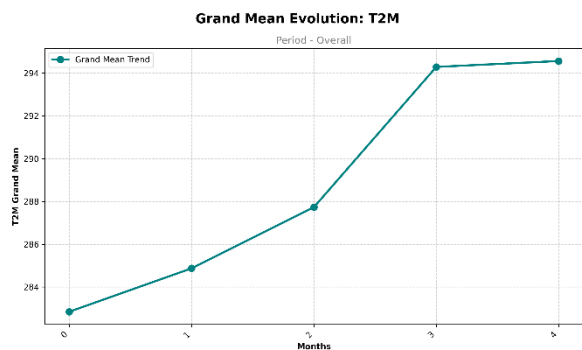


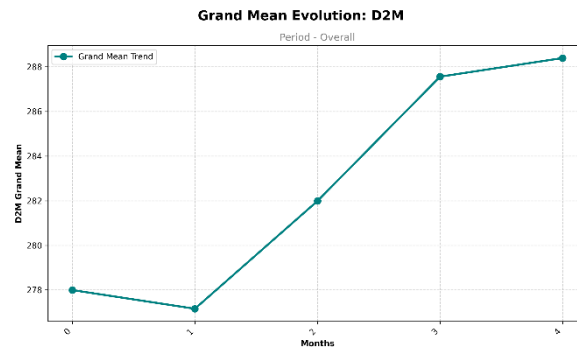




Yearly average plots



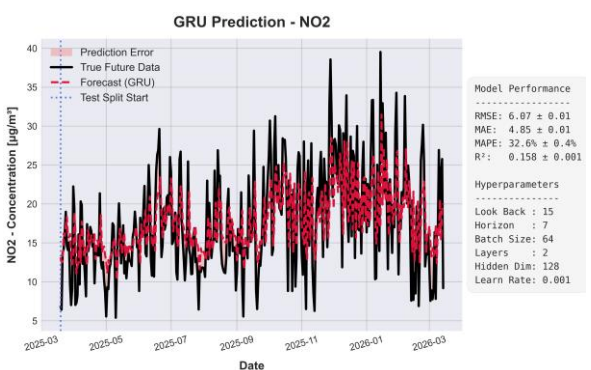
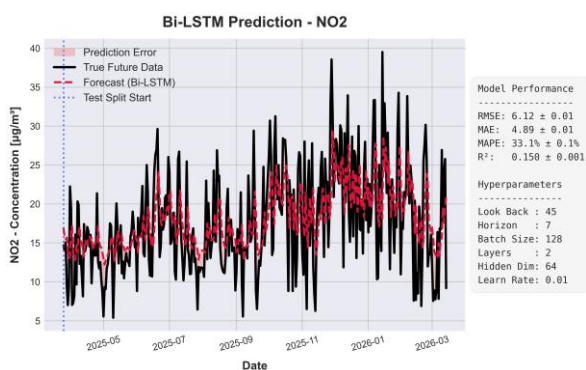
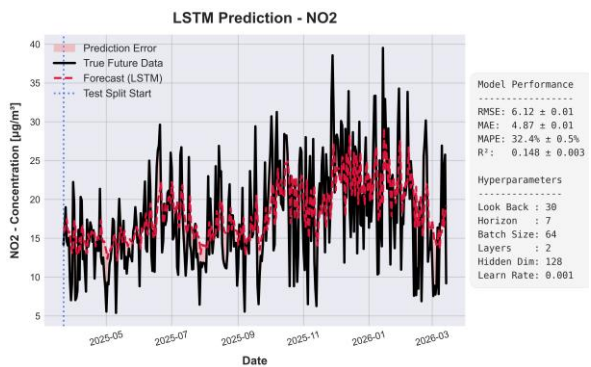
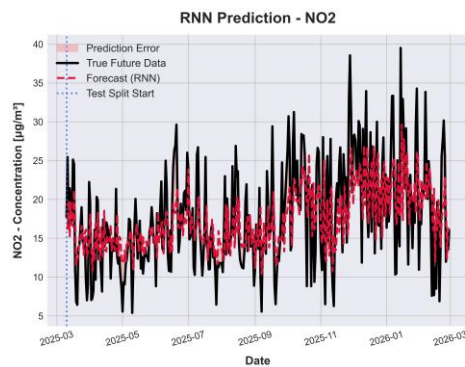


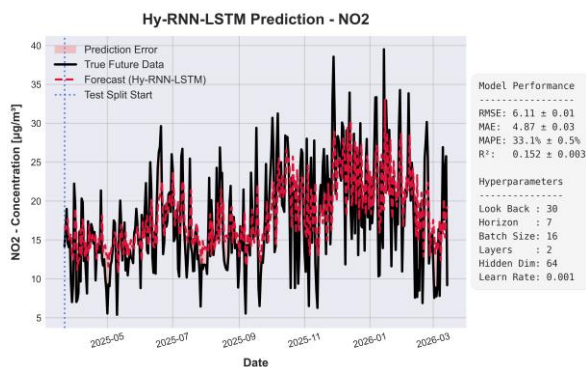


3. OTHER DEEP LEARNING PLOTS

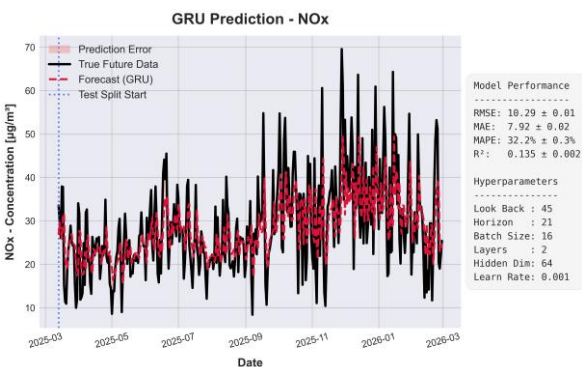
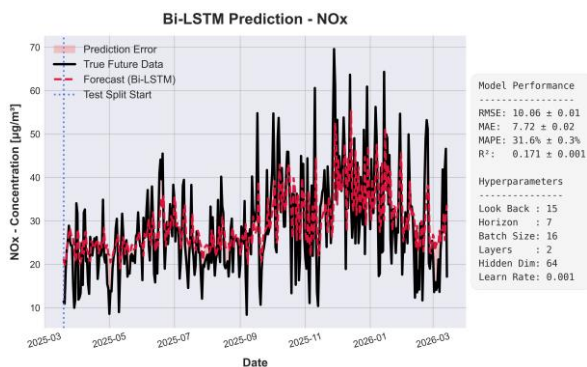
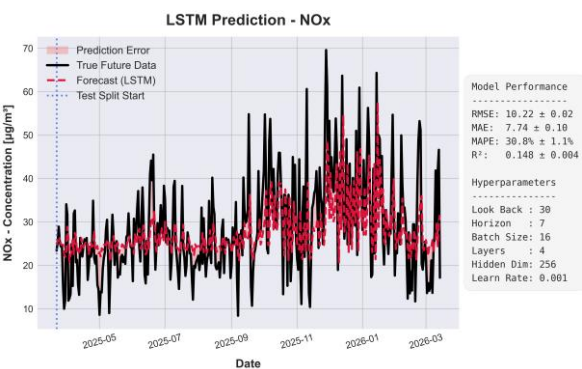
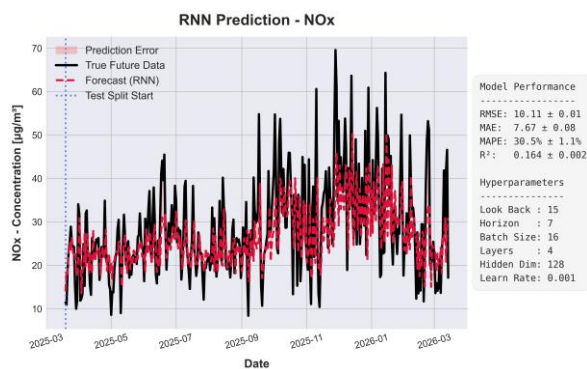
3.1 Univariate Models

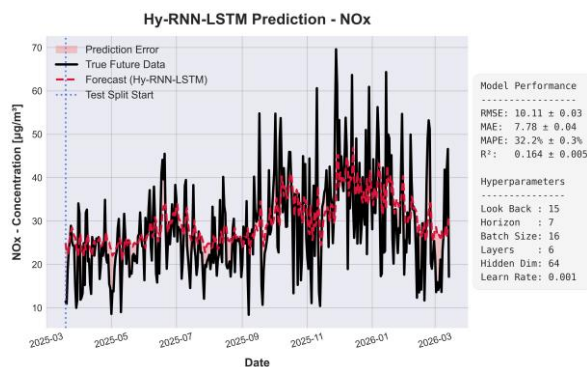
NO₂



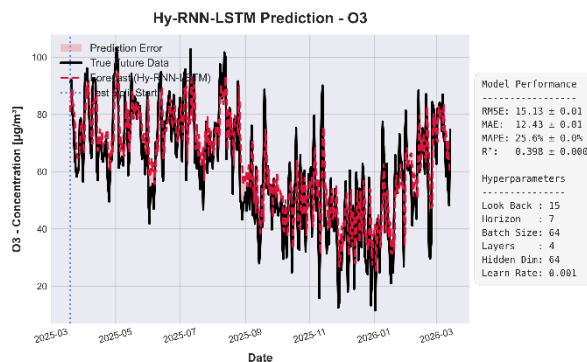
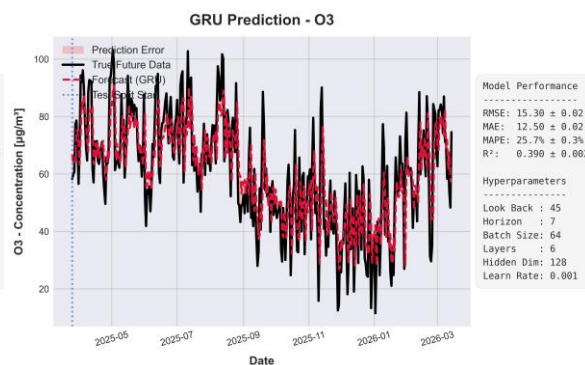
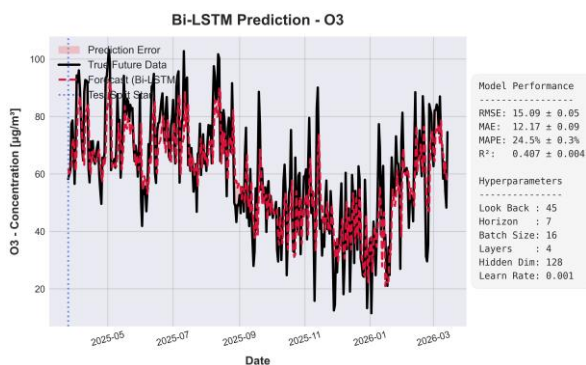
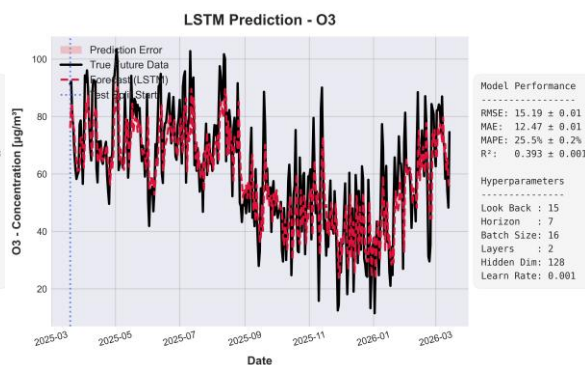
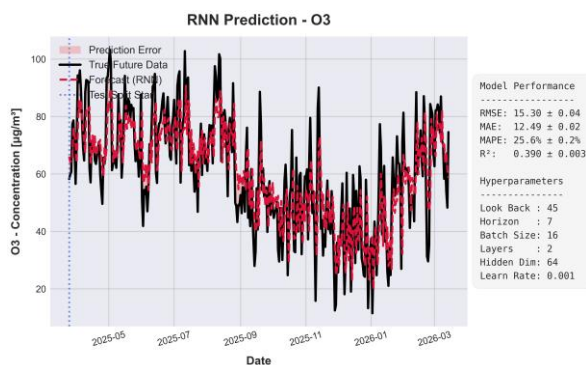


NO_x

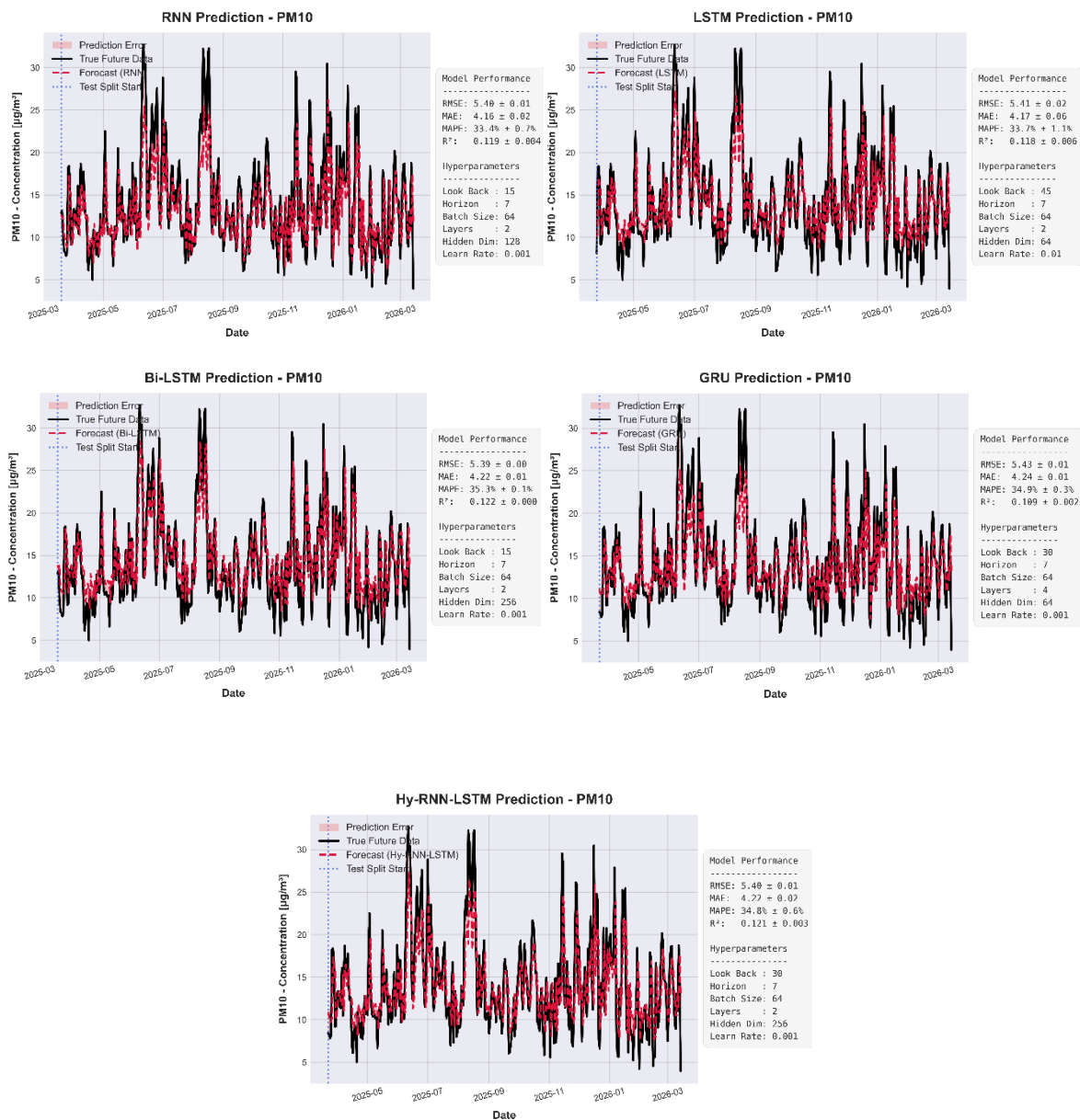




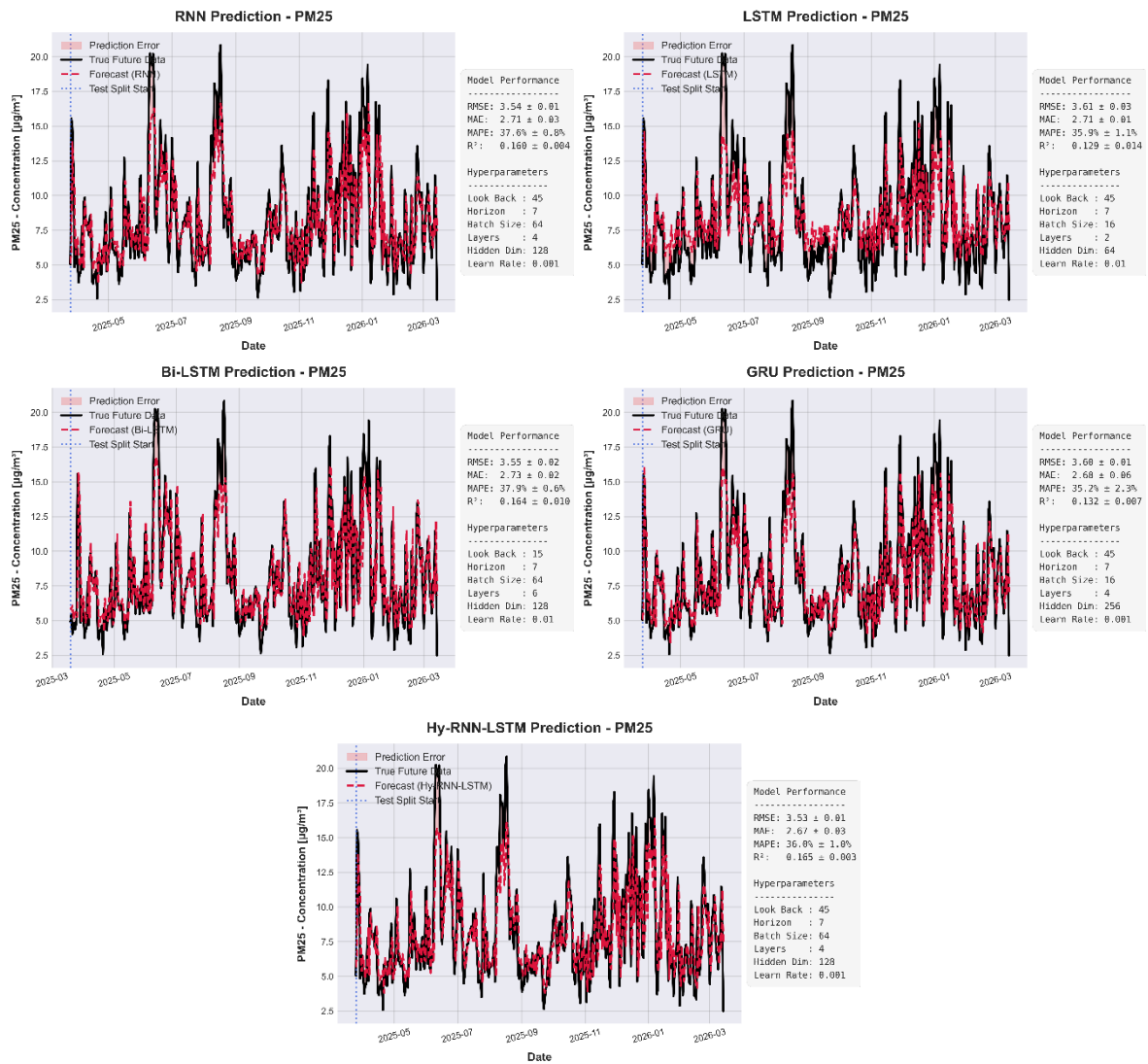
O₃



PM₁₀

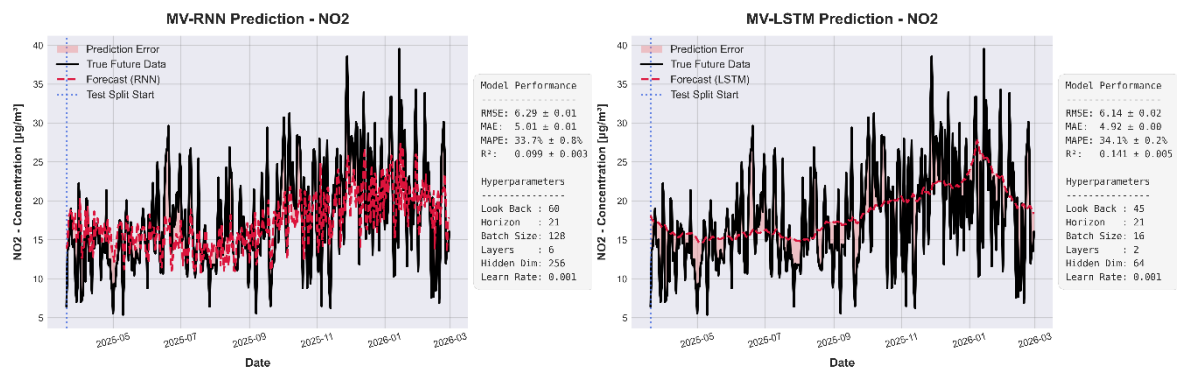


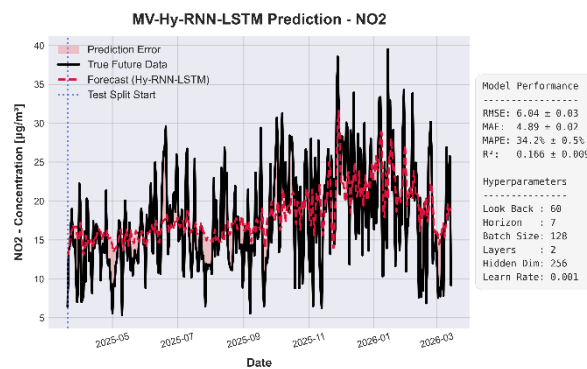
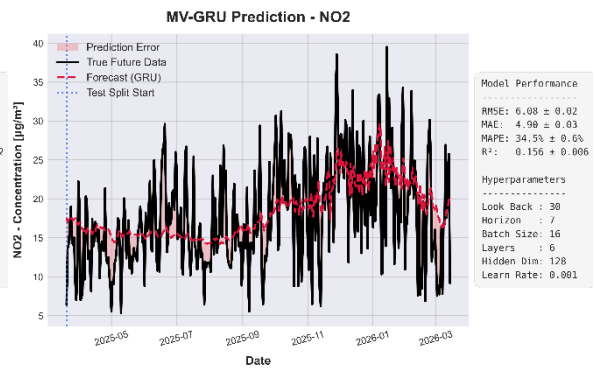
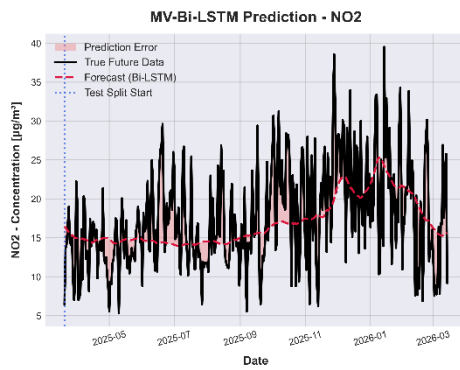
PM_{2.5}



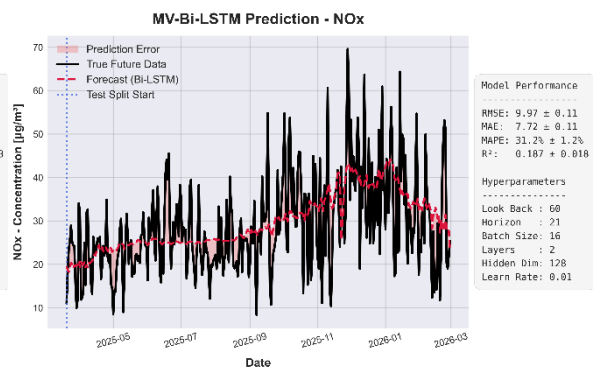
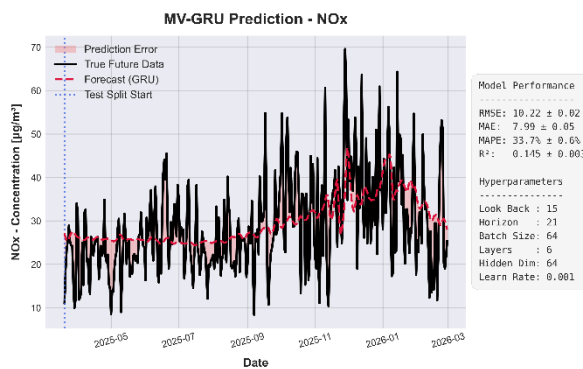
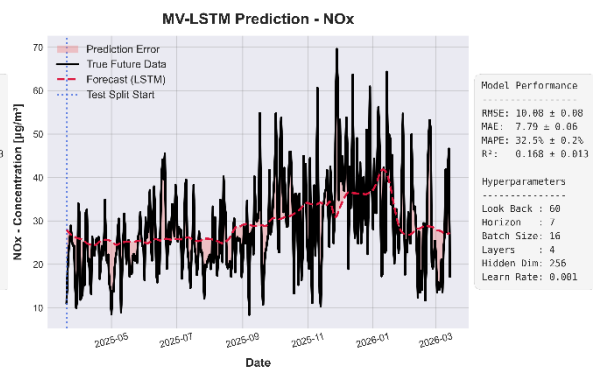
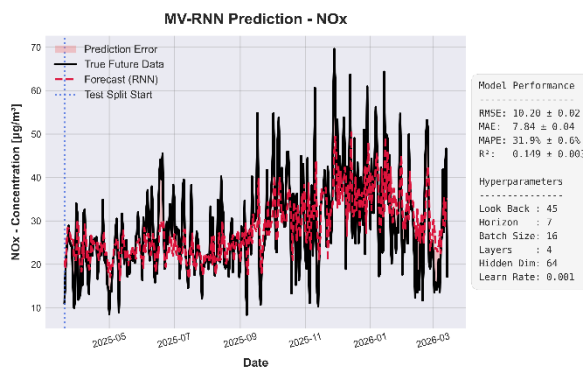
3.2 Multivariate Models

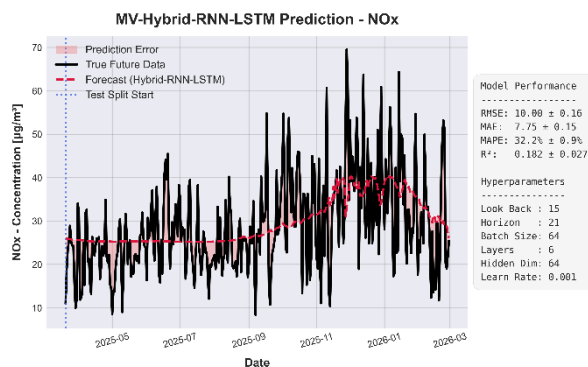
NO₂



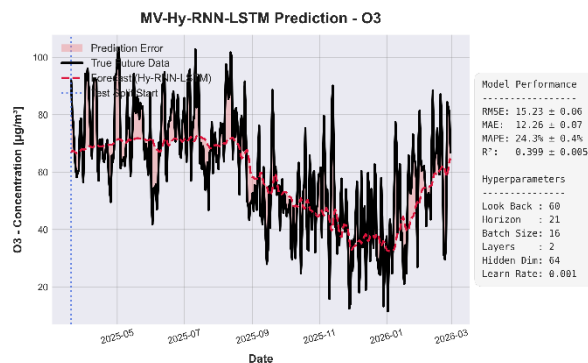
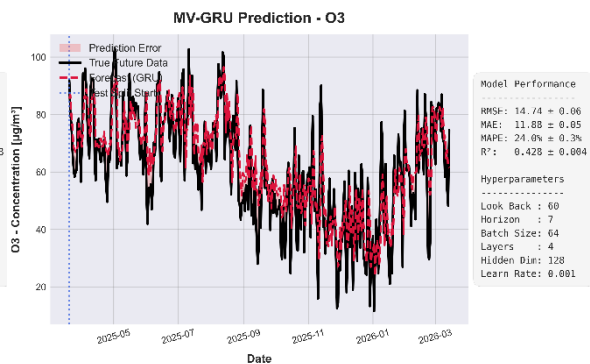
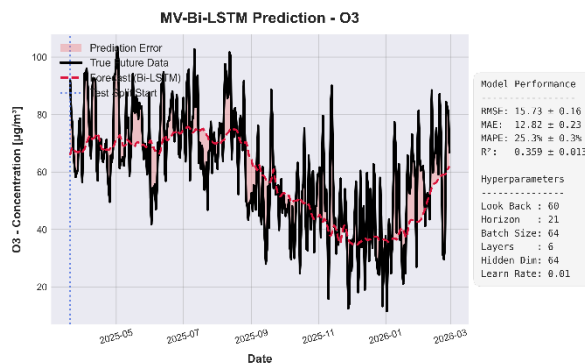
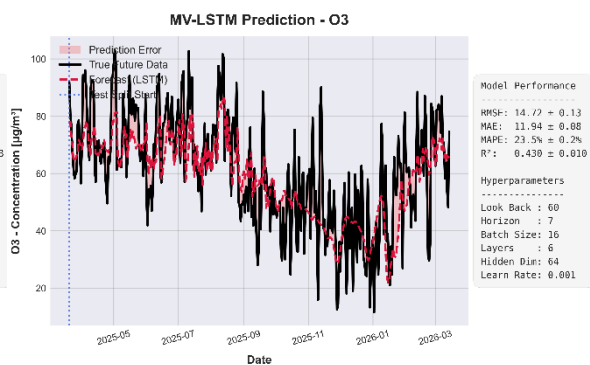
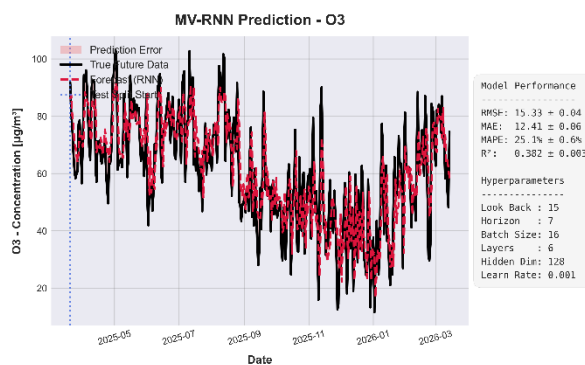


No_x

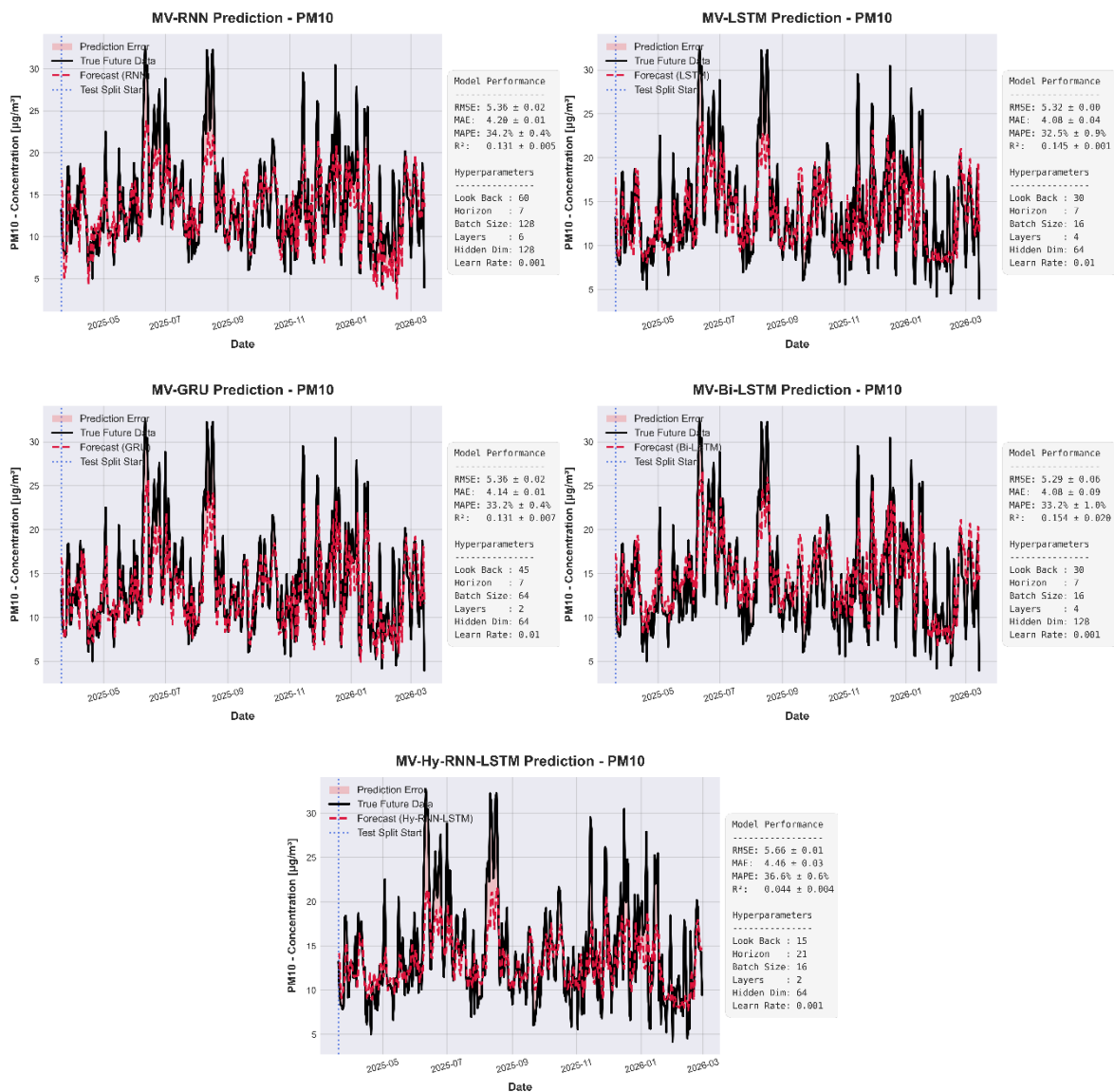




O₃



PM₁₀



PM_{2.5}

